

# Robust Structural Estimation under Misspecified Latent-State Dynamics\*

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## Abstract

Estimation and counterfactual analysis in dynamic structural models rely on assumptions about the dynamic process of latent variables, which may be misspecified. We propose a framework to quantify the sensitivity of scalar parameters of interest (e.g., welfare, elasticity) to such assumptions. We derive bounds on the scalar parameter by perturbing a reference dynamic process, while imposing a stationarity condition for time-homogeneous models or a Markovian condition for time-inhomogeneous models. The bounds are the solutions to optimization problems, for which we derive a computationally tractable dual formulation. We establish consistency, convergence rate, and asymptotic distribution for the estimator of the bounds. We demonstrate the approach with two applications: an infinite-horizon dynamic demand model for new cars in the United Kingdom, Germany, and France, and a finite-horizon dynamic labor supply model for taxi drivers in New York City. In the car application, perturbed price elasticities deviate by at most 15.24% from the reference elasticities, while perturbed estimates of consumer surplus from an additional \$3,000 electric vehicle subsidy vary by up to 102.75%. In the labor supply application, the perturbed Frisch labor supply elasticity deviates by at most 76.83% for weekday drivers and 42.84% for weekend drivers.

**Keywords:** Dynamic structural models, Sensitivity analysis, Misspecified dynamics

**JEL Codes:** C14, C18, C51, C61

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# 1 Introduction

Dynamic structural models are useful tools for counterfactual policy analysis in various fields of economics. The dynamic process of latent variables is a key feature of these models, as it captures the persistence of unobserved factors that affect agents' decisions over time. Examples of potentially serially correlated latent variables include product characteristics in demand estimation (Nair (2007); Schiraldi (2011); Gowrisankaran and Rysman (2012)), search costs in consumer search (Koulayev (2014)), firm productivity in trade (Piveteau (2021)), patent profitability in optimal stopping (Pakes (1986)), quality in technology adoption (De Groote and Verboven (2019)), health shocks in insurance (Fang and Kung (2021)), and beliefs about ability in labor economics (Miller (1984); Arcidiacono et al. (2025)).

Assumptions about the dynamic process governing the serial dependence of latent variables are central to the estimation and counterfactual analysis in dynamic structural models. These assumptions capture agents' uncertainty about the future, such as a consumer's uncertainty about future product characteristics in demand estimation. The misspecification of these assumptions can lead to biased estimates of future continuation values, which in turn bias counterfactual predictions (e.g., welfare and elasticity). However, the direction and magnitude of this bias are unclear, because the models are dynamic and nonlinear. This raises the need for sensitivity analysis of empirical results to these distributional assumptions.

In this paper, we propose a framework to quantify this sensitivity by perturbing a reference dynamic process to compute bounds on a scalar parameter of interest. The scalar parameter (e.g., welfare and elasticity) is a function of model primitives, such as model parameters, the transition distribution of latent variables, and the value function in dynamic structural models. In our proposed framework, the bounds on this scalar parameter are the solutions to constrained optimization problems whose feasible region (identified set) is defined by moment conditions for estimation, structural constraints (e.g., the Bellman equation), and the perturbation set around the reference transition distribution. The distribution that achieves the bound is called the worst-case distribution.

A central challenge is to define the perturbation set in a way that simplifies computation while maintaining key structural features of the dynamic process of latent variables. For time-homogeneous models, the structural feature is the stationarity condition, i.e., the perturbed dynamic process must be stationary. For finite-horizon, time-inhomogeneous models, the perturbed trajectory of latent variables must be Markovian. In addition, the terminal distribution is fixed because it can be nonparametrically point identified (Lewbel (2000); Matzkin (2007)). Because the stationarity and Markov conditions are functional constraints

imposed directly on the distribution being optimized, they are computationally difficult to impose. We contribute to the sensitivity analysis literature (e.g., [Christensen and Connault \(2023\)](#)) by providing a computationally tractable framework to deal with these constraints.

There are further practical challenges. First, the properties (e.g., closed-form, smoothness) of the worst-case distribution are typically unclear. Second, the expectations that define the scalar parameter, the model-implied moments, and the structural constraints are all calculated with respect to the perturbed distribution. Because this distribution is itself an optimization variable that changes during the optimization, the numerical integration can be difficult to implement. Third, the value function in dynamic structural models is infinite-dimensional due to the serial dependence of latent variables, which complicates the optimization problem further.

To address these challenges, we define the perturbation set as a Kullback–Leibler (KL) divergence ball around the reference distribution. For the time-homogeneous case, the reference distribution is a joint distribution of current and future latent variables. For the time-inhomogeneous case, it is the joint distribution of the entire trajectory of latent variables. The KL radius controls the size of the perturbation set. We then employ the Optimal Transport (OT) framework to impose structural constraints on the distribution of latent variables. In the time-homogeneous case, the stationarity condition requires that the marginal distributions of current and future latent variables of the perturbed joint distribution coincide, which can be imposed using OT. This marginal constraint, together with the KL divergence penalty from the problem’s Lagrangian, yields a computationally tractable Entropic Optimal Transport (EOT) problem. In the time-inhomogeneous case, the perturbed trajectory must be Markovian, and the terminal distribution is fixed. We also show how to formulate this as an EOT problem.

The EOT problem can be solved using its dual formulation, which has three key advantages over the primal formulation. First, it provides a closed-form expression for the worst-case distribution and characterizes its smoothness. During our proposed optimization algorithm, this closed-form expression is used to update the value function in dynamic structural models. Second, the Sinkhorn algorithm ([Sinkhorn and Knopp \(1967\)](#); [Cuturi \(2013\)](#)) allows us to solve the EOT problem and compute the worst-case distribution efficiently. Finally, because the expectations in the dual problem are taken with respect to the reference distribution, an appropriate numerical integration method can be chosen in advance.

Then we consider three complementary sensitivity measures to interpret the results. First, the *global sensitivity* approach computes the largest deviation from the reference value. It progressively increases the KL radius until a pre-specified maximum radius is reached. We

show that it provides a tractable approximation to the nonparametric bounds on the scalar parameter when the KL divergence constraint is removed. Moreover, we derive an explicit upper bound on the approximation error, which enables us to control the error within a desired level. Second, the *local sensitivity* approach analyzes the effect of small perturbations. It computes the right derivatives of the bounds with respect to the KL radius at zero, which serves as our local sensitivity measure. Finally, the *robustness metric* approach, inspired by [Spini \(2024\)](#), computes the smallest deviation from the reference distribution required to produce sensitive results. It is the smallest KL divergence from the reference distribution required for the scalar parameter’s value to fall below a user-specified threshold (e.g., 5% below the reference value). In addition to our three sensitivity measures, we can also estimate an alternative model and set the radius as the KL divergence from the alternative to reference models.

For large sample properties, we propose an estimator for the bound, establishing its consistency and convergence rate. To this end, we first establish the consistency and convergence rate of the estimator of the identified set, following [Chernozhukov et al. \(2007\)](#). We then derive the asymptotic distribution of the plug-in estimator by proving the Hadamard directional differentiability of the bound.

We apply our framework to an infinite-horizon dynamic demand model for new cars in the UK, Germany, and France. We consider the sensitivity of the price elasticity and consumer surplus from an additional \$3,000 electric vehicle subsidy. In the model, the indirect utility is a latent variable due to unobserved product characteristics, and its transition is typically modeled as an AR(1) process (e.g., [Schiraldi \(2011\)](#); [Gowrisankaran and Rysman \(2012\)](#)). For the price elasticity, we find that the French market is the least sensitive to the distributional assumption (at most 6.20% deviation from the reference elasticity), while the German market is the most sensitive (at most 15.24% deviation). We also find that this sensitivity is relatively stable over time for all three markets. For the consumer surplus, the German market is also the most sensitive (at most 102.75% deviation from the reference consumer surplus), while the UK and French markets are less sensitive (at most 25.17% and 24.73% deviation, respectively). Importantly, the results remain economically meaningful even under the worst-case distribution, with the consumer surplus remaining at least \$309 million for the German market, \$1,243 million for the French market, and \$2,584 million for the UK market.

We also apply our framework to a finite-horizon dynamic labor supply model for taxi drivers in New York City. We consider the sensitivity of the elasticity of stopping work with respect to wage and the Frisch elasticity of labor supply. In the model, the market-level supply shock is a latent variable, and its transition is also modeled as an AR(1) process.



For the elasticity of stopping work, we find that weekday drivers’ elasticity is more sensitive in the morning, while weekend drivers’ elasticity is more sensitive in the afternoon. For the Frisch elasticity, both weekday and weekend drivers’ elasticities are sensitive to the distributional assumption, with at most 76.83% and 42.84% deviation from the reference elasticity, respectively.

## 1.1 Related Literature

*Identification and estimation.* Many papers have focused on identification in a range of dynamic structural models including finite mixture models (Kasahara and Shimotsu (2009); Luo et al. (2022); Higgins and Jochmans (2023, 2025)), unobservable Markov processes (Hu and Shum (2012)), and counterfactual conditional choice probabilities in dynamic binary choice models (Norets and Tang (2014)). Berry and Compiani (2023) uses the generalized instrumental variable approach for unobserved state variables in dynamic discrete choice (DDC) models. In fixed effects DDC models, Aguirregabiria et al. (2021) considers identification of structural parameters using sufficient statistics, while Aguirregabiria and Carro (2024) studies identification of average marginal effects. Hwang (2024) employs proxy variables for the latent variables. Kalouptside et al. (2021c) proposes the Euler Equations in Conditional Choice Probabilities (ECCP) estimator. By leveraging finite-dependence properties and cross-sectional data, it identifies structural parameters in the presence of serially correlated market-level unobserved variables without distributional assumptions. Arcidiacono and Miller (2011) adapts the Expectation-Maximization algorithm to estimate DDC models with discrete latent types. Norets (2009) extends the Bayesian estimation of DDC models by Imai et al. (2009) to allow for serially correlated latent variables, while Blevins (2016) proposes a sequential Monte Carlo method. Chiong et al. (2016) shows identification in DDC models is an optimal transport problem under serial independence of utility shocks. In addition, Chen et al. (2011), Schennach (2014), and Fan et al. (2023, 2025) consider inference of finite-dimensional parameters in the presence of an infinite-dimensional parameter, namely the distribution of latent variables.

Our framework contributes to this literature in three ways. First, we focus directly on a scalar parameter (e.g., welfare, elasticity) rather than on the identified set of model primitives. Second, to analyze sensitivity, we consider a perturbation set around the reference distribution rather than the set of all distributions. Third, we complement identification strategies that do not rely on distributional assumptions. In the labor supply application, we use the ECCP estimator of Kalouptside et al. (2021c) to point identify the utility parameters and then conduct sensitivity analysis of the labor supply elasticity with respect to

assumptions about the dynamic process of the market-level supply shock.

*Sensitivity analysis and robustness.* In DDC models, [Kalouptsi et al. \(2021a,b\)](#) relax common normalizations on the utility function. [Bugni and Ura \(2019\)](#) considers the local misspecification of the transition density of observable variables, assuming the transition density is correctly specified in the limit (as the sample size goes to infinity) in DDC models. [Andrews et al. \(2017\)](#) considers a setting in which the moments are locally misspecified under the reference distribution. Subsequent work [Armstrong and Kolesár \(2021\)](#) constructs near-optimal confidence intervals in such models. [Kitamura et al. \(2013\)](#) considers the robust estimation under moment restrictions. [Bonhomme and Weidner \(2022\)](#) perturbs the reference model and assumes the size of the perturbation shrinks to zero as the sample size goes to infinity. [Chen et al. \(2024\)](#) relaxes the rational expectation assumption. [Gu and Russell \(2024\)](#) considers the identification of scalar counterfactual parameters using optimal transport. [Spini \(2024\)](#) studies the robustness of policy effects to changes in the distribution of covariates. [Armstrong \(2025\)](#) provides a selective review of misspecification in econometrics. Most closely related is [Christensen and Connault \(2023\)](#), who conducts sensitivity analysis with respect to parametric assumptions about the distribution of latent variables in structural models. However, they focus on relaxing the marginal distribution assumption while maintaining serial independence.

We complement this literature by providing a computationally tractable framework for robust structural estimation under misspecified latent-state dynamics. Our main contribution is to handle the key structural constraints that make such problems difficult: the stationarity condition in time-homogeneous models and the Markov property together with the fixed terminal distribution in time-inhomogeneous models. These constraints are not addressed in [Christensen and Connault \(2023\)](#), who focus on relaxing marginal distributional assumptions while maintaining serial independence. In addition, by leveraging EOT duality, we obtain a closed-form expression for the worst-case distribution, which allows us to overcome the computational challenges created by the infinite-dimensional value function arising from serial dependence in dynamic structural models.

*Distributionally robust optimization (DRO).* The literature on DRO ([Kuhn et al. \(2019\)](#); [Rahimian and Mehrotra \(2019\)](#); [Blanchet et al. \(2022\)](#); [Gao and Kleywegt \(2023\)](#); [Wang et al. \(2021\)](#)) usually studies the uncertainty due to limited observability of data, noisy measurements, or estimation errors.

Our work is distinct in that we study the misspecification due to assumptions about the serial dependence of latent variables—a problem of model specification rather than data limitation. Moreover, because our framework can be treated as a moment-constrained DRO

problem, we employ the minimax theorem (see [Fan \(1953\)](#) and [Ricceri and Simons \(1998\)](#), Theorem 1.3) to exchange the order of the supremum over Lagrangian multipliers (for the moment conditions and structural constraints) and the infimum over distributions in the perturbation set. This exchange allows a direct application of duality results from the DRO literature, which simplifies our proof of the dual formulation significantly.

*Applied work.* Building on the seminal work on the estimation of DDC models ([Wolpin \(1984\)](#); [Rust \(1987\)](#); [Hotz and Miller \(1993\)](#); [Aguirregabiria and Mira \(2002, 2007\)](#); [Petersendorfer and Schmidt-Dengler \(2008\)](#); [Arcidiacono and Miller \(2011\)](#)), most applied work assumes the serial independence of utility shocks. In the presence of serially correlated latent variables, parametric models are often used, as seen in the works of [Schiraldi \(2011\)](#); [Gowrisankaran and Rysman \(2012\)](#); [Blevins et al. \(2018\)](#); [Piveteau \(2021\)](#) and others mentioned in the introduction.

We contribute to this literature by developing a computationally tractable framework for sensitivity analysis of scalar parameters of interest to these distributional assumptions. We also demonstrate our framework through two empirical applications: an infinite-horizon dynamic demand model, and a finite-horizon dynamic labor supply model.

**Outline:** Sections 2 and 3 present our framework for time-homogeneous and time-inhomogeneous models. Section 4 establishes the large sample properties of our estimator. Section 5 introduces three sensitivity measures. Section 6 discusses practical implementation. Sections 7 and 8 present two empirical applications. Section 9 concludes. Appendix A presents additional examples. All proofs are in Appendix B.

**Notation:** Let  $U \in \mathcal{U} \subseteq \mathbb{R}^d$  be a vector of latent variables with support  $\mathcal{U}$  where  $\mathcal{U}$  is assumed to be Polish. Let  $\mathcal{P}(\mathcal{U})$  be the space of Borel probability measures on  $\mathcal{U}$  and  $\mathcal{B}(\mathcal{U})$  be the Borel  $\sigma$ -algebra on  $\mathcal{U}$ . In this paper, all measures are assumed to be absolutely continuous with respect to the Lebesgue measure. Denote by  $\mathbb{E}_F[\cdot]$ ,  $\mathbb{E}_x[\cdot]$  the expectations with respect to  $F \in \mathcal{P}(\mathcal{U})$  and the probability distribution of the random variable  $x$ , respectively. For variables in a stationary dynamic context, a prime (e.g.,  $x'$ ) denotes the variable's value in the next period. Let  $F_1, F_2 \in \mathcal{P}(\mathcal{U})$ , we write  $F_1 \ll F_2$  if  $F_1$  is absolutely continuous with respect to  $F_2$ , and  $F_1 \otimes F_2$  as the product measure. For a finite-dimensional vector, denote by  $\|\cdot\|_p$  the  $p$ -norm. Denote by  $L^p(F)$  the space of functions for which  $\int |f|^p dF < \infty$ . For a set  $\mathcal{A}$ , let  $\text{int}(\mathcal{A})$  denote its interior. Denote by  $\mathbb{R}_+$  the set of non-negative real numbers, and  $\mathbb{N}$  the set of natural numbers. Finally, let  $\mathcal{J} := \{1, \dots, J\}$ .

## 2 Methodology for Time-Homogeneous Models

This section presents our methodology for time-homogeneous models. Section 2.1 defines the perturbation set. Section 2.2 gives two examples. Section 2.3 presents the general framework and derives duality results. Section 2.4 discusses how to perturb the stationary distribution.

### 2.1 Definition of Perturbation Set

In time-homogeneous models, we perturb the reference latent variable dynamics while imposing that the perturbed process remains stationary. Consider an unobserved stationary first-order Markov process  $\{\xi_t\}_{t \in \mathbb{Z}}$  with state space  $\Xi \subseteq \mathbb{R}^{d_\xi}$ . The process is stationary with respect to  $\nu_0 \in \mathcal{P}(\Xi)$  if and only if:

$$\int F_0(d\xi'|\xi)\nu_0(d\xi) = \nu_0(d\xi') \quad \text{for all } \xi' \in \Xi$$

where  $F_0(d\xi'|\xi)$  is the reference transition kernel (e.g., conditional Gaussian distribution for an AR(1) process). This is equivalent to requiring that the marginal distributions of the joint distribution  $dF_0(\xi, \xi') := F_0(d\xi'|\xi)\nu_0(d\xi)$  are both  $\nu_0$ , i.e.,  $F_0 \in \Pi(\nu_0, \nu_0)$  where  $\Pi(\nu_0, \nu_0)$  is the set of joint distributions on  $\Xi^2$  with marginals  $\nu_0$ . Moreover, for any  $F \in \Pi(\nu_0, \nu_0)$ , its conditional density  $F(d\xi'|\xi)$  preserves  $\nu_0$  as a stationary distribution. Then, the perturbation set around the reference joint distribution  $F_0$  is defined as:

$$\mathcal{F} := \{F \in \mathcal{P}(\Xi^2) \mid \underbrace{F \in \Pi(\nu_0, \nu_0)}_{\text{Stationarity}}, \underbrace{D_{KL}(F\|F_0) \leq \delta}_{\text{Perturbation}}\}$$

where  $\delta \geq 0$  measures the size Kullback-Leibler (KL) divergence ball around  $F_0$ , and

$$D_{KL}(F\|F_0) := \begin{cases} \int \log\left(\frac{dF}{dF_0}\right) dF & \text{if } F \ll F_0 \\ +\infty & \text{otherwise} \end{cases}$$

This construction allows us to perturb the transition kernel of the Markov process while keeping its stationary distribution unchanged. It can introduce nonlinear dynamics into the latent variable process. For example, if the reference model is an AR(1) process, the perturbation set includes any nonlinear first-order Markov process with the same stationary distribution as the AR(1) process. Section 2.4 discusses how to perturb the stationary distribution. In this case, we replace the condition  $F \in \Pi(\nu_0, \nu_0)$  with  $F \in \Pi(\nu, \nu)$ , where  $\nu$  is the perturbed stationary distribution that is in a neighborhood of  $\nu_0$ .

The marginal constraints with the KL constraint form a computationally tractable EOT problem whose implementation depends on its dual formulation (see Section 2.3).<sup>1</sup>

We now turn to the general definition of the perturbation set. We partition a vector of latent variables  $U \in \mathcal{U} \subseteq \mathbb{R}^d$  into  $2 \leq k \leq d$  subvectors, i.e.,  $U = (U_1, \dots, U_k)$ .<sup>2</sup> Each subvector  $U_i \in \mathcal{U}_i \subseteq \mathbb{R}^{d_i}$  has a marginal distribution  $\nu_i \in \mathcal{P}(\mathcal{U}_i)$  for  $i = 1, \dots, k$ . Let  $F_0$  denote the reference distribution for  $U$ . The perturbation set is defined as:

$$\mathcal{F} := \{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu_1, \dots, \nu_k), D_{KL}(F \| F_0) \leq \delta\}$$

where  $\Pi(\nu_1, \dots, \nu_k)$  is the set of joint distributions on  $\mathcal{U}$  with marginals  $\{\nu_i\}_{i=1}^k$ .

**Remark 1.** (i) In the time-homogeneous example, we take  $U = (\xi, \xi')$ . This pair of latent variables can be further partitioned into some subvectors to analyze sensitivity to assumptions about cross-sectional dependence.

(ii) Higher-order Markov dynamics can be handled by expanding the state space. For example, if the perturbed process is second-order Markov, then perturbation set can be defined on the joint distribution of  $(\tilde{\xi}_t, \tilde{\xi}_{t+1})$  where  $\tilde{\xi}_t = (\xi_t, \xi_{t+1})$ .

## 2.2 Examples

Our framework applies to a variety of latent variables, including utility shocks, productivity characteristics, labor supply shocks, etc. This section focuses on a parametric model for latent variables beyond utility shocks in infinite and finite-horizon DDC models. Appendix A considers: (i) serial independence of utility shocks in DDC models, and (ii) serial independence of consumption shocks in dynamic discrete-continuous choice models. The bounds on a scalar parameter are solutions to constrained optimization problems over the perturbation set subject to structural constraints (e.g., Bellman equation) and moment conditions.

**Example 1** (Infinite Horizon Dynamic Discrete Choice Models with Serially Correlated Latent Variables). This example considers a parametric model for serially correlated latent variables in a single-agent DDC model as in Rust (1994). Let  $\xi \in \Xi$  be the exogenously evolving latent variable (e.g., unobserved productivity characteristics). Serial dependence captures the persistence in these unobserved characteristics. An AR(1) process is often used to model the transition of  $\xi$ . Let  $U := (\xi, \xi')$ , and the reference distribution be  $dF_{\theta_f}(U) := F_{\theta_f}(d\xi' | \xi) \nu_{\theta_f}(d\xi)$  where  $F_{\theta_f}(d\xi' | \xi)$  is the conditional distribution parameterized by  $\theta_f \in \Theta_f$

<sup>1</sup>For duality results of general divergence constrained OT problem, see Bayraktar et al. (2025).

<sup>2</sup>The vector  $U$  can also contain observable variables.

(e.g., the parameters of the AR(1) process), and  $\nu_{\theta_f}$  is its stationary distribution. The perturbation set for a given  $\theta_f$  is defined as:

$$\mathcal{F}_{\theta_f} := \{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu_{\theta_f}, \nu_{\theta_f}), D_{KL}(F \| F_{\theta_f}) \leq \delta\}$$

Agents solve the smoothed<sup>3</sup> Bellman equation for the conditional value function  $v \in \mathcal{V}$  where  $\mathcal{V}$  is a function class (e.g., the class of square integrable functions): for  $\forall (j, x, \xi) \in \mathcal{J} \times \mathcal{X} \times \Xi$ , it holds that

$$v_j(x, \xi) = u_j(x, \xi; \theta) + \beta \mathbb{E}_{\xi' | \xi} \mathbb{E}_{x' | x, j} \left[ \log \left( \sum_{j' \in \mathcal{J}} \exp(v_{j'}(x', \xi')) \right) \right] + \beta \gamma \quad (1)$$

where  $x \in \mathcal{X}$  is the observable state variable,  $\beta \in (0, 1)$  is the discount factor,  $\gamma$  is the Euler constant, and  $u_j(x, \xi; \theta)$  is the period utility of choosing  $j \in \mathcal{J}$  parameterized by  $\theta \in \Theta$ .

Suppose the scalar parameter of interest is the average elasticity of action  $j$  with respect to variable  $x_l$ , defined as:

$$\mathbb{E}_{\nu_{\theta_f}} \mathbb{E}_x \left[ \frac{\partial p(j|x, \xi)}{\partial x_l} \frac{x_l}{P_0(j|x)} \right]$$

where  $P_0(j|x)$  is the population CCP, the expectation is taken with respect to  $\nu_{\theta_f}$  for  $\xi$  and the stationary distribution of  $x$ , and  $p(j|x, \xi) = \frac{\exp(v_j(x, \xi))}{\sum_{j' \in \mathcal{J}} \exp(v_{j'}(x, \xi))}$  is the model-implied Conditional Choice Probability (CCP).

We convert the smoothed Bellman equation (1) into an unconditional moment restriction that depends on the joint distribution  $F$ . We assume<sup>4</sup> there exists a class of test functions  $\mathcal{G}$ <sup>5</sup> such that  $v$  solves the Bellman equation (1) if and only if:

$$\sup_{g \in \mathcal{G}} \mathbb{E}_F \mathbb{E}_{x, j, x'} \left[ g_j(x, \xi) \left( v_j(x, \xi) - u_j(x, \xi; \theta) - \beta \log \left( \sum_{j' \in \mathcal{J}} \exp(v_{j'}(x', \xi')) \right) - \beta \gamma \right) \right] = 0$$

where  $(\xi, \xi') \sim F$ , and the inner expectation is taken over the observable variables:  $x$  drawn from its stationary distribution,  $j \sim P_0(j|x)$ , and  $x'$  from the transition distribution conditional on  $(x, j)$ . Let  $g := (g_j)_{j \in \mathcal{J}}$ . Then, we rewrite the structural constraints as:

$$\sup_{g \in \mathcal{G}} \mathbb{E}_F [\psi(U; \theta, v, g)] = 0$$

<sup>3</sup>We assume that the utility shock is additively separable in the period utility function and follows an i.i.d. Extreme Value Type I distribution, leading to the log-sum-exp form of the value function Rust (1987).

<sup>4</sup>The test function converts the continuum of conditional moment restrictions into a single unconditional moment restriction (see for example Andrews and Shi (2013) and Schennach (2014)).

<sup>5</sup>For example, if  $\mathcal{V}$  is the class of square integrable functions,  $\mathcal{G}$  is the class of square integrable functions.

We consider the following moment conditions for estimation:

$$\mathbb{E}_{\nu_{\theta_f}} [p(j|x, \xi)] = P_0(j|x) \quad \forall (j, x) \in \mathcal{J} \times \mathcal{X}$$

We assume  $\mathcal{X}$  has discrete support, and rewrite the moment conditions as:

$$\mathbb{E}_F [m(U; v)] = P_0$$

where  $m(U; v)$  stacks the model-implied CCPs, and  $P_0$  stacks the population CCPs.

Then, the lower bound on the elasticity is given by:

$$\begin{aligned} \inf_{\theta_f \in \Theta_f} \inf_{(\theta, v, F) \in \Theta \times \mathcal{V} \times \mathcal{F}_{\theta_f}} \mathbb{E}_{\nu_{\theta_f}} \mathbb{E}_x \left[ \frac{\partial p(j|x, \xi)}{\partial x_l} \frac{x_l}{P_0(j|x)} \right] \\ \text{s.t.} \quad \mathbb{E}_F [m(U; v)] = P_0 \\ \sup_{g \in \mathcal{G}} \mathbb{E}_F [\psi(U; \theta, v, g)] = 0 \end{aligned}$$

In the next section, we will discuss the implementation for a fixed  $\theta_f$ . The overall lower bound requires optimizing over  $\theta_f \in \Theta_f$ . In practice, we can discretize the estimated AR(1) process and scale the grid points according to the candidate  $\theta_f$  during the optimization. The resulting optimization over  $\theta_f$  is then carried out using the algorithm proposed in Section 6.2.

**Example 2** (Finite Horizon Dynamic Discrete Choice Models with Serially Correlated Latent Variables). This example considers a finite-horizon DDC model where the latent variable  $\xi \in \Xi$  (e.g., health shocks) follows a first-order stationary Markov process. Let  $U := (\xi, \xi')$  be a vector of current and future latent variables. In practice, we may set the reference distribution as the estimated distribution from a parametric model, such as an AR(1) process. The reference distribution  $F_0$  is the product of the conditional distribution and its stationary distribution,  $\nu_0$ . The perturbation set is defined as:

$$\mathcal{F} := \{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu_0, \nu_0), D_{KL}(F \| F_0) \leq \delta\}$$

The conditional value function  $v_t \in \mathcal{V}$  at time period  $t \leq T < +\infty$  solves the smoothed Bellman equation: for  $\forall (x_t, \xi_t, j) \in \mathcal{X} \times \Xi \times \mathcal{J}$ ,

$$v_{jt}(x_t, \xi_t) = u_j(x_t, \xi_t; \theta) + \beta \mathbb{E}_{\xi_{t+1}|\xi_t} \mathbb{E}_{x_{t+1}|x_t, j} \left[ \log \left( \sum_{j' \in \mathcal{J}} \exp(v_{j't+1}(x_{t+1}, \xi_{t+1})) \right) \right] + \beta \gamma \quad (2)$$

where  $x \in \mathcal{X}$  is the observable state,  $\beta \in (0, 1)$  is the discount factor,  $\gamma$  is the Euler constant,



$u_j(x, \xi; \theta)$  is the period utility parameterized by  $\theta \in \Theta$ , and  $v_{jT}(x_T, \xi_T) = u_j(x_T, \xi_T; \theta)$ .

Let  $V_t(x_t, \xi_t) := \log \left( \sum_{j \in \mathcal{J}} \exp(v_{jt}(x_t, \xi_t)) \right)$  denote the ex-ante value function at  $t$ . Suppose the scalar parameter of interest is the consumer surplus derived from the choice set  $\mathcal{J}$  at period  $t$ :

$$\mathbb{E}_{\nu_0} \mathbb{E}_{x_t} \left[ \frac{1}{\alpha} V_t(x_t, \xi_t) \right]$$

where we assume  $u_j(x_t, \xi_t; \theta)$  is linear in price and  $\alpha$  is the price coefficient.

We convert the smoothed Bellman equation (2) into restrictions that depend on the joint distribution  $F \in \mathcal{F}$ . We assume there exists a class of test functions  $\mathcal{G}$  such that for each  $t \leq T - 1$ ,  $v_t$  solves (2) if and only if:

$$\sup_{g_t \in \mathcal{G}} \mathbb{E}_F \mathbb{E}_{x_t, j_t, x_{t+1}} \left[ g_{jt}(x_t, \xi_t) \times \left( v_{jt}(x_t, \xi_t) - u_{jt}(x_t, \xi_t; \theta) - \beta V_{t+1}(x_{t+1}, \xi_{t+1}) - \beta \gamma \right) \right] = 0$$

where  $(\xi_t, \xi_{t+1}) \sim F$ , and the inner expectation is taken over the observable variables:  $x_t$  drawn from its period- $t$  marginal distribution,  $j_t \sim P_{0t}(j|x_t)$  from the population CCPs, and  $x_{t+1}$  from the transition distribution conditional on  $(x_t, j_t)$ . Let  $g := (g_{jt})_{j \in \mathcal{J}, t \leq T-1}$  and  $v := (v_{jt})_{j \in \mathcal{J}, t \leq T-1}$ . Then, we rewrite the structural constraints as:

$$\sup_{g \in \mathcal{G}} \mathbb{E}_F [\psi(U; \theta, v, g)] = 0$$

where  $\psi$  is the sum of the objective functions in the above equation for each  $t \leq T - 1$ .

We consider the following moment conditions for estimation: for each  $t \leq T$ ,

$$\mathbb{E}_{\nu_0} [p_t(j|x_t, \xi)] = P_{0t}(j|x_t) \quad \forall (j, x_t) \in \mathcal{J} \times \mathcal{X}$$

where  $p_t(j|x, \xi) = \frac{\exp(v_{jt}(x, \xi))}{\sum_{j' \in \mathcal{J}} \exp(v_{j't}(x, \xi))}$  is the model-implied CCP at period  $t$ . We assume  $\mathcal{X}$  has discrete support, and rewrite the moment conditions as  $\mathbb{E}_F [m(U; v)] = P_0$  where  $m(U; v)$  and  $P_0$  stack the model-implied and population CCPs for all  $t \leq T$ .

Then, the lower bound on consumer surplus at period  $t$  is given by:

$$\begin{aligned} & \inf_{(\theta, v, F) \in \Theta \times \mathcal{V} \times \mathcal{F}} \mathbb{E}_{\nu_0} \mathbb{E}_{x_t} \left[ \frac{1}{\alpha} V_t(x_t, \xi_t) \right] \\ & \text{s.t.} \quad \mathbb{E}_F [m(U; v)] = P_0 \\ & \quad \sup_{g \in \mathcal{G}} \mathbb{E}_F [\psi(U; \theta, v, g)] = 0 \end{aligned}$$

## 2.3 Framework and Duality

We now present a general framework that nests the above examples. In general, the model is not point-identified when  $\mathcal{F}$  is not a singleton.<sup>6</sup> Therefore, we propose to compute upper and lower bounds on the outcome of interest. Let the scalar parameter of interest,  $s : \mathcal{U} \times \Theta \times \mathcal{V} \rightarrow \mathbb{R}$ , be a function of the latent variable  $U$ , and the model primitives  $(\theta, v) \in \Theta \times \mathcal{V}$ . The lower bound is the solution to the following optimization problem:

$$\begin{aligned} \kappa(\delta, P) &:= \inf_{(\theta, v, F) \in \Theta \times \mathcal{V} \times \mathcal{F}} \mathbb{E}_F [s(U; \theta, v)] \\ \text{s.t. } &\mathbb{E}_F [m(U; \theta, v)] = P \\ &\sup_{g \in \mathcal{G}} \mathbb{E}_F [\psi(U; \theta, v, g)] = 0 \end{aligned} \tag{Primal}$$

where  $\mathcal{F} := \{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu_1, \dots, \nu_k), D_{KL}(F \| F_0) \leq \delta\}$ .

The first constraint is a moment condition where the moment function  $m : \mathcal{U} \times \Theta \times \mathcal{V} \rightarrow \mathbb{R}^{d_P}$  is finite-dimensional as we assume the observable variable  $X \in \mathcal{X}$  has discrete support and stack the moment functions for each  $x \in \mathcal{X}$ . The second constraint is a structural constraint defined by  $\psi : \mathcal{U} \times \Theta \times \mathcal{V} \times \mathcal{G} \rightarrow \mathbb{R}$  that is linear in the test function  $g \in \mathcal{G}$  for a given  $(\theta, v)$ . Finally,  $v \in \mathcal{V}$  is the solution to the structural constraint.

- Remark 2.** (i) The upper bound can be obtained by replacing  $s(U; \theta, v)$  with  $-s(U; \theta, v)$ .
- (ii) The moment condition can contain restrictions linear in  $F$ , e.g., covariance restrictions.
- (iii) If some model primitives (e.g., a subvector of  $\theta$ ) are point-identified, they are treated as fixed values rather than optimized over.
- (iv) The framework can also be applied to models without structural constraints, such as static and panel discrete choice models.

The [Primal](#) problem can be intractable due to the optimization over  $\mathcal{F}$ . First, the properties (e.g., closed-form and smoothness) of the optimal  $F^*$  are typically unknown, complicating the choice of approximation methods. Second, the marginal distribution conditions are functional constraints imposed directly on the distribution being optimized, which are computationally difficult to impose. Third, expectations are taken with respect to the perturbed distribution, making numerical integration difficult.

To overcome these issues, we derive the [Dual](#) problem corresponding to the [Primal](#). The [Dual](#) provides the closed-form of the optimal  $F^*$ , and characterizes its smoothness. Moreover,

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<sup>6</sup>For example, see [Schennach \(2014\)](#); [Molinari \(2020\)](#).

in the dual, the expectation is taken with respect to the reference distribution  $F_0$ . Section 6.2 proposes a computationally tractable algorithm that utilizes the optimal  $F^*$ . To motivate the duality, consider the Lagrangian of the [Primal](#):

$$\kappa(\delta, P) = \inf_{\substack{(\theta, v) \in \Theta \times \mathcal{V} \\ F \in \Pi(\nu_1, \dots, \nu_k)}} \sup_{\substack{\lambda \in \mathbb{R}^{d_P} \\ \lambda_{KL} \geq 0, g \in \mathcal{G}}} \mathbb{E}_F [c(U; \theta, v, g, \lambda)] + \lambda_{KL} (D_{KL}(F \| F_0) - \delta) - \lambda^T P \quad (3)$$

where  $c(U; \theta, v, g, \lambda) := s(U; \theta, v) + \lambda^T m(U; \theta, v) + \psi(U; \theta, v, g)$ ,  $\lambda$  is the Lagrange multiplier for the moment condition,  $\lambda_{KL}$  is the Lagrange multiplier for the KL divergence constraint and  $g$  is the Lagrange multiplier function for the structural constraint under Assumption 1(iii). For a given  $(\theta, v)$ , under Assumption 1, we can swap the order of the infimum over  $F$  and the supremum over  $(\lambda, \lambda_{KL}, g)$ . Then, we can rewrite (3) as:

$$\inf_{(\theta, v) \in \Theta \times \mathcal{V}} \sup_{\substack{\lambda \in \mathbb{R}^{d_P} \\ \lambda_{KL} \geq 0, g \in \mathcal{G}}} \inf_{F \in \Pi(\nu_1, \dots, \nu_k)} \mathbb{E}_F [c(U; \theta, v, g, \lambda)] + \lambda_{KL} D_{KL}(F \| F_0) - \lambda_{KL} \delta - \lambda^T P$$

The inner infimum is known as the Entropic Optimal Transport (EOT) problem (for  $\lambda_{KL} > 0$ )<sup>7</sup>:

$$\mathcal{C}(\theta, v, g, \lambda, \lambda_{KL}) := \inf_{F \in \Pi(\nu_1, \dots, \nu_k)} \mathbb{E}_F [c(U; \theta, v, g, \lambda)] + \lambda_{KL} D_{KL}(F \| F_0)$$

where  $c(U; \theta, v, g, \lambda)$  is the cost function, and  $\mathcal{C}(\theta, v, g, \lambda, \lambda_{KL})$  is called the optimal EOT value. The EOT problem is computationally fast to solve using the Sinkhorn algorithm, which relies on the duality of the EOT problem. Section 6.1 reviews the EOT duality and the Sinkhorn algorithm. Moreover, the closed-form and smoothness of the unique solution  $F^*$  to the EOT problem can be derived from its duality (see Theorem 1 and Proposition 1). Next, we impose assumptions for the minimax theorem to swap the order of infimum and supremum, and the EOT duality to hold:

**Assumption 1.** *We assume:*

- (i) *The marginals  $\{\nu_i\}_{i=1}^k$  have finite  $p$ -th moment for some integer  $p \geq 1$ .*
- (ii)  *$F_0 \ll F_{\otimes} := \otimes_{i=1}^k \nu_i$ , and let  $\rho(U) := \log \frac{dF_{\otimes}(U)}{dF_0(U)}$ .*
- (iii)  *$\mathcal{G}$  is convex and symmetric, i.e., for  $g_1, g_2 \in \mathcal{G}$ ,  $\eta g_1 + (1 - \eta)g_2 \in \mathcal{G}$  for  $\forall \eta \in [0, 1]$ , and  $-g \in \mathcal{G}$  if  $g \in \mathcal{G}$ . Moreover, if  $g \in \mathcal{G}$ , then  $\eta g \in \mathcal{G}$  for  $\forall \eta \geq 0$ .*

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<sup>7</sup>See [Nutz \(2021\)](#) for a comprehensive introduction to the EOT problem. It is called the Optimal Transport (OT) problem when  $\lambda_{KL} = 0$  (see [Villani et al. \(2009\)](#)). The optimal value is called the optimal OT value.

- (iv) For  $\forall (\theta, v, g) \in \Theta \times \mathcal{V} \times \mathcal{G}$ , it holds that  $\rho(U), s(U; \theta, v), m(U; \theta, v), \psi(U; \theta, v, g)$  are lower semicontinuous in  $U$ .
- (v) For  $\forall (\theta, v, g) \in \Theta \times \mathcal{V} \times \mathcal{G}$ , there exist a finite positive constant  $C_{\theta, v, g}$  and  $\hat{U} \in \mathcal{U}$  such that for  $\forall U \in \mathcal{U}$ , it holds that  $|\rho(U)| + |s(U; \theta, v)| + \|m(U; \theta, v)\|_1 + |\psi(U; \theta, v, g)| \leq C_{\theta, v, g}(1 + d(U, \hat{U}))$  where  $d(U, \hat{U}) := \sum_{i=1}^k d_i(U_i, \hat{U}_i)^p$  and  $d_i$  is a metric on  $\mathcal{U}_i$ .

Assumptions 1(i)-1(iv) are mild. There is no particular necessity to write  $\rho(U)$  in log-density form in Assumption 1(ii). Our notation is chosen to simplify the expression in Assumptions 1(iv) and 1(v). Assumption 1(v) imposes the growth rate condition. It is satisfied for all Examples in Section 2.2 if we assume  $u$ ,  $g$ , and  $v$  satisfy the growth rate condition. It ensures that  $c(U; \theta, v, g, \lambda) \in L^1(F)$  for  $\forall F \in \mathcal{F}$ , and can also be used to show the convergence of the Sinkhorn algorithm and the convergence of optimal EOT value to optimal OT value as  $\lambda_{KL} \downarrow 0$  (Eckstein and Nutz (2022, 2024)).<sup>8</sup>

**Theorem 1.** Let  $c(U; \theta, v, g, \lambda) := s(U; \theta, v) + \lambda^T m(U; \theta, v) + \psi(U; \theta, v, g)$  where  $\lambda \in \mathbb{R}^{d_P}$ . Under Assumption 1, the following holds:

- (i) (Minimax Duality)

$$\kappa(\delta, P) = \inf_{(\theta, v) \in \Theta \times \mathcal{V}} \sup_{\lambda \in \mathbb{R}^{d_P}, \lambda_{KL} \geq 0, g \in \mathcal{G}} \mathcal{C}(\theta, v, g, \lambda, \lambda_{KL}) - \lambda_{KL} \delta - \lambda^T P \quad (\text{Dual})$$

where  $\mathcal{C}(\theta, v, g, \lambda, \lambda_{KL})$  is the EOT problem with regularization parameter  $\lambda_{KL}$ :

$$\mathcal{C}(\theta, v, g, \lambda, \lambda_{KL}) := \inf_{F \in \Pi(\nu_1, \dots, \nu_k)} \mathbb{E}_F [c(U; \theta, v, g, \lambda)] + \lambda_{KL} D_{KL}(F \| F_0)$$

- (ii) (Entropic Optimal Transport Duality) For  $\lambda_{KL} > 0$ , we have:<sup>9</sup>

$$\mathcal{C}(\theta, v, g, \lambda, \lambda_{KL}) = \sup_{\{\phi_i \in L^1(\nu_i)\}_{i=1}^k} \sum_{i=1}^k \mathbb{E}_{\nu_i} \phi_i(U_i) - \lambda_{KL} \mathbb{E}_{F_0} \exp \left( \frac{\sum_{i=1}^k \phi_i(U_i) - c(U; \theta, v, g, \lambda)}{\lambda_{KL}} \right) + \lambda_{KL}$$

Moreover, there are unique maximizers  $\{\phi_i^*\}_{i=1}^k$  up to additive constants  $F_0$ -almost surely, and the unique worst-case distribution  $F^*$  has the density of the form:

$$\frac{dF^*(U)}{dF_0(U)} = \exp \left( \frac{\sum_{i=1}^k \phi_i^*(U_i) - c(U; \theta, v, g, \lambda)}{\lambda_{KL}} \right) \quad F_0\text{-a.s.}$$

<sup>8</sup>For the convergence of optimal EOT value to optimal OT value, lower semicontinuity alone is not sufficient (Nutz (2021) Example 5.1). A sufficient condition is the continuity condition on the cost function.

<sup>9</sup>See Villani (2021) for optimal transport duality ( $\lambda_{KL} = 0$ ).

Furthermore, we have  $\mathcal{C}(\theta, v, g, \lambda, \lambda_{KL}) = \sum_{i=1}^k \mathbb{E}_{\nu_i} \phi_i^*(U_i)$ .

(iii) If the lower semicontinuity in Assumption 1(iv) is strengthened to continuity, then optimizing over  $\lambda_{KL} > 0$  is equivalent to optimizing over  $\lambda_{KL} \geq 0$ .

Theorem 1 establishes the duality and provides the closed-form of  $F^*$ .  $\phi_i$  is a test function for the marginal distribution condition  $\nu_i$  for  $i = 1, \dots, k$ . The optimal  $\{\phi_i^*\}_{i=1}^k$  are the optimal EOT potentials, which can be efficiently obtained using the Sinkhorn algorithm.

The [Dual](#) is computationally tractable. First, it provides the closed-form of the worst-case distribution  $F^*$ , which can be used in the [Primal](#) problem even if we want to solve it directly. In Section 6.2, we propose an iterative algorithm that alternates between solving the EOT to obtain the worst-case distribution and updating  $v$  by solving the structural constraint with the worst-case distribution. Second, for a given  $(\theta, v, g, \lambda, \lambda_{KL})$ , the optimal  $\{\phi_i^*\}_{i=1}^k$  (and thus  $F^*$ ) can be efficiently computed using the Sinkhorn algorithm, which is computationally very fast. Third, the expectations in the [Dual](#) are taken with respect to the marginal distributions and the reference distribution. Therefore, numerical integration methods can be determined in advance. Finally, we have:

**Proposition 1.** *If  $c(U; \theta, v, g, \lambda)$  is  $q$ -times continuously differentiable in  $U$ , and  $\lambda_{KL} > 0$ , then  $\{\phi_i^*\}_{i=1}^k$  are  $q$ -times continuously differentiable in  $U_i$ . Therefore,  $\frac{dF^*(U)}{dF_0(U)}$  is also  $q$ -times continuously differentiable in  $U$ .*

Proposition 1 shows the smoothness of  $\{\phi_i^*\}_{i=1}^k$  for  $\lambda_{KL} > 0$  (the EOT case). For  $\lambda_{KL} = 0$  (the OT case), it is not straightforward to obtain the smoothness of the worst-case distribution (see [Villani et al. \(2009\)](#) Chapter 12).

## 2.4 Perturbation of Stationary Distribution

This section discusses how to perturb the stationary distribution in the time-homogeneous setting. We consider the serial dependence of the latent variables, as detailed in the example in Section 2.1. Let  $U := (\xi, \xi')$  be the vector of current and future latent variables. We define the perturbation set for the stationary distribution as  $\mathcal{N} = \{\nu \in \mathcal{P}(\Xi) \mid D_{KL}(\nu \parallel \nu_0) \leq \delta_1\}$  where  $\delta_1 \geq 0$ . The perturbation set for the joint distribution is:

$$\mathcal{F} := \{F \in \mathcal{P}(\Xi^2) \mid F \in \Pi(\nu, \nu), \nu \in \mathcal{N}, D_{KL}(F \parallel F_0) \leq \delta\}$$

Let  $\kappa_{stationary}(\delta_1, \delta, P)$  denote the lower bound on the scalar parameter of interest under this perturbation set. Under regularity conditions, we can swap the order of infimum over  $F$  and

the supremum over  $(\lambda, \lambda_{KL}, g)$ :

$$\kappa_{stationary}(\delta_1, \delta, P) = \inf_{(\theta, v) \in \Theta \times \mathcal{V}} \sup_{\lambda \in \mathbb{R}^{d_P}, \lambda_{KL} \geq 0, g \in \mathcal{G}} \inf_{\nu \in \mathcal{N}} \mathcal{C}(\theta, v, g, \lambda, \lambda_{KL}, \nu) - \lambda_{KL} \delta - \lambda^T P$$

where  $\mathcal{C}(\theta, v, g, \lambda, \lambda_{KL}, \nu)$  is the EOT problem with respect to the perturbed stationary distribution  $\nu$ :  $\mathcal{C}(\theta, v, g, \lambda, \lambda_{KL}, \nu) := \inf_{F \in \Pi(\nu, \nu)} \mathbb{E}_F [c(\xi, \xi'; \theta, v, g, \lambda)] + \lambda_{KL} D_{KL}(F \| F_0)$ . Its dual formulation is:

$$\mathcal{C}(\theta, v, g, \lambda, \lambda_{KL}, \nu) = \sup_{\phi_1, \phi_2 \in L^1(\nu)} \mathbb{E}_\nu [\phi_1(\xi) + \phi_2(\xi')] - \lambda_{KL} \mathbb{E}_{F_0} \exp \left( \frac{\phi_1(\xi) + \phi_2(\xi') - c(\xi, \xi'; \theta, v, g, \lambda)}{\lambda_{KL}} \right) + \lambda_{KL}$$

Under regularity conditions, we can swap the order of infimum over  $\nu$  and the supremum over  $(\phi_1, \phi_2)$ . Then, the inner infimum over  $\nu$  is:

$$\inf_{\nu \in \mathcal{N}} \mathbb{E}_\nu [\phi_1(\xi) + \phi_2(\xi')]$$

which is a KL-divergence distributionally robust optimization problem (see [Hu and Hong \(2013\)](#); [Rahimian and Mehrotra \(2019\)](#)) whose dual formulation is:

$$\sup_{\eta \geq 0} -\eta \log \mathbb{E}_{\nu_0} \exp \left( -\frac{\phi_1(\xi) + \phi_2(\xi')}{\eta} \right) - \eta \delta_1$$

where  $\eta$  is the Lagrange multiplier for the KL divergence constraint on  $\nu$ . To summarize:

**Theorem 2.** *Suppose Assumption 1 holds for any  $\nu \in \mathcal{N}$ , and  $\Xi$  is compact. Then, we have:*

$$\begin{aligned} \kappa_{stationary}(\delta_1, \delta, P) = \inf_{(\theta, v) \in \Theta \times \mathcal{V}} \sup_{\substack{\lambda \in \mathbb{R}^{d_P}, \eta \geq 0 \\ \lambda_{KL} \geq 0, g \in \mathcal{G} \\ \phi_1, \phi_2 \in L^\infty(\Xi) \\ \phi_1, \phi_2 \text{ l.s.c.}}} & -\eta \log \mathbb{E}_{\nu_0} \exp \left( -\frac{\phi_1(\xi) + \phi_2(\xi')}{\eta} \right) - \eta \delta_1 + \lambda_{KL} - \lambda_{KL} \delta - \lambda^T P \\ & - \lambda_{KL} \mathbb{E}_{F_0} \exp \left( \frac{\phi_1(\xi) + \phi_2(\xi') - c(\xi, \xi'; \theta, v, g, \lambda)}{\lambda_{KL}} \right) \end{aligned}$$

Theorem 2 shows the dual formulation when the stationary distribution is perturbed. Although the Sinkhorn algorithm does not apply directly, we can sequentially update  $(\phi_1, \phi_2)$  using the first-order optimality conditions like the Sinkhorn algorithm (see Section 6.1 for a review of the Sinkhorn algorithm).

### 3 Methodology for Time-Inhomogeneous Models

This section extends our framework to the time-inhomogeneous setting. We begin by defining the perturbation set for these models in Section 3.1. Section 3.2 provides an example of a finite-horizon dynamic discrete choice model. Section 3.3 presents the duality result. Finally, Section 3.4 discusses how to perturb the initial distribution.

#### 3.1 Definition of Perturbation Set

Consider a sequence of latent variables over a finite horizon,  $U := (\xi_1, \xi_2, \dots, \xi_T)$ , that follows a first-order Markov chain. The reference joint distribution is the product of an initial distribution and transition kernels:

$$dF_0(U) = \nu_1(d\xi_1)F_1(d\xi_2|\xi_1) \cdots F_{T-1}(d\xi_T|\xi_{T-1})$$

where  $\nu_1(d\xi_1)$  is the initial distribution, and  $F_t(d\xi_{t+1}|\xi_t)$  is the transition kernel from period  $t$  to  $t+1$ . Let  $\nu_T(d\xi_T)$  be the terminal distribution implied by this process.

We consider perturbing the reference distribution while holding its initial and terminal distributions fixed, i.e.,

$$\mathcal{F}_{\text{Markov}} := \{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi_{\text{Markov}}(\nu_1, \nu_T), D_{KL}(F\|F_0) \leq \delta\}$$

where  $\Pi_{\text{Markov}}(\nu_1, \nu_T)$  is the set of all joint distributions over  $\mathcal{U}$  that satisfy the first-order Markov property<sup>10</sup> and have  $\nu_1$  and  $\nu_T$  as their initial and terminal marginal distributions. Our perturbation set allows for any transition dynamics of the latent variables between the initial and terminal periods, as long as the overall process remains Markovian and the initial and terminal distributions are fixed.

We will discuss how to perturb the initial distribution in Section 3.4. The terminal distribution can often be nonparametrically identified; therefore, we fix it. For instance, in a finite horizon DDC model, the terminal period is a static discrete choice problem where the distribution of the latent variable can be identified (Lewbel (2000); Matzkin (2007)). Moreover, if  $\xi_t$  is the market-level latent variable, and the model has the finite dependence property (see Arcidiacono and Miller (2011))<sup>11</sup>, then the utility parameters can be identified without a distributional assumption for the latent variables (see Kalouptsi et al. (2021c)),

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<sup>10</sup>That is, for any  $t \leq T-1$ ,  $F(d\xi_{t+1}|\xi_1, \dots, \xi_t) = F(d\xi_{t+1}|\xi_t)$  almost surely under  $F$ .

<sup>11</sup>For example, a model with a terminating action has the finite dependence property.

which in turn identifies the terminal distribution.

### 3.2 Example

**Example 3** (Finite Horizon DDC with Time-Inhomogeneous Transition of Latent Variables). This example considers a time-inhomogeneous transition for the latent variables,  $U := (\xi_1, \xi_2, \dots, \xi_T)$ , whose perturbation set is  $\mathcal{F}_{\text{Markov}}$ . The model is similar to Example 2 but with a time-inhomogeneous transition.

We convert the smoothed Bellman equation (2) into restrictions that depend on the sequence of two-period marginal distributions  $\{F_{t,t+1}\}_{t=1}^{T-1}$ , where:

$$dF_{t,t+1}(\xi_t, \xi_{t+1}) = \int_{\xi_1, \dots, \xi_{t-1}, \xi_{t+2}, \dots, \xi_T} dF(\xi_1, \dots, \xi_T)$$

We assume there exists a class of test functions  $\mathcal{G}$  such that for each  $t \leq T-1$ ,  $v_{jt}$  solves (2) if and only if:

$$\begin{aligned} \sup_{g_t \in \mathcal{G}} \mathbb{E}_{F_{t,t+1}} \mathbb{E}_{x_t, j_t, x_{t+1}} \left[ g_{jt}(x_t, \xi_t) \right. \\ \left. \times \left( v_{jt}(x_t, \xi_t) - u_{j_t}(x_t, \xi_t; \theta) - \beta \log \left( \sum_{j' \in \mathcal{J}} \exp(v_{j't+1}(x_{t+1}, \xi_{t+1})) \right) - \beta \gamma \right) \right] = 0 \end{aligned}$$

where  $(\xi_t, \xi_{t+1}) \sim F_{t,t+1}$ , and  $(x_t, j_t)$  is distributed according to the observed data at time  $t$ , and  $x_{t+1}$  follows the conditional distribution given  $(x_t, j_t)$ . Let  $g := (g_{jt})_{j \in \mathcal{J}, t \leq T-1}$  and  $v := (v_{jt})_{j \in \mathcal{J}, t \leq T-1}$ . We then rewrite the structural constraints as:

$$\sup_{g \in \mathcal{G}} \mathbb{E}_F [\psi(U; \theta, v, g)] = 0$$

where  $\psi$  is the sum over  $t \leq T-1$  of terms inside the expectation of the previous equation.

Then, the lower bound on consumer surplus at period  $t$  is given by:

$$\begin{aligned} \inf_{(\theta, v, F) \in \Theta \times \mathcal{V} \times \mathcal{F}_{\text{Markov}}} \mathbb{E}_{\nu_t} \mathbb{E}_{x_t} \left[ \frac{1}{\alpha} \log \left( \sum_{j \in \mathcal{J}} \exp(v_{jt}(x_t, \xi_t)) \right) \right] \\ \text{s.t. } \mathbb{E}_F [m(U; \theta, v)] = P_0 \\ \sup_{g \in \mathcal{G}} \mathbb{E}_F [\psi(U; \theta, v, g)] = 0 \end{aligned}$$

where  $\nu_t$  is the marginal distribution of  $\xi_t$  implied by  $F$ .



### 3.3 Framework and Duality

The bound for the time-inhomogeneous case,  $\kappa_{\text{TI}}(\delta, P)$ , is defined similarly to the time-homogeneous case, but with the optimization performed over  $\mathcal{F}_{\text{Markov}}$ :

$$\begin{aligned} \kappa_{\text{TI}}(\delta, P) &:= \inf_{(\theta, v, F) \in \Theta \times \mathcal{V} \times \mathcal{F}_{\text{Markov}}} \mathbb{E}_F [s(U; \theta, v)] \\ \text{s.t. } &\mathbb{E}_F [m(U; \theta, v)] = P \\ &\sup_{g \in \mathcal{G}} \mathbb{E}_F [\psi(U; \theta, v, g)] = 0 \end{aligned} \quad (\text{TI})$$

where “TI” stands for time-inhomogeneous. To solve [TI](#), we follow the procedure in Theorem 1. However, the set  $\mathcal{F}_{\text{Markov}}$  is not necessarily convex, which prevents the proof strategy of the minimax duality in Theorem 1 from being applied directly. Therefore, we propose to solve a relaxed problem where the Markov property condition is removed. We then show that, under certain reasonable conditions, the solution to the relaxed problem is Markovian, thereby also solving the original problem. The perturbation set for the relaxed problem is defined as:

$$\mathcal{F}_{\text{relaxed}} := \{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu_1, \nu_T), D_{KL}(F \| F_0) \leq \delta\}$$

where  $\Pi(\nu_1, \nu_T)$  is the set of joint distributions with initial distribution  $\nu_1$  and terminal distribution  $\nu_T$ . The relaxed problem is given by:

$$\begin{aligned} \tilde{\kappa}_{\text{TI}}(\delta, P) &:= \inf_{(\theta, v, F) \in \Theta \times \mathcal{V} \times \mathcal{F}_{\text{relaxed}}} \mathbb{E}_F [s(U; \theta, v)] \\ \text{s.t. } &\mathbb{E}_F [m(U; \theta, v)] = P \\ &\sup_{g \in \mathcal{G}} \mathbb{E}_F [\psi(U; \theta, v, g)] = 0 \end{aligned} \quad (\text{Relaxed})$$

whose Lagrangian is:

$$\tilde{\kappa}_{\text{TI}}(\delta, P) = \inf_{\substack{(\theta, v) \in \Theta \times \mathcal{V} \\ F \in \Pi(\nu_1, \nu_T)}} \sup_{\substack{\lambda \in \mathbb{R}^{d_P} \\ \lambda_{KL} \geq 0, g \in \mathcal{G}}} \mathbb{E}_F [c(U; \theta, v, g, \lambda)] + \lambda_{KL}(D_{KL}(F \| F_0) - \delta) - \lambda^T P \quad (4)$$

where  $c(U; \theta, v, g, \lambda) := s(U; \theta, v) + \lambda^T m(U; \theta, v) + \psi(U; \theta, v, g)$  and  $\lambda$ ,  $\lambda_{KL}$ , and  $g$  are the Lagrange multipliers for the moment condition, the KL-divergence constraint, and the structural constraint, respectively. For a given  $(\theta, v)$ , under Assumption 2, we can swap the order of the infimum over  $F$  and the supremum over  $(\lambda, \lambda_{KL}, g)$ . Then, we rewrite (4) as:

$$\inf_{(\theta, v) \in \Theta \times \mathcal{V}} \sup_{\substack{\lambda \in \mathbb{R}^{d_P} \\ \lambda_{KL} \geq 0, g \in \mathcal{G}}} \inf_{F \in \Pi(\nu_1, \nu_T)} \mathbb{E}_F [c(U; \theta, v, g, \lambda)] + \lambda_{KL} D_{KL}(F \| F_0) - \lambda_{KL} \delta - \lambda^T P$$

The inner infimum is:

$$\mathcal{C}_{\text{TI}}(\theta, v, g, \lambda, \lambda_{KL}) := \inf_{F \in \Pi(\nu_1, \nu_T)} \mathbb{E}_F [c(U; \theta, v, g, \lambda)] + \lambda_{KL} D_{KL}(F \| F_0)$$

which can be rewritten as the discrete-time dynamic Schrödinger Bridge (SB) problem (see [Léonard \(2013\)](#); [De Bortoli et al. \(2021\)](#)). Because it only restricts the initial and terminal distributions, we can decompose it into two parts: the two-period marginal distribution part (the first and last period) and the conditional distribution part (the intermediate variables conditional on the first and last period). The second part is unconstrained, thereby having a closed-form solution. The first part is the static SB (or EOT) problem whose duality is similar to Theorem 1. We impose the following assumptions for the minimax duality, decomposition, static SB duality, and the Markov property:

**Assumption 2.** Let  $dF_0^{1,T}(\xi_1, \xi_T) := \int_{\xi_2, \dots, \xi_{T-1}} dF_0(\xi_1, \xi_2, \dots, \xi_T)$  be the two-period marginal of  $F_0$  at periods 1 and  $T$ . We assume:

- (i)  $\mathcal{U}$  is compact.
- (ii)  $F_0^{1,T} \sim \nu_1 \otimes \nu_T$ , i.e.,  $F_0^{1,T}$  and  $\nu_1 \otimes \nu_T$  are mutually absolutely continuous. Moreover,  $\log \frac{d(\nu_1 \otimes \nu_T)}{dF_0^{1,T}} \in L^1(\nu_1 \otimes \nu_T)$ .
- (iii) For  $\forall (\theta, v, g) \in \Theta \times \mathcal{V} \times \mathcal{G}$ , it holds that  $|s(U; \theta, v)| + \|m(U; \theta, v)\|_1 + |\psi(U; \theta, v, g)| < \infty$ .
- (iv) The functionals  $s(U; \theta, v)$ ,  $m(U; \theta, v)$ , and  $\psi(U; \theta, v, g)$  are pairwise additive, i.e.,  $s(U; \theta, v) = \sum_{t=1}^{T-1} s_t(\xi_t, \xi_{t+1}; \theta, v_t, v_{t+1})$ ,  $m(U; \theta, v) = \sum_{t=1}^{T-1} m_t(\xi_t, \xi_{t+1}; \theta, v_t, v_{t+1})$ , and  $\psi(U; \theta, v, g) = \sum_{t=1}^{T-1} \psi_t(\xi_t, \xi_{t+1}; \theta, v_t, v_{t+1}, g_t, g_{t+1})$  for some functions  $s_t$ ,  $m_t$ , and  $\psi_t$ .

The boundedness condition in Assumption 2(iii) is stronger than Assumption 1(v), which does not guarantee that  $c(U; \theta, v, g, \lambda) \in L^1(F)$  for any  $F \in \mathcal{F}_{\text{relaxed}}$ . Assumption 2(ii) is a sufficient condition for the SB duality to hold. Finally, Assumption 2(iv) is the key to the Markov property of the solution to the [Relaxed](#) problem, and is satisfied in Example 3. It does not hold if the moment function depends on the entire path of the latent variables.

**Theorem 3.** Let  $c(U; \theta, v, g, \lambda) := s(U; \theta, v) + \lambda^T m(U; \theta, v) + \psi(U; \theta, v, g)$  where  $\lambda \in \mathbb{R}^{d_P}$ . Under Assumptions 1(i), 1(iii), 1(iv), and 2, the following holds:

- (i) (Minimax Duality)

$$\tilde{\kappa}_{\text{TI}}(\delta, P) = \inf_{(\theta, v) \in \Theta \times \mathcal{V}} \sup_{\lambda \in \mathbb{R}^{d_P}, \lambda_{KL} \geq 0, g \in \mathcal{G}} \mathcal{C}_{\text{TI}}(\theta, v, g, \lambda, \lambda_{KL}) - \lambda_{KL} \delta - \lambda^T P$$

where  $\mathcal{C}_{TI}(\theta, v, g, \lambda, \lambda_{KL})$  is defined as:

$$\mathcal{C}_{TI}(\theta, v, g, \lambda, \lambda_{KL}) := \inf_{F \in \Pi(\nu_1, \nu_T)} \mathbb{E}_F [c(U; \theta, v, g, \lambda)] + \lambda_{KL} D_{KL}(F \| F_0) \quad (\text{S}_{\text{dyn}})$$

(ii) For  $\lambda_{KL} > 0$ , the unique worst-case distribution to  $\text{S}_{\text{dyn}}$  has the density of the form:

$$\frac{dF^*(U)}{dF_0(U)} = \exp \left( \frac{\phi_1^*(\xi_1) + \phi_T^*(\xi_T) - c(U; \theta, v, g, \lambda)}{\lambda_{KL}} \right)$$

where  $\phi_1^*(\xi_1)$  and  $\phi_T^*(\xi_T)$  are the unique maximizers (up to an additive constant) to:

$$\sup_{\phi_1 \in L^1(\nu_1), \phi_T \in L^1(\nu_T)} \mathbb{E}_{\nu_1} \phi_1(\xi_1) + \mathbb{E}_{\nu_T} \phi_T(\xi_T) - \lambda_{KL} \mathbb{E}_{R_{1,T}} \exp \left( \frac{\phi_1(\xi_1) + \phi_T(\xi_T)}{\lambda_{KL}} \right) + \lambda_{KL}$$

where the auxiliary reference measure  $R_{1,T}$  is defined as:

$$dR_{1,T}(\xi_1, \xi_T) := \int_{\xi_2, \dots, \xi_{T-1}} \exp \left( \frac{-c(U; \theta, v, g, \lambda)}{\lambda_{KL}} \right) dF_0(\xi_1, \dots, \xi_T)$$

Furthermore, the solution  $F^*$  has the Markov property, i.e.,  $F^* \in \Pi_{\text{Markov}}(\nu_1, \nu_T)$ .

(iii) (Equivalence) Suppose there exists an optimal  $\lambda_{KL}^* > 0$ , then:  $\kappa_{TI}(\delta, P) = \tilde{\kappa}_{TI}(\delta, P)$ .

Theorem 3 shows the duality for the [Relaxed](#) problem. The difference between Theorem 3(i) and Theorem 1(i) is that  $(\text{S}_{\text{dyn}})$  does not fix the intermediate marginal distributions. Therefore, we can decompose  $(\text{S}_{\text{dyn}})$  into the sum of two-period marginal distribution  $(F_0^{1,T})$  part, and the conditional distribution (the distribution of  $(\xi_2, \dots, \xi_{T-1})$  given  $\xi_1$  and  $\xi_T$ ) part. The latter part is an unconstrained optimization problem, thereby having a closed-form solution. The first part is the static SB problem, whose duality is given by Theorem 3(ii).

Assumption 2(iv) is crucial for the solution to have the Markov property. Under this assumption, the cost function  $c(U; \theta, v, g, \lambda)$  is also pairwise additive. Therefore, the density ratio in Theorem 3(ii) has the Markov property.<sup>12</sup> If there exists one optimal  $\lambda_{KL}^* > 0$ , then the [TI](#) problem is equivalent to the [Relaxed](#) problem.

The [Relaxed](#) problem can also be solved using the iterative algorithm proposed in Section 6.2. There is an additional step to obtain the two-period auxiliary reference distribution  $R_{1,T}$ . The  $\text{S}_{\text{dyn}}$  can also be solved using the Sinkhorn algorithm.

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<sup>12</sup>It can be treated as the (unnormalized) pairwise Markov random field [Wainwright et al. \(2008\)](#).

### 3.4 Perturbation of Initial Distribution

This section extends the framework to allow sensitivity analysis with respect to the initial distribution. Let  $\mathcal{N}$  be a convex closed set around the initial distribution  $\nu_1$ , e.g.,  $\mathcal{N} = \{\nu \in \mathcal{P}(\Xi) \mid D_{KL}(\nu \parallel \nu_1) \leq \delta_1\}$  where  $\delta_1 \geq 0$ . Then, the perturbation set is defined as:

$$\mathcal{F}_{\mathcal{N}, \text{Relaxed}} := \{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu, \nu_T), \nu \in \mathcal{N}, D_{KL}(F \parallel F_0) \leq \delta\}$$

Let  $\tilde{\kappa}_{\text{TI}, \text{Initial}}(\delta, P)$  be the lower bound on the scalar parameter for the [Relaxed](#) problem with the perturbation set  $\mathcal{F}_{\mathcal{N}, \text{Relaxed}}$ . The following minimax duality similar to Theorem 3 holds:

**Theorem 4** (Minimax Duality with Perturbation of Initial Distribution). *Suppose  $\mathcal{N}$  is convex and closed, and that the assumptions in Theorem 3 hold for each  $\nu \in \mathcal{N}$ . Then,*

$$\tilde{\kappa}_{\text{TI}, \text{Initial}}(\delta, P) = \inf_{(\theta, v) \in \Theta \times \mathcal{V}} \sup_{\lambda \in \mathbb{R}^d, \lambda_{KL} \geq 0, g \in \mathcal{G}} \mathcal{C}_{\text{TI}}(\theta, v, g, \lambda, \lambda_{KL}) - \lambda_{KL} \delta - \lambda^T P$$

where  $\mathcal{C}_{\text{TI}}(\theta, v, g, \lambda, \lambda_{KL})$  is defined as follows:

$$\mathcal{C}_{\text{TI}, \text{Initial}}(\theta, v, g, \lambda, \lambda_{KL}) := \inf_{\nu \in \mathcal{N}} \inf_{F \in \Pi(\nu, \nu_T)} \mathbb{E}_F [c(U; \theta, v, g, \lambda)] + \lambda_{KL} D_{KL}(F \parallel F_0) \quad (5)$$

To solve (5), as shown in the proof of Theorem 3(ii) and Section 3.3, we can decompose the problem into two parts: the two-period marginal distribution part, and the conditional distribution part. The first part requires solving:

$$\inf_{\nu \in \mathcal{N}} \inf_{F_{1,T} \in \tilde{\Pi}(\nu, \nu_T)} \mathbb{E}_{F_{1,T}} [D_{KL}(F_{1,T} \parallel R_{1,T})]$$

where  $\tilde{\Pi}(\nu, \nu_T)$  is the set of distributions of  $(\xi_1, \xi_T)$  whose marginal distributions are  $\nu$  and  $\nu_T$ , respectively. It is equivalent to solving:

$$\inf_{\nu \in \mathcal{N}} \inf_{F_{1,T} \in \tilde{\Pi}(\nu, \nu_T)} \mathbb{E}_{F_{1,T}} \left[ \log \left( \frac{d(\nu \otimes \nu_T)}{dR_{1,T}} \right) \right] + D_{KL}(F_{1,T} \parallel \nu \otimes \nu_T)$$

The inner infimum is an EOT problem. Let  $\text{EOT}(\nu, \nu_T)$  be its optimal value.

**Lemma 1.** *Under the assumptions in Theorem 4,  $\text{EOT}(\nu, \nu_T)$  is convex in  $\nu$ . Its directional derivative with respect to  $\nu$  in the direction  $\nu'$  is given by:*

$$\lim_{\epsilon \downarrow 0} \frac{\text{EOT}(\nu + \epsilon(\nu' - \nu), \nu_T) - \text{EOT}(\nu, \nu_T)}{\epsilon} = \int \phi^* d(\nu' - \nu)$$

where  $\phi^*$  is the optimal EOT potential for  $\nu$ .

Lemma 1 shows that (5) is a convex optimization problem with respect to  $\nu$ . Moreover,  $\phi^*$  is a result of the Sinkhorn algorithm, which can be used to search the optimal  $\nu$  efficiently.

## 4 Large Sample Properties

This section establishes the large sample properties of the estimator for the bound. Section 4.1 proposes a consistent estimator and shows its convergence rate. Section 4.2 establishes the asymptotic distribution of the plug-in estimator for the bound.

### 4.1 Consistency and Convergence Rate

The bound on the scalar parameter is the projection of the identified set defined by the moment conditions and structural constraints onto the scalar parameter. We follow [Chernozhukov et al. \(2007\)](#) to propose an estimator for the identified set and show its consistency and convergence rate. Let  $P_n$  be an estimator for  $P_0$  where  $n$  is the sample size. Denote by  $\epsilon_n \in \mathbb{R}_+$  the tolerance level for the moment conditions that goes to zero at a suitable rate as  $n \rightarrow \infty$ . Our estimators  $\kappa(\delta, P_n, \epsilon_n)$ , and  $\tilde{\kappa}_{\text{TI}}(\delta, P_n, \epsilon_n)$  for the bounds replace the moment conditions by the approximate moment conditions.<sup>13</sup>

**Assumption 3.** Let  $\mathcal{A}$  be either  $\Theta \times \mathcal{F}$  or  $\Theta \times \mathcal{F}_{\text{relaxed}}$ , and  $\alpha := (\theta, F) \in \mathcal{A}$ . Assume:

- (i)  $\Theta \subseteq \mathbb{R}^{d_\theta}$  is convex and compact.
- (ii) If  $\mathcal{A} = \Theta \times \mathcal{F}_{\text{relaxed}}$ , then Assumption 2(i) holds.
- (iii) For  $\forall \alpha \in \mathcal{A}$ , the structural constraint  $F$ -a.s. has a unique solution  $v(\alpha) \in \mathcal{V}$ .
- (iv) The identified set  $\mathcal{A}_I := \{\alpha \in \mathcal{A} \mid \mathbb{E}_F[m(U; \theta, v(\alpha))] = P_0\}$  is nonempty.
- (v)  $\mathbb{E}_F[m(U; \theta, v(\alpha))]$  is continuous in  $\alpha \in \mathcal{A}$ , i.e., the preimages of closed sets are closed.

Assumption 3(i) is mild. Assumption 3(iii) holds in single-agent DDC models. It rules out dynamic games with multiple equilibria. Assumption 3(iv) is also mild as the identified set for  $\theta$  is usually nonempty under the reference distribution  $F_0$ . Moreover, if the identified

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<sup>13</sup>To compute the estimator, the number of moment conditions is doubled due to the use of approximate moment conditions. The duality is similar to Theorems 1 and 3, thus is omitted for brevity.

set for  $\delta = +\infty$ <sup>14</sup> is nonempty, then Assumption 3(iv) implicitly assumes the radius  $\delta$  is large enough. The smallest radius such that the identified set is nonempty can be estimated (see Remark 4). Assumption 3(v) implies that the identified set and its estimator are compact as  $\mathcal{F}$  is compact (see Lemma 8) and thus  $\mathcal{A}$  is compact (see Lemma 2).

Under Assumption 3, the estimator for  $\mathcal{A}_I$  is defined as:

$$\hat{\mathcal{A}}_I := \{\alpha \in \mathcal{A} \mid \|\mathbb{E}_F[m(U; \theta, v(\alpha))] - P_n\|_\infty \leq \epsilon_n\}$$

The analysis of the consistency and convergence rate uses the *Hausdorff Distance*:

$$d_H(\mathcal{A}_1, \mathcal{A}_2) := \max \left\{ \sup_{\alpha_1 \in \mathcal{A}_1} d(\alpha_1, \mathcal{A}_2), \sup_{\alpha_2 \in \mathcal{A}_2} d(\alpha_2, \mathcal{A}_1) \right\}$$

where  $d(\alpha_1, \mathcal{A}_1) := \inf_{\alpha_2 \in \mathcal{A}_2} d(\alpha_1, \alpha_2)$  and  $d(\alpha_1, \alpha_2)$  is a metric on  $\mathcal{A}$ .

**Assumption 4.** Assume  $P_n$  is a  $\sqrt{n}$ -consistent estimator for  $P_0$ , and there exists  $c_n$  such that  $\sqrt{n}\|P_0 - P_n\|_\infty \leq c_n$  with probability approaching 1 where  $c_n$  can be data-dependent. Let  $\epsilon_n = \frac{c_n}{\sqrt{n}}$ , and assume  $\epsilon_n \xrightarrow{p} 0$ .

Assumption 4 is mild as we assumed the observable variable has discrete support, e.g.,  $P_n$  can be the frequency estimator. In practice, we can set  $c_n \propto \log n$ . Then, the convergence rate in Theorem 5(ii) is  $\sqrt{n}$ -consistent up to a logarithmic factor. We show some properties of the identified set and its estimator:

**Lemma 2.** Under Assumptions 3 and 4,  $\mathcal{A}_I, \hat{\mathcal{A}}_I$  are closed and compact. Moreover,  $\hat{\mathcal{A}}_I$  is nonempty.

By the extreme value theorem, the infimum is achieved if the scalar parameter is continuous on  $\mathcal{A}$ , i.e., Assumption 6(i) holds. Therefore, the optimization problem has a solution. Next, we impose the polynomial minorant condition as in Chernozhukov et al. (2007) for the convergence rate of the estimator:

**Assumption 5** (Polynomial Minorant Condition). There exists positive constants  $C_1$  and  $C_2$  such that:  $\|\mathbb{E}_F[m(U; \theta, v(\alpha))] - P_0\|_\infty \geq C_1 \min \{C_2, d(\alpha, \mathcal{A}_I)\}$ .

**Theorem 5.** Under Assumptions 3 and 4, we have:

(i) (Consistency)  $d_H(\hat{\mathcal{A}}_I, \mathcal{A}_I) = o_p(1)$ .

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<sup>14</sup>In this case, the KL divergence constraint is replaced by the absolute continuity constraint, i.e.,  $F \ll F_0$ .

(ii) (Convergence Rate) Under Assumption 5,  $d_H(\hat{\mathcal{A}}_I, \mathcal{A}_I) = O_p(\frac{\max\{1, c_n\}}{\sqrt{n}})$ .

Theorem 5 establishes the  $\sqrt{n}$ -consistency up to a logarithmic factor (if  $c_n \propto \log n$ ). Then, we impose the following continuity assumption on the scalar parameter of interest:

**Assumption 6.** Let  $s(\alpha) := \mathbb{E}_F[s(U; \theta, v(\alpha))]$ , assume one of the following:

(i)  $s(\alpha)$  is continuous in  $\alpha \in \mathcal{A}$ .

(ii)  $s(\alpha)$  is Lipschitz continuous in  $\alpha \in \mathcal{A}$ .

**Theorem 6.** Under Assumptions 3 and 4, we have:

(i) (Consistency) Under Assumption 6(i),  $\kappa(\delta, P_n, \epsilon_n) \xrightarrow{p} \kappa(\delta, P_0)$ , and  $\tilde{\kappa}_{TI}(\delta, P_n, \epsilon_n) \xrightarrow{p} \tilde{\kappa}_{TI}(\delta, P_0)$ .

(ii) (Convergence Rate) Under Assumption 6(ii),  $|\kappa(\delta, P_n, \epsilon_n) - \kappa(\delta, P_0)| = O_p(\frac{\max\{1, c_n\}}{\sqrt{n}})$ , and  $|\tilde{\kappa}_{TI}(\delta, P_n, \epsilon_n) - \tilde{\kappa}_{TI}(\delta, P_0)| = O_p(\frac{\max\{1, c_n\}}{\sqrt{n}})$ .

## 4.2 Asymptotic Distribution

This section establishes the asymptotic distribution of  $\kappa(\delta, P_n)$  and  $\tilde{\kappa}_{TI}(\delta, P_n)$ . To this end, we first show the Hadamard directional differentiability of  $\kappa(\delta, P)$  and  $\tilde{\kappa}_{TI}(\delta, P)$  with respect to  $P$  at  $P_0$  similar to [Christensen and Connault \(2023\)](#). We begin with the definition of Hadamard directional differentiability:

**Definition 1.** The map  $f : \mathbb{R}^{d_P} \rightarrow \mathbb{R}$  is Hadamard directionally differentiable at  $P \in \mathbb{R}^{d_P}$ , if there exists a continuous map  $f' : \mathbb{R}^{d_P} \rightarrow \mathbb{R}$  such that for  $h \in \mathbb{R}^{d_P}$ , we have:

$$\lim_{i \rightarrow \infty} \frac{f(P + t_i h_i) - f(P)}{t_i} = f'(P; h)$$

for all sequences  $\{h_i\} \subseteq \mathbb{R}^{d_P}$ ,  $t_i \downarrow 0$ , and  $h_i \rightarrow h \in \mathbb{R}^{d_P}$  as  $i \rightarrow \infty$ .

Under Assumption 3, we can restate the optimization problem as:

$$\inf_{\alpha \in \mathcal{A}} s(\alpha) \quad \text{s.t.} \quad P(\alpha) = P_0$$

where  $P(\alpha) := \mathbb{E}_F[m(U; \theta, v(\alpha))]$ . Moreover, the identified set  $\mathcal{A}_I$  is nonempty, which means the feasible set for the optimization problem is nonempty. By Lemma 2,  $\mathcal{A}_I$  is compact. Under Assumption 6(i), the Extreme Value Theorem (see [Rudin et al. \(1976\)](#) Theorem

4.16.) implies that the infimum is attained. Denote by  $\mathcal{A}_{I,TH}^*$ ,  $\mathcal{A}_{I,TI}^*$  the nonempty sets of optimizers for the problems  $\kappa(\delta, P_0)$  and  $\tilde{\kappa}_{TI}(\delta, P_0)$ , respectively.

To establish Hadamard directional differentiability of  $\kappa(\delta, P)$  and  $\tilde{\kappa}_{TI}(\delta, P)$  at  $P_0$ , we impose assumptions similar to those in [Bonnans and Shapiro \(2013\)](#) Theorem 4.25.<sup>15</sup>

**Assumption 7.** Assume  $s(\alpha)$  and  $P(\alpha)$  are continuously differentiable on  $\mathcal{A}$ . That is, they are Gâteaux differentiable on  $\mathcal{A}$  and the corresponding derivatives  $Ds(\alpha)$  and  $DP(\alpha)$  are continuous on  $\mathcal{A}$  (in the operator norm topology).<sup>16</sup>

**Assumption 8.** Let  $\mathcal{A}_I^*$  be either  $\mathcal{A}_{I,TH}^*$  or  $\mathcal{A}_{I,TI}^*$ . Assume:

- (i)  $0 \in \text{int}\{DP(\alpha)(\mathcal{A} - \alpha)\}$  for  $\forall \alpha \in \mathcal{A}_I^*$ .
- (ii) For  $\forall h \in \mathbb{R}^{d_P}$ , it holds that for  $\forall P_t := P_0 + th + o(t)$  and  $t > 0$  small enough, the problem  $\kappa(\delta, P_t)$  has an  $o(t)$ -optimal solution  $\alpha(t)$  such that  $d(\alpha(t), \mathcal{A}_I^*) = O(t)$ .
- (iii) For  $\forall t_n \downarrow 0$  the sequence  $\{\alpha(t_n)\}$  has a limit point (in the norm topology)  $\alpha_0 \in \mathcal{A}_I^*$ .

**Theorem 7.** Under Assumptions 3, 7, and 8, the maps  $\kappa(\delta, P)$  and  $\tilde{\kappa}_{TI}(\delta, P)$  are Hadamard directionally differentiable at  $P_0$  in any direction  $h \in \mathbb{R}^{d_P}$ , and:

$$\kappa'(\delta, P_0; h) = \inf_{\alpha \in \mathcal{A}_{I,TH}^*} \sup_{\lambda \in \Lambda(\alpha, P_0)} -\lambda^T h, \quad \tilde{\kappa}'_{TI}(\delta, P_0; h) = \inf_{\alpha \in \mathcal{A}_{I,TI}^*} \sup_{\lambda \in \Lambda(\alpha, P_0)} -\lambda^T h$$

where  $\Lambda(\alpha, P_0)$  is the nonempty set of Lagrange multipliers corresponding to  $\alpha \in \mathcal{A}_I^*$ .<sup>17</sup>

Moreover, if  $\sqrt{n}(P_n - P_0) \xrightarrow{d} Z \sim \mathcal{N}(0, \Sigma)$ , then  $\sqrt{n}(\kappa(\delta, P_n) - \kappa(\delta, P_0)) \xrightarrow{d} \kappa'(\delta, P_0; Z)$  and  $\sqrt{n}(\tilde{\kappa}_{TI}(\delta, P_n) - \tilde{\kappa}_{TI}(\delta, P_0)) \xrightarrow{d} \tilde{\kappa}'_{TI}(\delta, P_0; Z)$ .

Theorem 7 shows the asymptotic distribution of the bound's estimator. To conduct inference, we may follow the procedure in [Fang and Santos \(2019\)](#). In addition, the numerical delta method [Hong and Li \(2018\)](#) combined with our practical implementation in Section 6 can be used to overcome the computational challenge.

<sup>15</sup>We work on the primal problem to show the Hadamard directional differentiability, while [Christensen and Connault \(2023\)](#) works on the dual problem (see their Theorem 6.2).

<sup>16</sup>For a given direction  $\alpha_1 \in \mathcal{A}$ , the Gâteaux derivatives are understood as  $Ds(\alpha)(\alpha_1 - \alpha)$  and  $DP(\alpha)(\alpha_1 - \alpha)$ . See [Bonnans and Shapiro \(2013\)](#) Page 35 for the definition of Gâteaux derivative.

<sup>17</sup>See [Bonnans and Shapiro \(2013\)](#) Definition 3.8 and Theorem 3.9. Robinson's constraint qualification is satisfied under Assumptions 3, 7, and 8 (see Section B.4.4). Therefore,  $\Lambda(\alpha^*, P_0)$  is nonempty.



## 5 Interpreting the Results

In practice, we can estimate an alternative (parametric) model and set the radius to be the KL divergence from the alternative to the reference distribution. In addition, this section considers three complementary sensitivity measures to interpret the results: global sensitivity, local sensitivity, and robustness metric.

### 5.1 Global Sensitivity

The global sensitivity<sup>18</sup> approach progressively increases the radius until a pre-specified maximum radius is reached. We focus on the time-homogeneous case and show that it provides a computationally tractable approximation to the nonparametric bound defined by:

$$\begin{aligned} \kappa_{\text{NP}}(P) &:= \inf_{(\theta, v, F) \in \Theta \times \mathcal{V} \times \mathcal{F}} \mathbb{E}_F [s(U; \theta, v)] \\ \text{s.t. } &\mathbb{E}_F [m(U; \theta, v)] = P \\ &\sup_{g \in \mathcal{G}} \mathbb{E}_F [\psi(U; \theta, v, g)] = 0 \end{aligned} \tag{Primal}$$

where  $\mathcal{F} := \{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu_1, \dots, \nu_k)\}$ . Moreover, we provide an explicit upper bound on the approximation error.

After applying the minimax duality, we need to solve the following problem:

$$\inf_{(\theta, v) \in \Theta \times \mathcal{V}} \sup_{\lambda \in \mathbb{R}^{d_P}, g \in \mathcal{G}} \inf_{F \in \Pi(\nu_1, \dots, \nu_k)} \mathbb{E}_F [c(U; \theta, v, g, \lambda)] - \lambda^T P$$

where the inner problem is an OT problem, which is computationally challenging in high-dimensional settings. The EOT is a computationally tractable approximation to the OT problem. Recall  $\mathcal{C}(\theta, v, g, \lambda, \lambda_{KL}) := \inf_{F \in \Pi(\nu_1, \dots, \nu_k)} \mathbb{E}_F [c(U; \theta, v, g, \lambda)] + \lambda_{KL} D_{KL}(F \| F_0)$ .

**Theorem 8** (Adapted from [Eckstein and Nutz \(2024\)](#) Theorem 3.1(i)). *Suppose Assumption 1 holds. Assume the marginals  $\{\nu_i\}_{i=1}^k$  have finite  $p + \eta$ -th moment for some  $\eta > 0$  and integer  $p \geq 1$ , and  $c(U; \theta, v, g, \lambda)$  satisfies the  $A_{L,C}$  condition in [Eckstein and Nutz \(2024\)](#) where  $L, C$  depend on  $(\theta, v, g, \lambda)$ . Let  $d_i$  be the dimension of  $U_i$ . Then, for any  $\lambda_{KL} \in (0, 1]$ ,*

$$0 \leq \mathcal{C}(\theta, v, g, \lambda, \lambda_{KL}) - \mathcal{C}(\theta, v, g, \lambda, 0) \leq \left( \sum_{i=2}^k d_i \right) \lambda_{KL} \log \left( \frac{1}{\lambda_{KL}} \right) + (k-1)^{\frac{1}{p}} LC \lambda_{KL}$$

<sup>18</sup>See [Christensen and Connault \(2023\)](#) Theorem 2.1 for similar results. However, their results are silent about how large the radius should be so that the bounds are close to the nonparametric bounds.

Theorem 8 provides an explicit upper bound on the approximation error of  $\mathcal{C}(\theta, v, g, \lambda, \lambda_{KL})$  to  $\mathcal{C}(\theta, v, g, \lambda, 0)$ . For DDC models, the constants  $L$  and  $C$  can be explicitly characterized under additional conditions (see [Eckstein and Nutz \(2022\)](#) Lemma 3.5, and [Eckstein and Nutz \(2024\)](#) Remark 2.1.) The upper bound strictly decreases to zero as  $\lambda_{KL} \downarrow 0$ . Therefore, we can choose a sufficiently small  $\lambda_{KL}$  (or sufficiently large  $\delta$ ) to achieve a desired accuracy for the approximation. Our framework thus approximates the nonparametric bounds in a computationally tractable way with an explicitly quantifiable approximation error.

**Remark 3.** Under certain conditions, one can establish the convergence of the EOT worst-case distribution to the OT worst-case distribution as the regularization parameter  $\lambda_{KL} \downarrow 0$ . [Nutz \(2021\)](#) Theorem 5.5 provides one sufficient condition: the existence of a solution  $F^*$  to the OT problem such that  $D_{KL}(F^* \| F_0) < +\infty$ .

## 5.2 Local Sensitivity

The local sensitivity<sup>19</sup> approach computes the right derivative of the bounds at  $\delta = 0$ , which measures the effect of a small perturbation of the reference distribution on the bounds. We show the right differentiability of the bounds with respect to  $\delta$ . Define:

$$\begin{aligned}\Pi_{TH} &:= \{F \in \mathcal{P}(\mathcal{U}) \mid \Pi(\nu_0, \dots, \nu_k), C_3 \leq dF \leq C_4, \|dF\|_{Lip} \leq L\} \\ \Pi_{TI} &:= \{F \in \mathcal{P}(\mathcal{U}) \mid \Pi(\nu_0, \nu_T), C_3 \leq dF \leq C_4, \|dF\|_{Lip} \leq L\}\end{aligned}$$

where  $\|\cdot\|_{Lip}$  is the Lipschitz constant, and  $C_3, C_4, L$  are positive constants. We assume:

**Assumption 9.** Let  $\mathcal{A}_{I,Lip}^\delta$  be either  $\mathcal{A}_{I,TH}^{\delta,Lip}$  or  $\mathcal{A}_{I,TI}^{\delta,Lip}$  defined as:

$$\begin{aligned}\mathcal{A}_{I,TH}^{\delta,Lip} &:= \{\alpha \in \Theta \times \Pi_{TH} \mid \mathbb{E}_F[m(U; \theta, v(\alpha))] = P_0, D_{KL}(F \| F_0) \leq \delta\} \\ \mathcal{A}_{I,TI}^{\delta,Lip} &:= \{\alpha \in \Theta \times \Pi_{TI} \mid \mathbb{E}_F[m(U; \theta, v(\alpha))] = P_0, D_{KL}(F \| F_0) \leq \delta\}\end{aligned}$$

and  $\mathcal{A}_{I,Lip}^{\delta,*}$  be either  $\mathcal{A}_{I,TH}^{\delta,Lip,*}$  or  $\mathcal{A}_{I,TI}^{\delta,Lip,*}$  that are the sets of solutions to the optimization problems  $\kappa(\delta, P_0)$  and  $\tilde{\kappa}_{TI}(\delta, P_0)$  over  $\mathcal{A}_{I,TH}^{\delta,Lip}$  and  $\mathcal{A}_{I,TI}^{\delta,Lip}$ . Assume:

(i)  $\delta^* := \inf\{\delta \geq 0 \mid \mathcal{A}_{I,Lip}^\delta \neq \emptyset\}$  is finite.<sup>20</sup>

(ii)  $\mathcal{U}$  is compact.

(iii)  $\|dF_0\|_{Lip} \leq L$ .

<sup>19</sup>See [Bartl et al. \(2021\)](#) Theorem 2.2 and [Christensen and Connault \(2023\)](#) Page 276 for similar results.

<sup>20</sup>The smallest radius can be computed, see Remark 4.

- (iv)  $0 \in \text{int}\{DP(\alpha)(\Theta \times \Pi - \alpha)\}$  for  $\forall \alpha \in \mathcal{A}_{I,Lip}^{\delta,*}$ .
- (v) For  $\forall \delta_t := \delta + t + o(t)$  and  $t > 0$  small enough, the optimization problem corresponding to  $\delta_t$  has an  $o(t)$ -optimal solution  $\alpha(t)$  such that  $d(\alpha(t), \mathcal{A}_{I,Lip}^{\delta,*}) = O(t)$ .
- (vi) For  $\forall t_n \downarrow 0$  the sequence  $\{\alpha(t_n)\}$  has a limit point (in the norm topology)  $\alpha_0 \in \mathcal{A}_{I,Lip}^{\delta,*}$ .

**Theorem 9.** Under Assumptions 3, 7, and 9,  $\kappa(\delta, P_0)$  and  $\tilde{\kappa}_{TI}(\delta, P_0)$  are right differentiable at  $\delta \geq \delta^*$  and their right derivatives are given by:

$$\begin{aligned} \lim_{\epsilon \downarrow 0} \frac{\kappa(\delta + \epsilon, P_0) - \kappa(\delta, P_0)}{\epsilon} &= \inf_{\alpha \in \mathcal{A}_{I,TH}^{\delta,Lip,*}} \sup_{\lambda_{KL} \in \Lambda_{KL}(\alpha, \delta)} -\lambda_{KL} \\ \lim_{\epsilon \downarrow 0} \frac{\tilde{\kappa}_{TI}(\delta + \epsilon, P_0) - \tilde{\kappa}_{TI}(\delta, P_0)}{\epsilon} &= \inf_{\alpha \in \mathcal{A}_{I,TI}^{\delta,Lip,*}} \sup_{\lambda_{KL} \in \Lambda_{KL}(\alpha, \delta)} -\lambda_{KL} \end{aligned}$$

where  $\Lambda_{KL}(\alpha, \delta)$  is the nonempty set of Lagrange multipliers corresponding to  $(\alpha, \delta)$ .

Theorem 9 shows the right differentiability. We can also compute the derivative of the length of the bounds. In practice, we may need to compute it numerically due to the optimization over the set of optimizers and the Lagrange multipliers.

### 5.3 The Robustness Metric

The robustness metric is the smallest deviation from the reference distribution that can lead to sensitive results (Spini (2024)). In practice, we begin by estimating a reference scalar parameter,  $\hat{s}_{F_0}$ , under the reference distribution. If the perturbed scalar parameter  $s_F$  is below a certain threshold, e.g.,  $\bar{s} = 0.95 \cdot \hat{s}_{F_0}$ , then we may be concerned about the robustness of the results. The robustness metric is defined as:

$$\begin{aligned} \delta(\bar{s}, P) &:= \inf_{(\theta, v, F) \in \Theta \times \mathcal{V} \times \Pi(\nu_1, \dots, \nu_k)} D_{KL}(F \| F_0) & \tilde{\delta}_{TI}(\bar{s}, P) &:= \inf_{(\theta, v, F) \in \Theta \times \mathcal{V} \times \Pi(\nu_1, \nu_T)} D_{KL}(F \| F_0) \\ \text{s.t. } \mathbb{E}_F[m(U; \theta, v)] &= P & \text{s.t. } \mathbb{E}_F[m(U; \theta, v)] &= P \\ \mathbb{E}_F[s(U; \theta, v)] &\leq \bar{s} & \mathbb{E}_F[s(U; \theta, v)] &\leq \bar{s} \\ \sup_{g \in \mathcal{G}} \mathbb{E}_F[\psi(U; \theta, v, g)] &= 0 & \sup_{g \in \mathcal{G}} \mathbb{E}_F[\psi(U; \theta, v, g)] &= 0 \end{aligned} \tag{6}$$

where  $\bar{s} \in \mathbb{R}$  is a user-specified threshold. The optimization problem searches for a distribution  $F$  in the identified set that results in  $\mathbb{E}_F[s(U; \theta, v)] \leq \bar{s}$  and is the closest to the reference distribution  $F_0$  in terms of KL divergence.

**Remark 4.** The  $\delta^*$  in Section 5 can be obtained by removing the constraint for  $\bar{s}$  in (6). That is, we seek the smallest radius  $\delta^*$  such that the identified set is nonempty. See [Schennach \(2014\)](#) Page 356 and [Christensen and Connault \(2023\)](#) Section 3.3 for similar definitions.

We can plot the bounds against  $\delta$  and then find the radius corresponding to  $\bar{s}$ . Alternatively, we can compute it directly by solving (6) whose duality results are given by:

**Theorem 10.** Let  $c(U; \theta, v, g, \lambda, \lambda_s) := \lambda^T m(U; \theta, v) + \lambda_s s(U; \theta, v) + \psi(U; \theta, v, g)$  where  $\lambda \in \mathbb{R}^{d_P}$ . Under Assumption 1, the following holds:

(i) (Minimax Duality)

$$\delta(\bar{s}, P) = \inf_{(\theta, v) \in \Theta \times \mathcal{V}} \sup_{\lambda \in \mathbb{R}^{d_P}, \lambda_s \geq 0, g \in \mathcal{G}} \mathcal{C}(\theta, v, g, \lambda, \lambda_s) - \lambda^T P - \lambda_s \bar{s}$$

where  $\mathcal{C}(\theta, v, g, \lambda, \lambda_s)$  is the EOT problem with regularization parameter 1:

$$\mathcal{C}(\theta, v, g, \lambda, \lambda_s) := \inf_{F \in \Pi(\nu_1, \dots, \nu_k)} \mathbb{E}_F [c(U; \theta, v, g, \lambda, \lambda_s)] + D_{KL}(F \| F_0)$$

(ii) (Entropic Optimal Transport Duality) We have:

$$\mathcal{C}(\theta, v, g, \lambda, \lambda_s) = \sup_{\{\phi_i \in L^1(\nu_i)\}_{i=1}^k} \sum_{i=1}^k \mathbb{E}_{\nu_i} \phi_i(U_i) - \mathbb{E}_{F_0} \exp \left( \sum_{i=1}^k \phi_i(U_i) - c(U; \theta, v, g, \lambda, \lambda_s) \right) + 1$$

Moreover, there are unique maximizers  $\{\phi_i^*\}_{i=1}^k$  up to additive constants  $F_0$  almost surely, and the unique worst-case distribution  $F^*$  has the density of the form:

$$\frac{dF^*(U)}{dF_0(U)} = \exp \left( \sum_{i=1}^k \phi_i^*(U_i) - c(U; \theta, v, g, \lambda, \lambda_s) \right) \quad F_0\text{-a.s.}$$

**Theorem 11.** Let  $c(U; \theta, v, g, \lambda, \lambda_s) := \lambda^T m(U; \theta, v) + \lambda_s s(U; \theta, v) + \psi(U; \theta, v, g)$  where  $\lambda \in \mathbb{R}^{d_P}$ . Under Assumptions 1(i), 1(iii), 1(iv), and 2, the following holds:

(i) (Minimax Duality)

$$\delta_{TI}(\bar{s}, P) = \inf_{(\theta, v) \in \Theta \times \mathcal{V}} \sup_{\lambda \in \mathbb{R}^{d_P}, \lambda_s \geq 0, g \in \mathcal{G}} \mathcal{C}_{TI}(\theta, v, g, \lambda, \lambda_s) - \lambda^T P - \lambda_s \bar{s}$$

where  $\mathcal{C}_{TI}(\theta, v, g, \lambda, \lambda_s)$  is defined as follows:

$$\mathcal{C}_{TI}(\theta, v, g, \lambda, \lambda_s) := \inf_{F \in \Pi(\nu_1, \nu_T)} \mathbb{E}_F [c(U; \theta, v, g, \lambda, \lambda_s)] + D_{KL}(F \| F_0) \quad (7)$$

(ii) The unique worst-case distribution to (7) has the form:

$$\frac{dF^*(U)}{dF_0(U)} = \exp(\phi_1^*(\xi_1) + \phi_T^*(\xi_T) - c(U; \theta, v, g, \lambda, \lambda_s))$$

where  $\phi_1^*(\xi_1)$  and  $\phi_T^*(\xi_T)$  are the unique maximizers (up to an additive constant) to:

$$\sup_{\phi_1 \in L^1(\nu_1), \phi_T \in L^1(\nu_T)} \mathbb{E}_{\nu_1} \phi_1(\xi_1) + \mathbb{E}_{\nu_T} \phi_T(\xi_T) - \mathbb{E}_{R_{1,T}} \exp(\phi_1(\xi_1) + \phi_T(\xi_T)) + 1$$

where the auxiliary reference measure  $R_{1,T}$  is defined as:

$$dR_{1,T}(\xi_1, \xi_T) := \int_{\xi_2, \dots, \xi_{T-1}} \exp(-c(U; \theta, v, g, \lambda, \lambda_s)) dF_0(\xi_1, \dots, \xi_T)$$

Furthermore, the solution  $F^*$  has the Markov property, i.e.,  $F^* \in \Pi_{\text{Markov}}(\nu_1, \nu_T)$ .

(iii) (Equivalence)  $\tilde{\delta}_{TI}(\bar{s}, P) = \delta_{TI}(\bar{s}, P)$  where  $\delta_{TI}(\bar{s}, P)$  is the optimal value of the optimization problem in (6) for the time-inhomogeneous case with the first-order Markov property constraint on  $F$ .

Theorem 11 provides the dual formulation for computing the smallest radius in the time-inhomogeneous case. The equivalence holds as the regularization parameter is 1.

## 6 Practical Implementation

This section presents the practical implementation of the proposed framework. Section 6.1 reviews the entropic optimal transport problem and the Sinkhorn algorithm. Section 6.2 proposes a computationally feasible algorithm.

### 6.1 Entropic Optimal Transport and Sinkhorn Algorithm

This section reviews the Sinkhorn algorithm for the entropic optimal transport problem.<sup>21</sup> Let  $(\mathcal{U}_i, \nu_i)$  for  $i = 1, \dots, k$  be probability spaces, where  $\mathcal{U}_i$  is the support for the random variable  $U_i$ . For a cost function  $c : \mathcal{U}_1 \times \dots \times \mathcal{U}_k \rightarrow \mathbb{R}$ , the entropic optimal transport

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<sup>21</sup>See [Sinkhorn and Knopp \(1967\)](#), [Cuturi \(2013\)](#) and [Nutz \(2021\)](#).

problem<sup>22</sup> with regularization parameter  $\lambda_{KL} > 0$  is defined as:

$$\mathcal{C}_{\lambda_{KL}} := \inf_{F \in \Pi(\nu_1, \dots, \nu_k)} \mathbb{E}_F [c(U_1, \dots, U_k)] + \lambda_{KL} D_{KL}(F \| F_{\otimes}) \quad (\text{EOT})$$

whose dual is given by:

$$\mathcal{C}_{\lambda_{KL}} = \sup_{\{\phi_i \in L^1(\nu_i)\}_{i=1}^k} \sum_{i=1}^k \mathbb{E}_{\nu_i} \phi_i(U_i) - \lambda_{KL} \mathbb{E}_{F_{\otimes}} \exp \left( \frac{\sum_{i=1}^k \phi_i(U_i) - c(U_1, \dots, U_k)}{\lambda_{KL}} \right) + \lambda_{KL} \quad (8)$$

where  $\phi_i$  is the test function for the marginal distribution constraint  $\nu_i$ . The dual problem is a concave maximization problem over  $\{\phi_i\}_{i=1}^k$ . The worst-case distribution is given by:

$$\frac{dF^*(U)}{dF_{\otimes}(U)} = \exp \left( \frac{\sum_{i=1}^k \phi_i^*(U_i) - c(U_1, \dots, U_k)}{\lambda_{KL}} \right)$$

where the optimizers  $(\phi_1^*, \dots, \phi_k^*)$  are known as the optimal EOT potentials (also called Schrödinger potentials), which are the solutions to the Schrödinger equation (SE):

$$\phi_1(U_1) = -\lambda_{KL} \log \left( \mathbb{E}_{F_{\otimes, -1}} \exp \left( \frac{\sum_{i=2}^k \phi_i(U_i) - c(U_1, \dots, U_k)}{\lambda_{KL}} \right) \right) \quad \nu_1\text{-a.s.} \quad (\text{SE1})$$

$\vdots$

$$\phi_k(U_k) = -\lambda_{KL} \log \left( \mathbb{E}_{F_{\otimes, -k}} \exp \left( \frac{\sum_{i=1}^{k-1} \phi_i(U_i) - c(U_1, \dots, U_k)}{\lambda_{KL}} \right) \right) \quad \nu_k\text{-a.s.} \quad (\text{SEk})$$

where  $F_{\otimes, -i}$  is the product measure of all marginals except for the  $i$ -th marginal. The Schrödinger equations (SE1) to (SEk) can be interpreted as the variational first-order conditions for optimality (see [Nutz \(2021\)](#) Remark 3.4). Moreover, they also characterize the marginal constraints. To see this, define:

$$dF(U) = \exp \left( \frac{\sum_{i=1}^k \phi_i(U_i) - c(U_1, \dots, U_k)}{\lambda_{KL}} \right) dF_{\otimes}(U)$$

The marginal density can be obtained by integrating out the other marginals; therefore:

$$(\text{SEi}) \Leftrightarrow \text{the } i\text{-th marginal of } F \text{ is } \nu_i$$

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<sup>22</sup>If  $F_0 \neq F_{\otimes}$ , then Lemma 12 reformulates the problem as the [EOT](#) problem.

The Sinkhorn algorithm can be interpreted as a coordinate ascent scheme for the optimization problem (8). It is a computationally fast<sup>23</sup>, iterative method for solving (SE1)-(SEk).

**Algorithm 1** (Sinkhorn Algorithm). Initialize  $\phi_i^{(0)} := 0$  for  $i = 1, \dots, k$ . For iteration  $t$ , sequentially update for  $j = 1, \dots, k$  by:

$$\phi_j^{(t+1)}(U_j) := -\lambda_{KL} \log \int \exp \left( \frac{\sum_{i \neq j} \phi_i^{(t)}(U_i) - c(U_1, \dots, U_k)}{\lambda_{KL}} \right) dF_{\otimes, -j}.$$

Stop if  $\sup_j \|\phi_j^{(t+1)} - \phi_j^{(t)}\|_2 < \epsilon$  for a tolerance  $\epsilon > 0$ .

## 6.2 Proposed Algorithm

For a given  $(\theta, v, g, \lambda, \lambda_{KL})$ , the EOT problem provides the worst-case distribution,  $F^*$ . This allows us to update  $v$  by solving the structural constraint with  $F^*$ . We therefore propose an iterative algorithm to solve the minimax problem. The algorithm proceeds by alternating between updating  $(\theta, v)$  and the dual (Lagrange multiplier) variables,  $(g, \lambda, \lambda_{KL})$ . After initializing all parameters, each iteration  $t$  involves the following steps:

**Algorithm 2.** Initialize  $(\theta^{(0)}, v^{(0)}, g^{(0)}, \lambda^{(0)}, \lambda_{KL}^{(0)})$ . At iteration  $t$ ,

1. **Update Model Primitives  $(\theta, v)$ :** For  $(\theta^{(t)}, v^{(t)}, g^{(t)}, \lambda^{(t)}, \lambda_{KL}^{(t)})$ , update<sup>24</sup> the model parameters  $(\theta, v)$  by:
  - (a) Propose a new candidate  $\theta^{(t+1)}$ .
  - (b) Solve the EOT problem with  $(\theta^{(t+1)}, v^{(t)}, g^{(t)}, \lambda^{(t)}, \lambda_{KL}^{(t)})$  and obtain  $F^*$ .
  - (c) Update  $v^{(t+1)}$  by solving the structural constraint with  $F^*$ .
  - (d) Accept/reject the proposed  $(\theta^{(t+1)}, v^{(t+1)})$ .
2. **Update Dual Variables  $(g, \lambda, \lambda_{KL})$ :** For  $(\theta^{(t+1)}, v^{(t+1)}, g^{(t)}, \lambda^{(t)}, \lambda_{KL}^{(t)})$ , update  $(g, \lambda, \lambda_{KL})$  following the same procedure as in the previous step.
3. Iterate until convergence, or a pre-specified number of iterations is reached.

The computational cost per iteration mainly comes from solving the EOT problem and the structural constraint, which are both computationally fast. However, the number of

<sup>23</sup>For its convergence rate, see [Peyré et al. \(2019\)](#), [Carlier \(2022\)](#), and [Eckstein and Nutz \(2022\)](#).

<sup>24</sup>If gradient-based methods are used, then we smooth the non-differentiable components (e.g., the indicator function in Example 4) using a smooth approximation.

iterations required for convergence can be much larger, as our optimization problem is a minimax problem that is potentially non-differentiable.

## 7 Empirical Application: Infinite Horizon DDC

This section applies our framework to an infinite-horizon dynamic demand for new cars in the UK, France, and Germany. Due to the unobserved product characteristics, the indirect utility of purchasing is the latent variable. To estimate the price elasticity and conduct welfare analysis of electric vehicles (EV) subsidy, we require a distributional assumption for the latent variable to solve the Bellman equation. Existing literature often uses an AR(1) process (e.g., [Schiraldi \(2011\)](#); [Gowrisankaran and Rysman \(2012\)](#)), which may be misspecified. For example, the indirect utility may exhibit nonlinear dynamics. Therefore, we conduct a sensitivity analysis with respect to this reference distribution.

### 7.1 Data

We use the trim-level<sup>25</sup> data from IHS Markit during the period from 2014 to 2023. The monthly level dataset contains sales, list price, and characteristics of car models in the UK, France and Germany, which are treated as three independent markets in our analysis. To construct the final dataset, we first aggregate data from the trim-level to the model-level. Then, we aggregate fuel types into: petrol, diesel, electric, and hybrid. We remove models whose total sales during the data period are less than 20,000.<sup>26</sup> Finally, we adjust list prices by subtracting EV subsidies. The initial market size for January 2014 is calculated by subtracting the number of registered cars from the total population of each country. The market size is then updated each subsequent month by subtracting the total number of cars sold in the preceding period.

Table 1 presents the summary statistics. The three markets offer around 141–215 products from around 23 to 30 brands per month. In terms of average sales per model, Germany has 81,401 units, closely followed by France with 81,357 units, and the UK with 70,682 units. The average price is around \$33,444 in the UK, \$27,830 in France, and \$36,117 in Germany.

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<sup>25</sup>In the automobile industry, a trim-level refers to a specific version of a vehicle model that comes with a particular set of features, options, and styling elements.

<sup>26</sup>In addition, we exclude car-month observations with sales below 150 units in the German market and below 5 units in the French market.



Table 1: Summary Statistics by Country (Monthly, 2014-2023)

|         | <u>Avg #</u> | <u>Avg #</u> | <u>Avg #</u> | <u>Price (USD)</u> | <u>Horsepower</u> | <u>Weight (kg)</u> |
|---------|--------------|--------------|--------------|--------------------|-------------------|--------------------|
| Country | Products     | Brands       | Sales        | Mean               | Mean              | Mean               |
| UK      | 196          | 30           | 70,682       | 33,444             | 140               | 1,836              |
| France  | 141          | 23           | 81,357       | 27,830             | 112               | 1,702              |
| Germany | 215          | 26           | 81,401       | 36,117             | 151               | 1,952              |

**Note:** The price is adjusted for EV subsidies. First two columns are average number of products and brands per month. Average sales is the average number of cars sold per model. Mean price, horsepower, and weight are weighted by total sales.

## 7.2 The Model

The model is infinite horizon. At each month  $t$ , a consumer  $i$  chooses  $j \in \mathcal{J}_t \cup \{0\}$  where  $\mathcal{J}_t$  is the set of available cars at  $t$ , and 0 is the outside option of not purchasing. Each car  $j \in \mathcal{J}_t$  is characterized by a vector of observable characteristics  $x_{jt}$ , price  $p_{jt}$ , and an unobserved characteristic  $\xi_{jt}$ . The period utility of choosing  $j$  is given by:

$$u(j, x_t, p_t, \xi_t, \varepsilon_{it}) = \begin{cases} \alpha p_{jt} + x_{jt}^T \theta + \xi_{jt} + \varepsilon_{ijt} & \text{if } j \in \mathcal{J}_t \\ \varepsilon_{0it} & \text{if } j = 0 \end{cases}$$

where  $\varepsilon_{it}$  is a vector of i.i.d. type I extreme value utility shocks, and  $x_t, p_t, \xi_t$  are the vectors of observable characteristics, prices, and unobserved characteristics for all cars in  $\mathcal{J}_t$ .

We assume a purchase is a terminating decision, i.e., consumers exit the market after the purchase. The conditional value function of purchasing car  $j$  can be written as the sum of the current period utility and the flow utility after purchase:

$$v_j(x_t, p_t, \xi_t) = \frac{x_{jt}^T \theta + \xi_{jt}}{1 - \beta} + \alpha p_{jt}$$

where  $\beta = 0.975$  is the discount factor. The inclusive value of purchasing is defined as:

$$\omega_t = \log \sum_{j \in \mathcal{J}_t} \exp \left( \frac{x_{jt}^T \theta + \xi_{jt}}{1 - \beta} + \alpha p_{jt} \right)$$

Following [Schiraldi \(2011\)](#) and [Gowrisankaran and Rysman \(2012\)](#), we assume:

**Assumption 10** (Inclusive Value Sufficiency<sup>27</sup> (IVS)).  $G(\omega_{t+1}|x_t, p_t, \xi_t)$  can be summarized by  $G(\omega_{t+1}|\omega_t)$  where  $G$  is the conditional distribution function.

<sup>27</sup>The IVS assumption has also been used in [Hendel and Nevo \(2006\)](#); [Melnikov \(2013\)](#); [Osborne \(2018\)](#).

Under the IVS assumption,  $\omega_t$  is the only state variable, and the value function  $V(\omega)$  is the solution to the smoothed Bellman equation:

$$V(\omega) = \log (\exp (v_0(\omega)) + \exp (v_1(\omega))) \quad (9)$$

where  $v_0(\omega) = \beta \mathbb{E}[V(\omega')|\omega]$  and  $v_1(\omega) = \omega$  are the conditional value functions of not purchasing and purchasing, respectively. The market share of car  $j$  at time  $t$  is given by:

$$s_{jt}(x_t, p_t, \xi_t) = \underbrace{\frac{\exp(\omega_t)}{\exp(V(\omega_t))}}_{\text{Probability of Purchasing a Car}} \times \underbrace{\frac{\exp(v_j(x_t, p_t, \xi_t))}{\exp(\omega_t)}}_{\text{Conditional Probability of Purchasing Car } j} \quad (10)$$

### 7.3 First-Stage Estimation

In each market, the car with the highest total sales is set as the reference product, denoted by  $r$ .<sup>28</sup> Taking the log-odds ratio for cars  $j$  and  $r$  at time  $t$  yields:

$$\log \left( \frac{s_{jt}}{s_{rt}} \right) = \alpha \Delta p_{jt} + \frac{\Delta x_{jt}^T \theta}{1 - \beta} + \frac{\Delta \xi_{jt}}{1 - \beta} \quad (11)$$

where  $\Delta p_{jt} := p_{jt} - p_{rt}$ ,  $\Delta x_{jt} := x_{jt} - x_{rt}$ , and  $\Delta \xi_{jt} := \xi_{jt} - \xi_{rt}$ . The parameters  $(\alpha, \theta)$  are identified<sup>29</sup> by the BLP instruments [Berry et al. \(1993\)](#). Moreover,  $\Delta \xi_{jt}$  can be recovered by fitting the relative market share  $\frac{s_{jt}}{s_{rt}}$ , while the unobserved characteristic of the reference car,  $\xi_{rt}$ , cannot.

The exogenous characteristics,  $x_{jt}$ , include vehicle log-weight, log-horsepower, brand fixed effects, SUV fixed effect, and fuel type fixed effects.<sup>30</sup> Table 2 presents the regression results. Price coefficients are negative and significant in all markets, ranging from -0.158 in France to -0.192 in Germany. Relative to petrol vehicles, EVs are preferred in the UK (0.019) and France (0.004), but not in Germany (0.000). Hybrid vehicles are valued positively across all three markets, with coefficients ranging from 0.029 in Germany to 0.037 in France. In contrast, diesel has a negative coefficient in the UK (-0.015) but positive effects in Germany (0.008) and France (0.006).

<sup>28</sup>The reference cars are Volkswagen Golf (Petrol) in Germany, Peugeot 208 (Petrol) in France, and Ford Fiesta (Petrol) in the UK. The reference car for each country is always available in that country's market.

<sup>29</sup>The intercept and reference fixed effects are not identified from (11); they are absorbed into  $\xi_{rt}$ .

<sup>30</sup>The instruments are the exogenous product characteristics, average log-weight and log-horsepower of competitors' products, the proportion of competitors' products, the proportion of hybrid cars squared, and the number of brands. A competitor product is defined as a car whose brand is not that of  $r$  or  $j$ .

Table 2: Instrumental Variable Regression Results

|                       | UK     |           | Germany |           | France |           |
|-----------------------|--------|-----------|---------|-----------|--------|-----------|
|                       | Coef.  | Std. Err. | Coef.   | Std. Err. | Coef.  | Std. Err. |
| <b>Price</b>          | −0.178 | (0.043)   | −0.192  | (0.004)   | −0.158 | (0.042)   |
| <b>Log horsepower</b> | 0.140  | (0.038)   | 0.165   | (0.003)   | 0.065  | (0.031)   |
| <b>Log weight</b>     | −0.008 | (0.003)   | 0.046   | (0.003)   | 0.002  | (0.001)   |
| <b>SUV</b>            | 0.018  | (0.002)   | −0.003  | (0.001)   | 0.013  | (0.002)   |
| <b>Diesel</b>         | −0.015 | (0.003)   | 0.008   | (0.001)   | 0.006  | (0.004)   |
| <b>Electric</b>       | 0.019  | (0.004)   | 0.000   | (0.001)   | 0.004  | (0.002)   |
| <b>Hybrid</b>         | 0.035  | (0.010)   | 0.029   | (0.001)   | 0.037  | (0.009)   |
| Adjusted $R^2$        | 0.549  |           | 0.533   |           | 0.803  |           |
| # of Month-Years      | 120    |           | 120     |           | 120    |           |
| # of Obs.             | 23,451 |           | 25,715  |           | 16,862 |           |

**Note:** An observation is a pair of  $(j, t)$  where  $j$  is a car model other than the reference car  $r$  and  $t$  is the time period. Standard errors are in parentheses. Brand fixed effects are not reported.

## 7.4 The Reference Distribution and Scalar Parameters of Interest

We first define the reference distribution and then introduce the scalar parameters of interest. After the first stage estimation, we can calculate  $\omega_t$  up to  $\xi_{rt}$  as:

$$\omega_t = \log \sum_{j \in \mathcal{J}_t} \exp \left( \frac{x_{jt}^T \theta + \Delta \xi_{jt}}{1 - \beta} + \alpha p_{jt} \right) + \frac{\xi_{rt}}{1 - \beta} \quad (12)$$

As we have identified the utility parameters, the potential sensitivity of the empirical results solely arises from the distributional assumption on  $\omega_t$ .

The reference transition of  $\omega_t$  to solve the Bellman equation is an AR(1) process:

$$\omega_t = \gamma_0 + \gamma_1 \omega_{t-1} + \eta_t \quad (13)$$

where  $\eta_t$  follows an i.i.d. normal distribution with mean 0 and variance  $\sigma^2$ .

The parameters  $(\gamma_0, \gamma_1, \sigma^2)$  are estimated using an iterative procedure. We begin with an initial guess of  $(\gamma_0, \gamma_1, \sigma^2)$  and circulate between: (i) solving the Bellman equation (9), (ii) recovering  $\{\omega_t\}_{t=1}^T$  from the market share of purchasing (the first part of (10)), (iii) updating  $(\gamma_0, \gamma_1, \sigma^2)$  by refitting an AR(1) process (13) until we find a fixed point. The reference distribution  $F_0$  for  $(\omega, \omega')$  is the product of the transition kernel of the estimated AR(1) process and its stationary distribution  $\nu_0$ . The perturbation set is defined as:

$$\mathcal{F} := \{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu_0, \nu_0), D_{KL}(F \| F_0) \leq \delta\}$$

We consider two scalar parameters: (i) the industrywide price elasticity of demand, (ii) the welfare analysis of an additional EV subsidy. For both cases, the transition of  $\omega_t$  is unchanged, i.e., we assume consumers' beliefs about the transition of  $\omega_t$  stay the same.

For industrywide price elasticity at period  $t_1$ , we consider a 1% increase in the price of all cars. The future  $\omega_{t_1+1}$  is conditional on:

$$\omega'_{t_1} = \log \sum_{j \in \mathcal{J}_{t_1}} \exp \left( \frac{x_{jt_1}^T \theta + \Delta \xi_{jt_1}}{1 - \beta} + 1.01 \cdot \alpha p_{jt_1} \right) + \frac{\xi_{rt_1}}{1 - \beta}$$

and the industrywide price elasticity at time  $t_1$  is:

$$\frac{s_{0t_1} - s_0(\omega'_{t_1})}{1 - s_{0t_1}} \times 100$$

where  $s_0(\omega'_{t_1})$  is the model-implied market share of not purchasing.

For EV subsidy at  $t_1$ , we consider an additional 3,000 USD subsidy. Denote by  $\mathcal{J}_{t_1, \text{EV}}$  the set of EVs at  $t_1$ . The future  $\omega_{t_1+1}$  is conditional on:

$$\omega_{t_1}^{\text{EV}} = \log \sum_{j \in \mathcal{J}_{t_1}} \exp \left( \frac{x_{jt_1}^T \theta + \Delta \xi_{jt_1}}{1 - \beta} + \alpha (p_{jt_1} - \mathbb{1}(j \text{ is an EV}) \cdot 3000) \right) + \frac{\xi_{rt_1}}{1 - \beta}$$

and the consumer surplus from the subsidy is given by:

$$\text{Consumer Surplus} = \frac{V(\omega_{t_1}^{\text{EV}}) - V(\omega_{t_1})}{-\alpha} \times M_{t_1}$$

where  $M_{t_1}$  is the market size at time  $t_1$ .<sup>31</sup>

## 7.5 Sensitivity Analysis

Our framework requires the constraints to be linear in  $F$ , while the Bellman equation (9) is not. We first reformulate it to fit into our framework. By the *Hotz-Miller Inversion Lemma* (Hotz and Miller (1993)), we have:

$$V(\omega) = \omega - \log s_1(\omega) \tag{14}$$

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<sup>31</sup>The cost of the subsidy is  $\text{Cost} = 3000 \cdot M_{t_1} \cdot \sum_{j \in \mathcal{J}_{t_1, \text{EV}}} \frac{\exp(v_j(x_{jt_1}, p_{jt_1} - 3000, \xi_{jt_1}))}{\exp(V(\omega_{t_1}^{\text{EV}}))}$ .

Taking the log-odds ratio of purchasing and not purchasing, and using (14), we have:

$$\log \left( \frac{1 - s_0(\omega)}{s_0(\omega)} \right) = \omega - \beta \mathbb{E} [\omega' - \log(1 - s_0(\omega')) | \omega] \quad (15)$$

The above constraint is the fixed point problem on the market share space (see [Aguirregabiria and Mira \(2002\)](#)). The right-hand side is linear in the conditional distribution. The following lemma establishes the relationship between the fixed point problems in (9) and (15).

**Assumption 11.** *Assume the inclusive value  $\omega$  has compact support  $\Omega \subset \mathbb{R}$  with nonempty interior equipped with the sup-norm and  $\mathbb{E} [f(\omega') | \omega] \in C(\Omega)$  for any  $f \in C(\Omega)$  where  $C(\Omega)$  is the space of continuous functions on  $\Omega$ .*

**Lemma 3.** *The following holds:*

- (i) *Under Assumption 11, the fixed point problem (9) has a unique solution on  $C(\Omega)$ .*
- (ii) *The fixed point problem (15) has a unique solution if and only if the fixed point problem (9) has a unique solution.*
- (iii) *If (9) and (15) both have unique solutions, then it holds that  $1 - s_0(\omega) = \exp(\omega - V(\omega))$  where  $s_0(\omega)$  and  $V(\omega)$  are the solutions to (9) and (15), respectively.*

Lemma 3 shows that solving the Bellman equation (9) is equivalent to solving (15). We further convert (15) into an unconditional moment constraint by assuming that  $s_0(\omega)$  is the solution to (15) if and only if:

$$\sup_{g \in C(\Omega)} \mathbb{E}_F \left[ g(\omega) \left( \log \left( \frac{1 - s_0(\omega)}{s_0(\omega)} \right) - \omega + \beta \omega' - \beta \log(1 - s_0(\omega')) \right) \right] = 0 \quad (16)$$

**Assumption 12.** *For  $\forall F \in \mathcal{F}$ , the solution  $s_0(\omega)$  corresponding to (15) satisfies the following: for all  $t = 1, \dots, T$ , there exists a unique  $\omega_t \in \Omega$  such that  $s_0(\omega_t) = s_{0t}$ .*

Assumption 12 allows us to profile out  $\{\omega_t\}_{t=1}^T$ , which is useful for the implementation. A sufficient condition is that  $s_0(\omega)$  is continuous and strictly decreasing, and its smallest and largest values are small and large enough. Under Assumption 11, we have  $s_0(\omega) \in C(\Omega)$ . Moreover, we can expect that a higher inclusive value  $\omega$  corresponds to a lower market share of not purchasing, i.e.,  $s_0(\omega)$  is decreasing in  $\omega$ . Therefore, Assumption 12 is mild.

The last condition is the fixed point constraint similar to the procedure to estimate the AR(1) process. Suppose the distribution  $F$  is used in (16), and  $\{\omega_t\}_{t=1}^T$  is the sequence of

recovered inclusive values. Denote by  $\hat{F}$  the estimator of the joint distribution for the pairs  $\{(\omega_t, \omega_{t+1})\}_{t=1}^{T-1}$ . Then, our fixed point constraint is:

$$D_{KL}(F\|\hat{F}) \leq \epsilon_T$$

where  $\epsilon_T$  is the tolerance level. To choose  $\epsilon_T$ , we estimate the joint distribution of inclusive values recovered from the AR(1) process by the kernel density estimator with Gaussian kernel and bandwidth selected by the 5-fold cross-validation. Then, we set  $\epsilon_T$  to be the KL divergence between the kernel density estimator and the reference distribution.

For EV subsidy, by (14), the consumer surplus (CS) is given by:

$$V(\omega_{t_1}^{\text{EV}}) - V(\omega_{t_1}) = \omega_{t_1}^{\text{EV}} - \omega_{t_1} + \log\left(1 + \frac{s_1(\omega_{t_1}^{\text{EV}}) - s_{1t_1}}{s_{1t_1}}\right)$$

where  $\omega_{t_1}^{\text{EV}} - \omega_{t_1}$  does not depend on  $F$ . Therefore, to bound CS, it is equivalent to bounding the change in the market share of purchase.

Putting everything together, the lower bound on the elasticity at  $t_1$  is given by:

$$\begin{aligned} & \inf_{s_0(\omega) \in C(\Omega)} \inf_{F \in \mathcal{F}} \frac{s_{0t_1} - s_0(\omega'_{t_1})}{1 - s_{0t_1}} \times 100 \\ & \text{s.t. } s_0(\omega_t) = s_{0t} \text{ for } t = 1, \dots, T \\ & \sup_{g \in C(\Omega)} \mathbb{E}_F \left[ g(\omega) \left( \log\left(\frac{1 - s_0(\omega)}{s_0(\omega)}\right) - \omega + \beta\omega' - \beta \log(1 - s_0(\omega')) \right) \right] = 0 \\ & D_{KL}(F\|\hat{F}) \leq \epsilon_T \end{aligned}$$

where  $\omega'_{t_1}$  is replaced by  $\omega_{t_1}^{\text{EV}}$  for the EV subsidy case. For  $\delta = 0$ , the reference distribution is the unique solution to the above problem. The corresponding elasticity is the reference industrywide elasticity.

## 7.6 Implementation

We adapt Algorithm 2 proposed in Section 6.2. We will use numerical integration to discretize the support of the AR(1) process. Therefore, the recovered inclusive values  $\{\omega_t\}_{t=1}^T$  are not differentiable with respect to the discretized market share function  $s_0(\omega)$ . To address this issue, we employ the MCMC optimization method. We alternate between solving the EOT problem to obtain the worst-case distribution, solving the Bellman equation to update  $s_0(\omega)$ , recovering the inclusive values, and checking the fixed point constraint. To derive a

tractable dual formulation, we handle the fixed point constraint in a specific way. Because this constraint depends on an estimator,  $\hat{F}$ , that changes during optimization—potentially causing numerical instability—we first derive the dual formulation without it.<sup>32</sup> Then, the fixed point constraint determines the acceptance/rejection of the candidate parameters in the Metropolis-Hastings step. Consider the following optimization problem:

$$\begin{aligned} \inf_{s_0(\omega) \in C(\Omega)} \inf_{F \in \mathcal{F}} \quad & \frac{s_{0t_1} - s_0(\omega'_{t_1})}{1 - s_{0t_1}} \times 100 \\ \text{s.t.} \quad & s_0(\omega_t) = s_{0t} \text{ for } t = 1, \dots, T \\ & \sup_{g \in C(\Omega)} \mathbb{E}_F \left[ g(\omega) \left( \log\left(\frac{1 - s_0(\omega)}{s_0(\omega)}\right) - \omega + \beta\omega' - \beta \log(1 - s_0(\omega')) \right) \right] = 0 \end{aligned}$$

Applying Theorem 1, its dual is:

$$\begin{aligned} \inf_{s_0(\omega) \in C(\Omega)} \sup_{g \in C(\Omega), \lambda_{KL} \geq 0} \quad & \frac{s_{0t_1} - s_0(\omega'_{t_1})}{1 - s_{0t_1}} \times 100 + \mathcal{C}(s_0, g, \lambda_{KL}) - \lambda_{KL}\delta \\ \text{s.t.} \quad & s_0(\omega_t) = s_{0t} \text{ for } t = 1, \dots, T \end{aligned}$$

where  $\mathcal{C}(s_0, g, \lambda_{KL})$  is the EOT problem:  $\mathcal{C}(s_0, g, \lambda_{KL}) := \sup_{F \in \Pi(\nu_0, \nu_0)} \mathbb{E}_F [c(\omega, \omega'; s_0, g)] + \lambda_{KL} D_{KL}(F \| F_0)$  whose cost function is  $c(\omega, \omega'; s_0, g) = g(\omega) \left( \log\left(\frac{1 - s_0(\omega)}{s_0(\omega)}\right) - \omega + \beta\omega' - \beta \log(1 - s_0(\omega')) \right)$  and the worst-case conditional distribution  $F^*$  is given by:

$$dF^*(\omega'|\omega) = \exp\left(\frac{\phi_1^*(\omega) + \phi_2^*(\omega') - c(\omega, \omega'; s_0, g)}{\lambda_{KL}}\right) dF_0(\omega'|\omega) \quad F_0\text{-a.s.}$$

where  $\phi_1^*(\omega)$  and  $\phi_2^*(\omega')$  are the optimal EOT potentials. During the optimization process,  $dF^*(\omega'|\omega)$  is used to update  $s_0(\omega)$  by solving (15) using fixed point iteration.

As shown in Theorem 1, the expectation in the dual is taken with respect to the reference distribution. Therefore, we discretize the estimated AR(1) process, which results in three approximation errors: (i) the Bellman equation is solved on the discretized support, (ii)  $\{\omega_t\}_{t=1}^T$  are recovered approximately, and (iii) the elasticity is computed approximately. That is, we approximate  $s_0(\omega)$  by the market share of the nearest grid point to  $\omega$ . Therefore, there is a trade-off between approximation error and computational cost. A finer discretization reduces the approximation error, while increasing the number of optimization parameters.

However, our dual formulation significantly improves computational efficiency. Suppose we discretize the AR(1) process into  $N$  grid points. If we directly solve (16), the number of

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<sup>32</sup>In principle, we can choose other constraints like the integral probability metrics and adapt the minimax theorem in Theorem 1. However, it can lead to more optimization parameters.

optimization parameters is  $O(N^2)$  due to the transition matrix. In contrast, the number of optimization parameters is  $O(N)$  in the dual formulation.

Algorithm 3 summarizes the simulated annealing MCMC optimization algorithm (Kirkpatrick et al. (1983)). It starts with the reference market share  $s^{(0)}$ , and alternate between proposing new parameters  $(g', \lambda'_{KL})$ , solving the EOT problem, solving the Bellman equation using the worst-case distribution, and accepting or rejecting the proposed parameters using the Metropolis-Hastings step based on the change in the elasticity penalized by the violation of the market share and fixed-point constraints. At each improvement step, it pools previous results across all radii for initialization. We choose 51 grid points, 5,000 MCMC steps, 5 optimization steps, 14 radii (the last is  $10^{10}$ ), and 100 as the simulated annealing multiplier.

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**Algorithm 3:** Simulated Annealing MCMC Optimization Algorithm

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**Parameters:**  $N$ : Number of grid points;  $T$ : MCMC steps per optimization run;  $J$ : Optimization steps;  $m$ : Simulated annealing multiplier;  $l$ : Number of radii;

```

for  $j = 1$  to  $J$                                      /* Optimization Step  $j$  */
do
  for  $i = 0$  to  $l - 1$ , set  $\delta_i = 10^{-3+i \cdot 0.25}$ 
  do
    if  $i = 0$  then
      If  $j = 0$ : Set  $s^{(0)}$  as the reference market share. Initialize  $g^{(0)}, \lambda_{KL}^{(0)}$ .
      If  $j > 0$ : Set  $(s^{(0)}, g^{(0)}, \lambda_{KL}^{(0)})$  to the optimal solution from the previous step's stored
        results with upper bound  $10^{-3}$  on the KL divergence to the reference distribution.
    else
      If  $j = 0$ : Set  $(g^{(0)}, \lambda_{KL}^{(0)}, s^{(0)})$  to the optimal solution from the previous step.
      If  $j > 0$ : Set  $(s^{(0)}, g^{(0)}, \lambda_{KL}^{(0)})$  to the optimal solution from the previous stored results
        with upper bound  $10^{-3+i \cdot 0.25}$  on the KL divergence to the reference distribution.
    end
  end
  for  $t = 1, \dots, T$                                 /* Simulated Annealing MCMC optimization */
  do
    // 1. Propose New Parameters
    Propose  $(g', \lambda'_{KL})$  from the random walk, solve the EOT problem  $\mathcal{C}(s_0^t, g', \lambda'_{KL})$  and
    obtain  $F^*$ , solve the Bellman equation (15) with  $F^*$ , recover  $\{\omega_t\}_{t=1}^T$ , estimate the
    distribution of  $\{(\omega_t, \omega_{t+1})\}_{t=1}^T$  by kernel density estimator, and compute the elasticity.
    // 2. Check Constraints and Apply Penalty
    Calculate the sum of violations from the market share and fixed-point constraints. The
    market share violation is defined as  $\max\{0, \text{vio}_F - \text{vio}_{ref}\}$  where  $\text{vio}_{ref}$  is the
    violation of the reference model. If the total violation exceeds 0.005, add a large
    penalty (100). If  $D_{KL}(F^* \| F_0) > \delta_i$ , add a large penalty (100).
    // 3. Accept/Reject (Metropolis-Hastings)
    Apply a Metropolis-Hastings step based on the change in the (penalized) elasticity
    multiplied by  $\frac{10 \cdot (1 + (s-1) \cdot (m-1))}{(T-1)}$  where the prior is  $\mathcal{N}(0, 100)$ .
    // 4. Adapt (Andrieu and Thoms (2008) Algorithm 4)
    Update the random walk via vanishing adaptation scheme.
  end
end
end
end

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## 7.7 Results

We estimate two alternative transition densities for each market. The first assumes that the inclusive values are i.i.d. normally distributed. The second estimates a nonlinear AR(1) process using a cubic spline<sup>33</sup>. In the following figures, we plot the KL divergence from the alternative models to the reference model. The independent model is closer to the reference model with KL divergence between 0.04 and 0.67, while the nonlinear AR(1) process is farther away, with KL divergence between 3.94 and 10.09.

Figures 1-3 plot the bounds on the industrywide elasticities<sup>34</sup> for the UK, Germany, and France in December 2023. The French market is the least elastic (reference elasticity: -4.048), while the German market is the most elastic (reference elasticity: -6.073). The UK market's reference elasticity is -5.336. Based on our three sensitivity measures, we define the local (global) sensitivity as the ratio of the local (global) interval length to the reference value. For local sensitivity, we set  $\delta = 0.001$ , while for global sensitivity, we set  $\delta = 10^{10}$ . The robustness metric is defined as the smallest deviation from the reference distribution such that the elasticity can deviate by, for example, 2.5% from the reference elasticity.

The French market is the least sensitive in terms of local and global sensitivity, with 1.16% local deviation and 6.20% global deviation from the reference elasticity. The UK market is less sensitive locally (1.66%) than the German market (3.52%). They are both more sensitive globally (15.16% for the UK vs. 15.24% for Germany) than the French market. The bounds for the UK market flatten around 0.178, while the French and German markets flatten around 0.056. For the robustness metric, we consider 2.5% deviation from the reference elasticity. The UK market's robustness metric is around 0.018 for the upper bound and 0.008 for the lower bound. The French market's robustness metric is around 0.025 for the upper bound and 0.018 for the lower bound. The German market's robustness metric is around 0.002 for the lower bound and 0.003 for the upper bound. Therefore, in terms of the robustness metric, the French market is also the most robust, while the German

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<sup>33</sup>The nonlinear AR(1) process is specified as:  $g(\omega) = \sum_{k=1}^{N+3} \rho_k \Phi\left(\frac{\omega-a}{h} - (k-2)\right)$  where  $a$  is the minimum of the discretized support of  $\omega$ ,  $h$  is the distance between two adjacent grid points,  $\rho_k$  are parameters to be estimated, and  $\Phi(t) = \begin{cases} 4 - 6t^2 + 3|t|^3 & \text{if } |t| \leq 1 \\ (2 - |t|)^3 & \text{if } 1 < |t| \leq 2 \\ 0 & \text{otherwise} \end{cases}$ . For this model, we set  $N = 4$  to avoid overfitting.

Then, we discretize the reference AR(1) process into 4 grid points, and compute the KL divergence from the nonlinear AR(1) process to the reference AR(1) process.

<sup>34</sup>Schiraldi (2011) finds a average long-run price elasticities ranging from -3.54 to -4.34 across different car segments for the Italian market. D'Haultfœuille et al. (2019) reports average elasticities between -3.94 and -6.40 across consumer groups for the French market. Reynaert and Sallee (2021) finds a mean own-price elasticity of -5.45 for the European market. Grieco et al. (2024) estimates an average elasticity of -5.36 for the U.S. market in 2015. Remmy (2025) reports a mean price elasticity of -4.043 for the German market.

market is the least robust.

Figures 1-3 also plot the bounds on the consumer surplus from an additional \$3,000 EV subsidy. The subsidy is implemented between July and December 2023, when the reference consumer surplus is maximized. They are November for the UK (reference CS: \$2,880 million), October for France (reference CS: \$1,432 million), and September for Germany (reference CS: \$856 million). Overall, the EV subsidy is beneficial as the lower bounds are \$2,584 million for the UK, \$1,243 million for France, and \$309 million for Germany. The corresponding costs for subsidy are around \$12 million for the UK, \$23 million for France, and \$41 million for Germany. The costs are insensitive to the misspecification, as they only depend on the absolute change in the market share of purchases, and the conditional market share of EVs, instead of percentage change used for consumer surplus.

In terms of local sensitivity, the UK market is the least sensitive (2.52 % local deviation), the French market exhibits similar local sensitivity (4.72% local deviation), while the German market is the most sensitive (24.79% local deviation). In terms of global sensitivity, the UK market and French markets share similar sensitivity (25.17 % global deviation for the UK, and 24.73 % for France), while the German market is the most sensitive (102.75 % global deviation). For the robustness metric, if we consider 10% deviation from the reference CS, the UK and French markets' robustness metric are more than 0.018, while the German market's robustness metric is around 0.001. Therefore, the German market is also the least robust in terms of robustness metric.

Figure 4 plots the time series of bounds on the industrywide elasticities. We set  $\delta = 0.001$  for the local deviation and  $\delta = 1$  for the global deviation, as the bounds flatten at a maximum of 0.178 in December 2023. Large points in the figure indicate that the KL divergence constraint is binding—specifically, when the KL divergence from the worst-case distribution to the reference distribution exceeds  $0.95 \cdot \delta$ . When two consecutive points align horizontally, this indicates that increasing the radius does not affect the bounds, as exemplified by the UK market in November 2023. In terms of both local and global sensitivity, the UK and French markets are less sensitive than the German market. All three markets exhibit some sensitive periods. For the UK market, the upper bound's global deviation in February 2022 is around 30% away from the reference elasticity. For the French market, the upper bounds' global deviations in April 2021, and July 2022 are around 50% away from the reference elasticity. For the German market, the lower bound's global deviation in March 2022 and April 2023 is around 50% away from the reference elasticity. In terms of local sensitivity, the lower bounds' of German market in April 2021 is around 50% away from the reference elasticity.

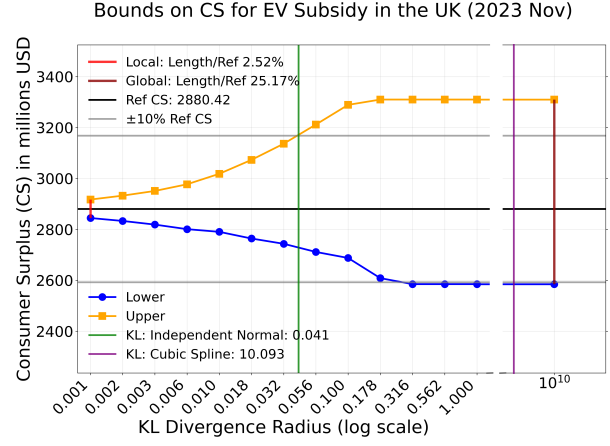
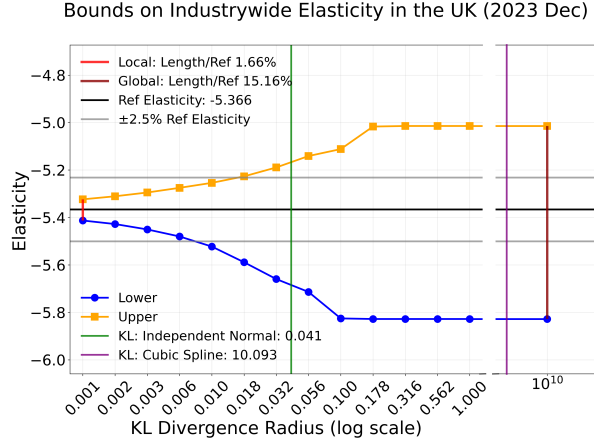


Figure 1: Bounds on Industrywide Elasticity and Consumer Surplus for the UK

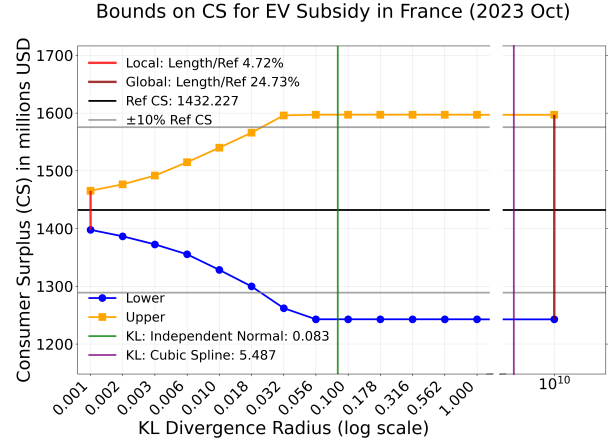
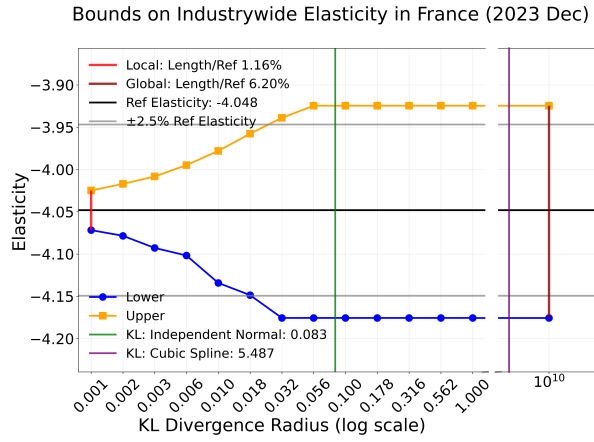


Figure 2: Bounds on Industrywide Elasticity and Consumer Surplus for France

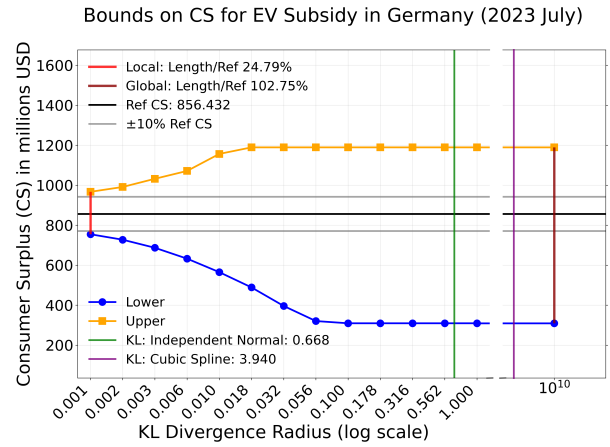
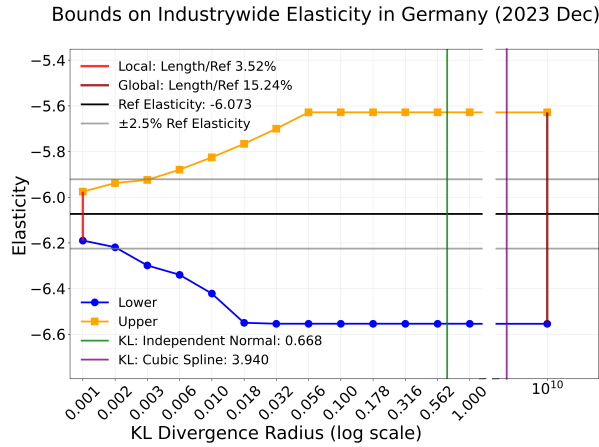


Figure 3: Bounds on Industrywide Elasticity and Consumer Surplus for Germany

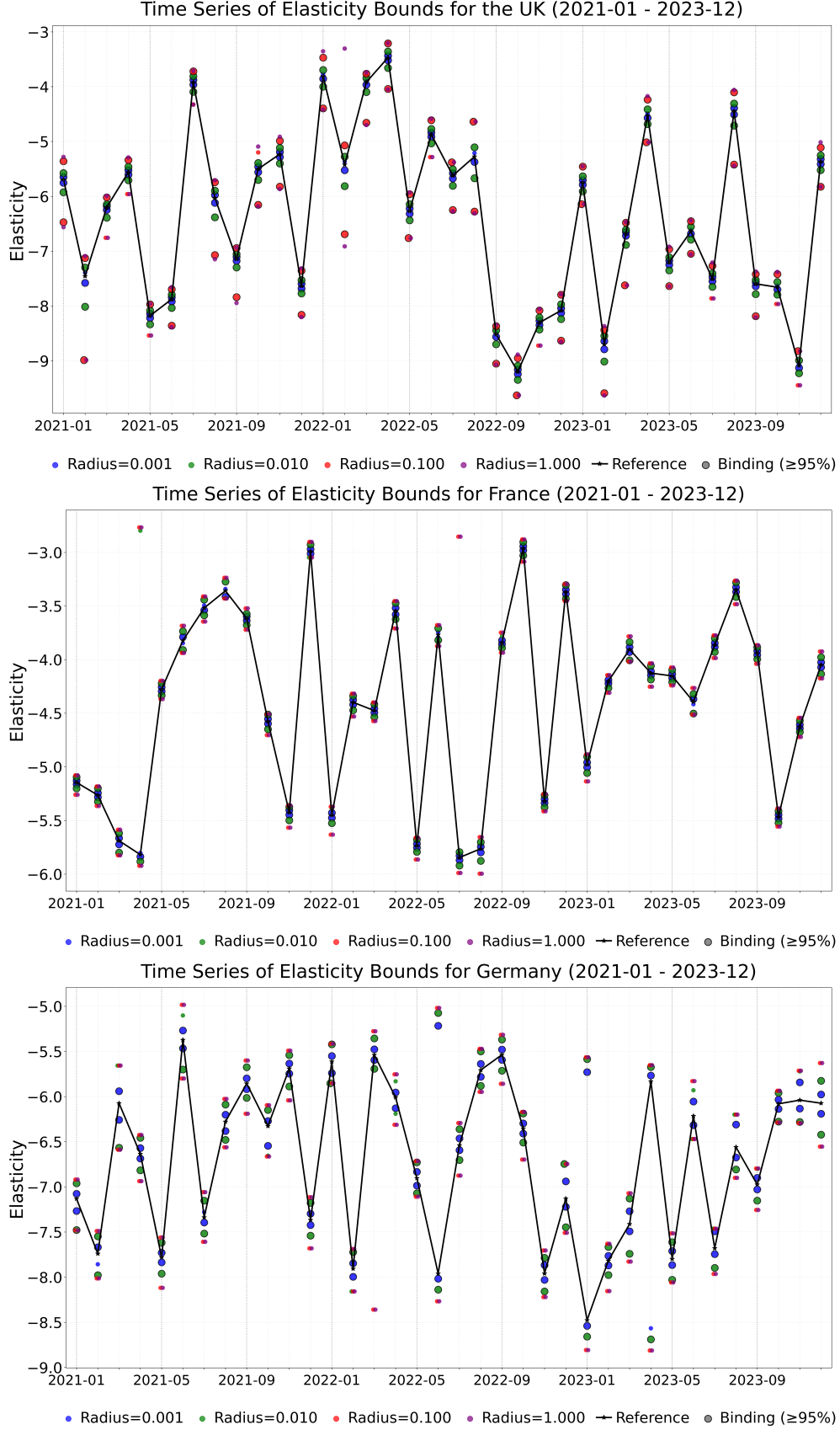


Figure 4: Time Series of Bounds on Industrywide Elasticity for UK, Germany, and France

## 8 Empirical Application: Finite Horizon DDC

This section applies our framework to a finite-horizon dynamic labor supply model for taxi drivers in New York City (NYC). In the model, the market-level supply shock is the latent variable. Our reference dynamic process is an AR(1) process. We consider the sensitivity analysis of the labor supply elasticity with respect to such distributional assumption.

### 8.1 The Data

We use data from New York City’s Taxi and Limousine Commission’s (TLC) Taxi Passenger Enhancement Project (TPEP). The TPEP data contain a complete record of all trips operated by licensed drivers. The day shift starts at 5 AM and ends at 5 PM, and the night shift starts at 5 PM and ends at 5 AM. We choose a sample of 10,500 drivers that were active in 2013 as in [Kalouptside et al. \(2021c\)](#). We restrict the sample to day shift drivers who were only working during the day shifts. We aggregate the transaction-level data to the driver-hour level. In addition, we create 10 uniformly spaced bins for weekdays (Monday-Thursday) and 4 bins<sup>35</sup> for weekends (Friday-Sunday) between the lowest and highest hourly earnings and calculate the average hourly earnings in each bin. Then, we remove important days (i.e., Memorial Day, the Fourth of July, and New Year’s Eve). Finally, we restrict the sample to shifts that started between 5 AM and 8 AM, which accounts for 86.84% of shifts for weekdays, and 71.84% for weekends. The final sample contains 3,562 drivers and 206 days for weekdays, and 3,322 drivers and 156 days for weekends.

Table 3 presents the hourly summary statistics. The average hourly earnings range from \$24.22 at 4:00 PM to \$37.00 at 8:00 AM. The share of drivers who continue working is high in the early morning, with 100% of drivers working at 6:00 AM and 7:00 AM. This share starts to decline after 2 PM, and drops substantially to 52.24% and 53.59% by 4:00 PM. Therefore, we assume that drivers can only choose to stop working between 8 AM and 4 PM.

### 8.2 The Model

At the beginning of hour  $t$  of day  $m$ , a taxi driver  $i$  decides whether to continue working ( $a = 1$ ) or not ( $a = 0$ ). The decision to stop working is a terminating action, meaning the

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<sup>35</sup>The number of bins is chosen so that a Gaussian distribution approximates the stationary distribution of the market-level supply shock recovered from the last period (see Figure 5).

Table 3: Summary statistics.

| Hour         | Share of Drivers that |         | Hourly Earnings (\$) |         |         |         |
|--------------|-----------------------|---------|----------------------|---------|---------|---------|
|              | Continue Working (%)  |         | Weekday              |         | Weekend |         |
|              | Weekday               | Weekend | Mean                 | Std Dev | Mean    | Std Dev |
| 6:00 AM      | 100.00                | 100.00  | 32.66                | 3.00    | 31.47   | 3.86    |
| 7:00 AM      | 100.00                | 100.00  | 35.04                | 2.88    | 30.11   | 4.49    |
| 8:00 AM      | 98.18                 | 96.21   | 37.00                | 2.84    | 31.21   | 5.68    |
| 9:00 AM      | 97.64                 | 96.34   | 34.73                | 2.37    | 30.56   | 4.39    |
| 10:00 AM     | 96.57                 | 96.58   | 30.02                | 2.19    | 30.72   | 3.33    |
| 11:00 AM     | 95.36                 | 95.80   | 29.12                | 2.30    | 31.48   | 3.11    |
| 12:00 PM     | 95.53                 | 93.98   | 30.86                | 2.26    | 32.88   | 2.70    |
| 1:00 PM      | 95.13                 | 94.80   | 30.62                | 2.34    | 33.55   | 2.65    |
| 2:00 PM      | 92.60                 | 92.84   | 33.80                | 2.32    | 35.02   | 2.62    |
| 3:00 PM      | 80.61                 | 82.09   | 34.67                | 2.02    | 35.45   | 2.64    |
| 4:00 PM      | 52.24                 | 53.59   | 24.22                | 2.08    | 25.61   | 2.40    |
| 5:00 PM      | 0.00                  | 0.00    | —                    | —       | —       | —       |
| <hr/>        |                       |         |                      |         |         |         |
| # of Drivers | 3,562                 | 3,322   |                      |         |         |         |
| # of Days    | 206                   | 156     |                      |         |         |         |

**Note:** The table uses TPEP Data from January 1, 2013 to December 31, 2013. An observation is defined by a driver-hour.

driver exits the market upon stopping. The period utility of working is given by:

$$u(a_{imt}, k_{imt}, w_{mt}, \xi_{mt}, \varepsilon_{imt}; \theta) = \begin{cases} \theta_0 + \theta_1 k_{imt} + \theta_2 k_{imt}^2 + \theta_3 w_{mt} + \xi_{mt} + \varepsilon_{i1mt} & \text{if } a_{imt} = 1 \\ \varepsilon_{i0mt} & \text{if } a_{imt} = 0 \end{cases}$$

where  $k_{imt}$  is the number of hours worked,  $w_{mt}$  is the average hourly earnings,  $\varepsilon_{imt} := (\varepsilon_{i1mt}, \varepsilon_{i0mt})$  is i.i.d. type I extreme value utility shocks, and  $\xi_{mt}$  is an exogenously evolved stationary unobserved market-level supply shock. It captures the market-level time-variant unobserved heterogeneity such as weather, congestion, or city events.

We assume markets are i.i.d, and suppress the subscript  $(i, m)$  for brevity. Let  $u(k_t, w_t; \theta)$  be the deterministic part of period of utility of working up to  $(\xi_t, \varepsilon_t)$ . Let  $\beta := 0.999999$  be the discount factor. The smoothed Bellman equation at time  $t$  is given by:

$$V_t(k_t, w_t, \xi_t) = \log (\exp (v_{0t}(k_t, w_t, \xi_t)) + \exp (v_{1t}(k_t, w_t, \xi_t)))$$

where the conditional value functions of working and not working are given by:

$$\begin{aligned} v_{1t}(k_t, w_t, \xi_t) &= u(k_t, w_t; \theta) + \xi_t + \beta \mathbb{E}_{\xi_{t+1}|\xi_t} \mathbb{E}_{w_{t+1}|w_t} [V_{t+1}(k_t + 1, w_{t+1}, \xi_{t+1})] \\ v_{0t}(k_t, w_t, \xi_t) &= 0 \end{aligned} \tag{17}$$

### 8.3 First-Stage Estimation

With only one terminating action, the utility parameters cannot be identified using (11). Therefore, we estimate the utility parameters using the Euler Equations in Conditional Choice Probabilities (ECCP) estimator introduced in [Kalouptside et al. \(2021c\)](#). By the *Hotz-Miller Inversion Lemma* ([Hotz and Miller \(1993\)](#)), we have:

$$V_t(k_t, w_t, \xi_t) = -\log p_t(k_t, w_t, \xi_t) = v_{1t}(k_t, w_t, \xi_t) - \log(1 - p_t(k_t, w_t, \xi_t)) \quad (18)$$

where  $p_t(k_t, w_t)$  is the CCP of not working. Combining (17) and (18) gives:

$$\log \left( \frac{1 - p_t(k_t, w_t, \xi_t)}{p_t(k_t, w_t, \xi_t)} \right) = u(k_t, w_t; \theta) + \xi_t - \beta \mathbb{E}_{\xi_{t+1}|\xi_t} \mathbb{E}_{w_{t+1}|w_t} [\log p_{t+1}(k_t + 1, w_{t+1}, \xi_{t+1})] \quad (19)$$

Without a distributional assumption for  $\xi_t$ , we cannot calculate the conditional expectation in (19). However, the cross-sectional data allows us to estimate CCPs, denoted as  $\hat{p}_t(k, w)$ . We estimate  $\hat{p}_t(k, w)$  by a flexible logit for each  $t$ . Let the expectational error be:

$$\hat{e}(k_t, w_t, k_{t+1}, w_{t+1}, \xi_t) := \beta \log \hat{p}_{t+1}(k_{t+1}, w_{t+1}) - \beta \mathbb{E}_{\xi_{t+1}|\xi_t} \mathbb{E}_{w_{t+1}|w_t} [\log p_{t+1}(k_{t+1}, w_{t+1}, \xi_{t+1})]$$

where  $k_{t+1} = k_t + 1$ . Then, we can rewrite (19) as:

$$\log \left( \frac{1 - \hat{p}_t(k_t, w_t)}{\hat{p}_t(k_t, w_t)} \right) + \beta \log \hat{p}_{t+1}(k_{t+1}, w_{t+1}) = u(k_t, w_t; \theta) + \xi_t + \hat{e}(k_t, w_t, k_{t+1}, w_{t+1}, \xi_t)$$

Therefore  $\theta$  can be identified using an instrument for  $\xi_t + \hat{e}(k_t, w_t, k_{t+1}, w_{t+1}, \xi_t)$ . Denote by  $\mathcal{K}_t$  the set of possible hours worked at  $t$ .<sup>36</sup> The ECCP estimator stacks all  $k \in \mathcal{K}_t$ . A unit of observation is defined by day-hour.

Following [Kalouptside et al. \(2021c\)](#), we use the previous day's average hourly earnings for the same hour as the IV. Table 4 shows the estimation results. The implied marginal value of time defined by  $-\frac{\theta_1 + 2\theta_2 k}{\theta_3}$  ranges from \$0 at around  $k = 7$  hours to \$5.35 at  $k = 11$  hours for weekdays, and from \$0 at around  $k = 8$  hours to \$9.95 at  $k = 11$  hours for weekends.

### 8.4 The Reference Distribution and Scalar Parameters of Interest

We define the reference distribution and introduce the scalar parameters of interest. Let  $N_{tk}$  be the number of drivers who has worked  $k$  hours at hour  $t$ . The market-level  $\xi_t$  equates the

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<sup>36</sup>Note that  $|\mathcal{K}_t| = 4$  for  $t \geq 9$  AM and  $|\mathcal{K}_t| = 3$  for  $t = 8$  AM.

Table 4: ECCP Estimation Results

| Variable                       | Model Estimates |           |         |           |
|--------------------------------|-----------------|-----------|---------|-----------|
|                                | Weekday         |           | Weekend |           |
|                                | Coef.           | Std. Err. | Coef.   | Std. Err. |
| <b>Constant</b>                | -2.1720         | (0.0304)  | -0.5631 | (0.0724)  |
| <b>Hours worked</b>            | 0.4035          | (0.0025)  | 0.1866  | (0.0032)  |
| <b>Hours worked (squared)</b>  | -0.0274         | (0.0002)  | -0.0116 | (0.0003)  |
| <b>Average hourly earnings</b> | 0.0373          | (0.0010)  | 0.0069  | (0.0023)  |
| # of Drivers                   | 3,562           |           | 3,322   |           |
| # of Days                      | 206             |           | 156     |           |

**Notes:** A unit of observation is defined by day-hour. Each observation stacks all  $k_t \in \mathcal{K}_t$ . The model is estimated using 2SLS with the previous day's average hourly earnings as the IV. The standard errors are clustered at the day- $k$  level.

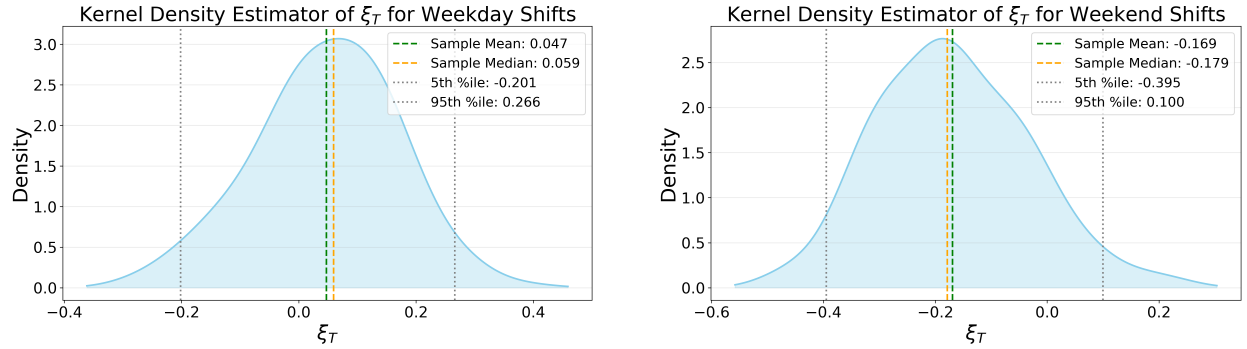
model-implied weighted average CCP of not working with the observed weighted average:

$$\sum_{k \in \mathcal{K}_t} \frac{N_{tk}}{\sum_{k \in \mathcal{K}_t} N_{tk}} p_t(k, w_t, \xi_t) = \sum_{k \in \mathcal{K}_t} \frac{N_{tk}}{\sum_{k \in \mathcal{K}_t} N_{tk}} \hat{p}_t(k, w_t) \quad (20)$$

As we assume  $\xi_t$  is stationary, its marginal distribution at  $T$  is its stationary distribution. At hour  $T$ , drivers solve a static problem, and the CCP of not working is:

$$p_T(k_T, w_T, \xi_T) = \frac{1}{1 + \exp(u(k_T, w_T, \xi_T))}$$

Therefore, the stationary distribution is identified by recovering  $\xi_T$  to satisfy (20) at  $T$ . Figure 5 plots the kernel density estimator of  $\xi_T$ . We fit a Gaussian distribution to the last period  $\xi_T$ . Denote its mean and standard deviation as  $\mu_\xi$  and  $\sigma_\xi$ , respectively.

Figure 5: Kernel Density Estimator of the Last Period  $\xi_T$



The reference model for  $\xi_t$  is an AR(1) process:

$$\xi_t = \mu + \rho\xi_{t-1} + \eta_t \quad (21)$$

where  $\eta_t$  follows i.i.d normal distribution with mean 0 and variance  $\sigma^2$ . As we have identified its stationary distribution, we only need to solve the fixed point problem for  $\rho$ : we begin with an initial guess of  $\rho$  and circulate between: (i) setting  $\mu = \mu_\xi \cdot (1 - \rho)$  and  $\sigma = \sigma_\xi \cdot \sqrt{1 - \rho^2}$ , (ii) solving the Bellman equation using backward induction, (iii) recovering  $\xi_t$  using (20), and (iv) updating  $\rho$  by refitting the AR(1) process to the recovered  $\xi_t$  until convergence. The reference distribution  $F_0$  for  $(\xi, \xi')$  is the product of the transition kernel of the estimated AR(1) process and its stationary distribution  $\nu_0$ . The perturbation set is defined as:

$$\mathcal{F} := \{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu_0, \nu_0), D_{KL}(F \| F_0) \leq \delta\}$$

We consider two scalar parameters of interest: the elasticity of stopping working with respect to average hourly earnings, and the Frisch elasticity of labor supply. Both elasticities are at the individual level, meaning the demand side remains unchanged. Therefore, we keep the transitions of  $w_t$  and  $\xi_t$  fixed.

For the elasticity of stopping working, we increase the average hourly earnings from the current bin,  $w_{mt}$ , to the next bin,  $w'_{mt}$ . The weighted average of the elasticity at hour  $t$  is:

$$\sum_m \sum_{k \in \mathcal{K}_{mt}} \frac{N_{mtk}}{\sum_{m,k \in \mathcal{K}_{mt}} N_{mtk}} \frac{p_t(k, w'_{mt}, \xi_{mt}) - \hat{p}_{kmt}}{\hat{p}_{kmt}} \times \frac{w'_{mt} - w_{mt}}{w_{mt}}$$

As shown in Table 3, the share of drivers who continue working begins to decline around 11 AM. Moreover, our model does not endogenize the initial entry decision (i.e., the choice of when to start a shift). To compute the Frisch elasticity, we assume 11 AM is the first hour drivers can choose to stop working, and consider a 1% increase in average hourly earnings beginning at 11 AM. The total expected hours worked at day  $m$  is:

$$H(m, \boldsymbol{\xi}_m, \mathbf{w}_m, p) := \sum_{k \in \{3,4,5,6\}} N_{mk} \sum_{t=12}^{16} (t-11) p_{mt}(k, w_{mt}, \xi_{mt}) \prod_{t_1=11}^{t-1} (1 - p_{t_1}(k, w_{mt_1}, \xi_{mt_1}))$$

where  $N_{mk}$  is the number of drivers whose hours worked is  $k$  at 11 AM of day  $m$ , and  $\boldsymbol{\xi}_m, \mathbf{w}_m$  are the vectors of  $\xi_{mt}, w_{mt}$  for  $t = 11, \dots, 16$ . Then, the Frisch elasticity is:

$$\frac{\sum_m H(m, \boldsymbol{\xi}_m, \mathbf{w}'_m, p') - \sum_m H(m, \boldsymbol{\xi}_m, \mathbf{w}_m, p)}{\sum_m H(m, \boldsymbol{\xi}_m, \mathbf{w}_m, p)} \times 100$$

where  $\mathbf{w}'_m = 1.01 \cdot \mathbf{w}_m$ , and  $p'$  is derived from the Bellman equation with  $\mathbf{w}'_m$ .

## 8.5 Sensitivity Analysis

We convert (19) into an unconditional moment constraint by assuming that  $p_t(k, w, \xi)$  is the solution to (19) for a given  $p_{t+1}(k+1, w, \xi)$  if and only if:

$$\sup_{g_{tk} \in C(\mathcal{W} \times \Xi)} \mathbb{E}_F \mathbb{E}_{w_{t+1}, w_t} \left[ g_{tk}(w_t, \xi_t) \left( \log \left( \frac{1 - p_t(k, w_t, \xi_t)}{p_t(k, w_t, \xi_t)} \right) - u(k, w_t) - \xi_t + \beta \log p_{t+1}(k+1, w_{t+1}, \xi_{t+1}) \right) \right] \quad (22)$$

Let the term inside the expectation in (22) be  $\psi_t(k, w_t, w_{t+1}, \xi_t, \xi_{t+1}; u, p_t, p_{t+1}, g_{tk})$ .

To profile out  $\xi_{mt}$  from recovering the probability of stop working, we assume:

**Assumption 13.** For  $\forall F \in \mathcal{F}$ , the solution  $p_{tk}(w, \xi)$  corresponding to (22) satisfies the following: for all  $m, t$ , there exists a unique  $\xi_{mt} \in \Xi$  that satisfies (20).

The final constraint is a fixed point constraint similar to the procedure used to estimate the AR(1) process. Suppose  $F$  is used in (22). Denote by  $\hat{F}$  an estimator of the distribution of the pair  $(\xi, \xi')$  using the recovered  $\{\xi_{mt}\}_{m=1, t=1}^{M, T-1}$  from (20). Then, our fixed point constraint is:

$$D_{KL}(F \| \hat{F}) \leq \epsilon_M$$

where  $\epsilon_M$  is the tolerance level. As the sample size is large, we use Scott's Rule to initialize the bandwidth and then use 5-fold cross-validation to select bandwidth candidates around Scott's estimate that maximizes the log-likelihood. To choose  $\epsilon_M$ , we estimate the joint distribution of supply shocks recovered from the AR(1) process by the kernel density estimator with Gaussian kernel. Then, we set  $\epsilon_M$  to be the KL divergence from the kernel density estimator to the reference distribution.

Then, the lower bound on the elasticity of stopping working at  $t$  is:

$$\begin{aligned} & \inf_{\{p_t\}_{t=1}^{T-1}} \inf_{F \in \mathcal{F}} \sum_m \sum_{k \in \mathcal{K}_{mt}} \frac{N_{mtk}}{\sum_{m, k \in \mathcal{K}_{mt}} N_{mtk}} \frac{p_t(k, w'_{mt}, \xi_{mt}) - \hat{p}_{kmt}}{\hat{p}_{kmt}} \times \frac{w'_{mt} - w_{mt}}{w_{mt}} \\ & \text{s.t.} \quad \sum_{k \in \mathcal{K}_{mt}} \frac{N_{mtk}}{\sum_{k \in \mathcal{K}_{mt}} N_{mtk}} p_t(k, w_{mt}, \xi_{mt}) = \sum_{k \in \mathcal{K}_{mt}} \frac{N_{mtk}}{\sum_{k \in \mathcal{K}_{mt}} N_{mtk}} \hat{p}_t(k, w_{mt}) \text{ for } \forall m, t \\ & \sup_{g_{tk}} \mathbb{E}_F \mathbb{E}_{w_{t+1}, w_t} [\psi_t(k, \xi_t, \xi_{t+1}, w_t, w_{t+1}; u, p_t, p_{t+1}, g_{tk})] \text{ for } \forall t \leq T-1, k \quad (23) \\ & D_{KL}(F \| \hat{F}) \leq \epsilon_M \end{aligned}$$

The last period  $T$  is a static problem, therefore  $p_T$  is not an optimization parameter. For  $\delta = 0$ , the reference distribution is the unique solution to the above problem. The corresponding elasticity is the reference elasticity.

For the Frisch elasticity, we increase the earnings coefficient by 1% from 11 AM. This allows us to leave the discretized state space and its transition unchanged at the cost of solving additional Bellman equations. The lower bound on the Frisch elasticity is:

$$\begin{aligned}
& \inf_{\{p_t, p'_t\}_{t=1}^{T-1}} \inf_{F \in \mathcal{F}} \frac{\sum_m H(m, \boldsymbol{\xi}_m, \mathbf{w}'_m, p') - \sum_m H(m, \boldsymbol{\xi}_m, \mathbf{w}_m, p)}{\sum_m H(m, \boldsymbol{\xi}_m, \mathbf{w}_m, p)} \times 100 \\
& \text{s.t.} \quad \sum_{k \in \mathcal{K}_{mt}} \frac{N_{mtk}}{\sum_{k \in \mathcal{K}_{mt}} N_{mtk}} p_t(k, w_{mt}, \xi_{mt}) = \sum_{k \in \mathcal{K}_{mt}} \frac{N_{mtk}}{\sum_{k \in \mathcal{K}_{mt}} N_{mtk}} \hat{p}_t(k, w_{mt}) \text{ for } \forall m, t \\
& \quad \sup_{g_{tk}^1} \mathbb{E}_F \mathbb{E}_{w_{t+1}, w_t} [\psi_t(k, \xi_t, \xi_{t+1}, w_t, w_{t+1}; u, p_t, p_{t+1}, g_{tk}^1)] \text{ for } \forall t \leq T-1, k \\
& \quad \sup_{g_{tk}^2} \mathbb{E}_F \mathbb{E}_{w_{t+1}, w_t} [\psi_t(k, \xi_t, \xi_{t+1}, w_t, w_{t+1}; u', p'_t, p'_{t+1}, g_{tk}^2)] \text{ for } \forall t \leq T-1, k \\
& \quad D_{KL}(F \| \hat{F}) \leq \epsilon_M
\end{aligned}$$

where  $u'$  is the utility function with the earning coefficient increased by 1% from 11 AM.

## 8.6 Implementation

We adjust the procedure in Section 7.6 to account for the finite horizon model. The main difference is that we solve the Bellman equation (19) by backward induction. The number of Bellman equations (23) is 31.<sup>37</sup> The number of optimization parameters is  $31 \cdot N_w N_\xi + 1$  for the elasticity of stopping working, and  $62 \cdot N_w N_\xi + 1$  for the Frisch elasticity, where  $N_w$  is the number of bins for average hourly earnings,  $N_\xi$  is the number of grid points for  $\xi$ , and 1 is for the KL divergence constraint. We choose  $N_\xi = 99$ , 5,000 MCMC steps for Frisch elasticities, 2,500 MCMC steps for elasticities of stopping working, 5 optimization steps, 14 radii (the last is  $10^{10}$ ), and 100 as the simulated annealing multiplier. The covariance matrix for the random walk in Algorithm 3 is restricted to be diagonal.

## 8.7 Results

The alternative model is an independent model where the supply shocks are i.i.d. and follow the stationary distribution identified at  $T$ . The independent model is closer to the reference

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<sup>37</sup>Note that  $\mathcal{K}_8 = \{1, 2, 3\}, \mathcal{K}_t = \{t-8, t-7, t-6, t-5\}$  for  $t = 9, \dots, 15$ .

model for weekends than for weekdays. The KL divergence from the independent model to the reference model is 0.115 for weekends and 3.673 for weekdays.

Figure 6 plots the elasticity of stopping working from 9 AM to 3 PM for weekdays and weekends. We set  $\delta = 0.001$  for local deviation and  $\delta = 1$  for global deviation. Large points indicate that the KL divergence constraint is binding. When two consecutive points align horizontally, this indicates that increasing the radius does not affect the bounds. The elasticity of stopping working is negative and decreases over time. In particular, the labor supply is inelastic before 11 AM on weekdays and 12 PM on weekends, and elastic after that. At 3 PM, the elasticity of stopping working is around -2 for weekdays and -2.5 for weekends. Overall, both weekday and weekend elasticities are not sensitive to the distributional assumption. For weekdays, elasticity in the morning is more sensitive than in the afternoon in terms of both local and global sensitivity, while it is the opposite for weekends.

Figure 7 plots the Frisch elasticity bounds for weekdays and weekends. The reference Frisch elasticity is 0.472 for weekends and 0.698 for weekdays. Our reference estimates are consistent with labor supply literature. For example, [Buchholz et al. \(2023\)](#) reports a Frisch elasticity ranging from 0.47-0.54 for NYC taxi drivers. For both weekdays and weekends, the bounds flatten around 0.032-0.056. In terms of local sensitivity, the results appear sensitive to the distributional assumption, with a deviation of 28.37% for weekdays and 21.76% for weekends. In terms of global sensitivity, the deviation is larger, with 76.83% for weekdays and 42.84% for weekends. For the robustness metric approach, we consider a 15% deviation from the reference Frisch elasticity. The weekday's robustness metric is around 0.001 for the upper bound, while the lower bound never reaches the 15% deviation. The weekend's robustness metric is around 0.008 for the upper bound, and 0.01 for the lower bound. Therefore, the weekday's lower bound is more robust than the weekend's lower bound, while the weekday's upper bound is less robust than the weekend's upper bound.

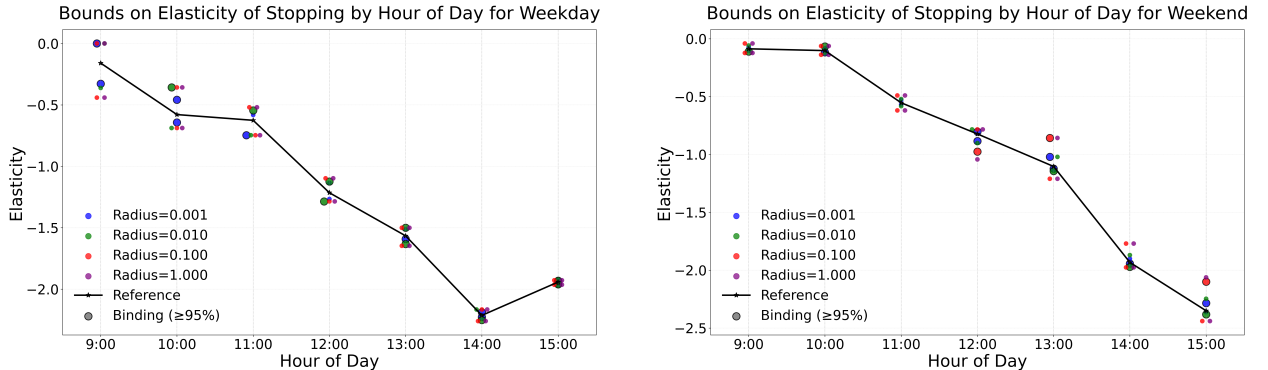


Figure 6: Elasticity of Stopping Working by Hour of Day for NYC Taxi Drivers

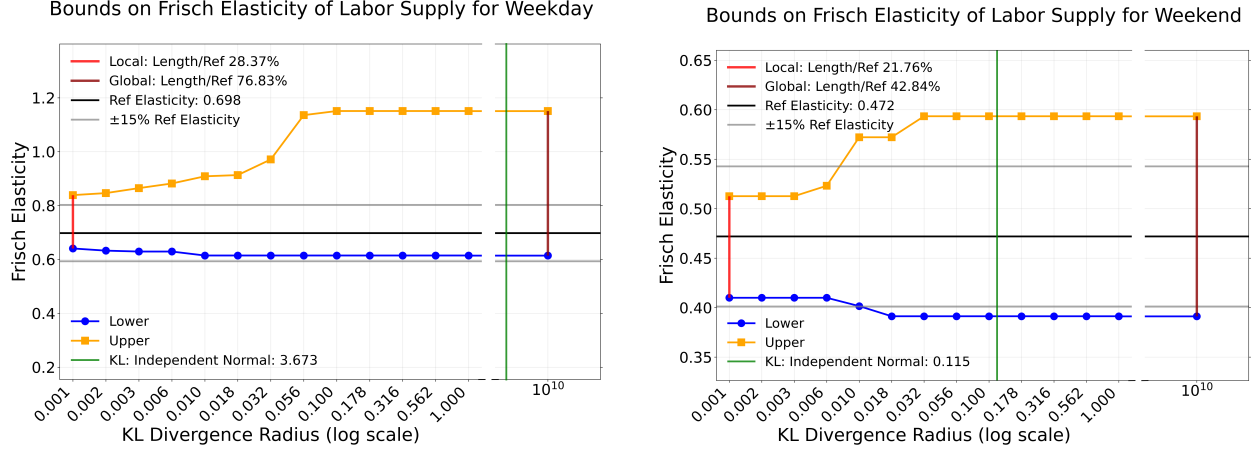


Figure 7: Frisch Elasticity Bounds by Hour of Day for NYC Taxi Drivers

## 9 Conclusion

We propose a computationally tractable framework to quantify the sensitivity of outcomes of interest to misspecified latent-state dynamics in structural models. We derive bounds on a scalar parameter of interest by perturbing a reference dynamic process, while imposing a stationarity condition for time-homogeneous models or a Markovian condition for time-inhomogeneous models. We apply the approach to an infinite-horizon dynamic demand model for new cars in the UK, Germany, and France, and a finite-horizon dynamic labor supply model for taxi drivers in New York City.

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## A Additional Examples

**Example 4** (Infinite Horizon Dynamic Discrete Choice Models). This example considers the serial independence assumption on utility shocks in a single-agent DDC model, as in [Rust \(1987\)](#). Agents solve the Bellman equation for the conditional value function  $v \in \mathcal{V}$ :

$$v_j(x, \varepsilon) = u_j(x, \varepsilon; \theta) + \beta \mathbb{E}_{\varepsilon' | \varepsilon} \mathbb{E}_{x' | x, j} \max_{j' \in \mathcal{J}} v_{j'}(x', \varepsilon') \quad (24)$$

where  $\varepsilon \in \mathbb{R}^J$  is a vector of utility shocks for each action  $j \in \mathcal{J}$ ,  $x \in \mathcal{X}$  is the observable state variable,  $\beta \in (0, 1)$  is the discount factor,  $u_j(x, \varepsilon; \theta)$  is the period utility parameterized by  $\theta \in \Theta$ , and  $\mathcal{V}$  is the class of conditional value functions.

Let  $U := (\varepsilon, \varepsilon')$  be a vector of current and future utility shocks. The serial independence assumption is often imposed on utility shocks, which implies a reference distribution:

$$dF_0(U) := \nu_0(d\varepsilon)\nu_0(d\varepsilon')$$

whose perturbation set is defined as:

$$\mathcal{F} := \{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu_0, \nu_0), D_{KL}(F \| F_0) \leq \delta\}$$

Suppose the scalar parameter of interest is the social welfare, defined as:

$$\mathbb{E}_{\nu_0} \mathbb{E}_x \max_{j \in \mathcal{J}} v_j(x, \varepsilon)$$

We convert the Bellman equation (24) into a restriction that depends on the joint distribution  $F \in \mathcal{F}$ . We assume there exists a class of test functions  $\mathcal{G}$  such that  $v$  solves the Bellman equation (24) if and only if:

$$\sup_{g \in \mathcal{G}} \mathbb{E}_F \mathbb{E}_{x,j,x'} \left[ g_j(x, \varepsilon) \left( v_j(x, \varepsilon) - u_j(x, \varepsilon; \theta) - \beta \max_{j' \in \mathcal{J}} v_{j'}(x', \varepsilon') \right) \right] = 0$$

We can rewrite the structural constraint as:

$$\sup_{g \in \mathcal{G}} \mathbb{E}_F [\psi(U; \theta, v, g)] = 0$$

where  $\psi(U; \theta, v, g) := \mathbb{E}_{x,j,x'} (g_j(x, \varepsilon) (v_j(x, \varepsilon) - u_j(x, \varepsilon; \theta) - \beta \max_{j' \in \mathcal{J}} v_{j'}(x', \varepsilon')))$ .

We consider the following moment conditions for estimation:

$$\mathbb{E}_F \mathbb{1}(v_j(x, \varepsilon) = \max_{j' \in \mathcal{J}} v_{j'}(x, \varepsilon)) = P_0(j|x) \quad \forall (j, x) \in \mathcal{J} \times \mathcal{X}$$

where  $\mathbb{1}$  is the indicator function, and  $P_0(j|x)$  is the population CCP. We assume  $\mathcal{X}$  has discrete support, and rewrite the moment conditions as:

$$\mathbb{E}_F [m(U; v)] = P_0$$

where  $m(U; v)$  stacks the indicator functions for each  $(x, j)$  given  $v$ , and  $P_0$  stacks the CCPs.

Then, the lower bound on the social welfare is given by:

$$\begin{aligned} \inf_{(\theta, v, F) \in \Theta \times \mathcal{V} \times \mathcal{F}} & \mathbb{E}_{\nu_0} \mathbb{E}_x \max_{j \in \mathcal{J}} v_j(x, \varepsilon) \\ \text{s.t.} & \quad \mathbb{E}_F [m(U; v)] = P_0 \\ & \quad \sup_{g \in \mathcal{G}} \mathbb{E}_F [\psi(U; \theta, v, g)] = 0 \end{aligned}$$

**Example 5** (Infinite Horizon Dynamic Discrete-Continuous Choice Models). This example

considers the serial independence assumption on the consumption shock in a single-agent dynamic discrete-continuous choice model, as in [Iskhakov et al. \(2017\)](#) and [Levy and Schiraldi \(2022\)](#). At each period, individuals make a discrete choice  $j \in \mathcal{J}$ , and a continuous choice  $q \in \mathbb{R}$ . The value function solves the Bellman equation:

$$V(x, I, \xi, \varepsilon) = \max_{j, q} \{u_j(q, x, I, \xi; \theta) + \varepsilon_j + \beta \mathbb{E}_{\xi'|\xi} \mathbb{E}_{x'|x, j} V(x', I', \xi', \varepsilon')\}$$

where  $I$  is the resource constraint (e.g., inventory) evolving deterministically over time according to  $I' = L(x, I, q, j)$ ,  $x \in \mathcal{X}$  is the observable state variable,  $\varepsilon \in \mathbb{R}^J$  is a vector of i.i.d. Extreme Value Type I utility shocks,  $\xi \in \Xi$  is the consumption shock,  $u_j(q, x, I, \xi; \theta)$  is the period utility parameterized by  $\theta \in \Theta$ , and  $\beta \in (0, 1)$  is the discount factor.

Let  $q_j^* := q_j^*(x, I, \xi)$  be the conditional optimal continuous choice for  $(j, x, I, \xi)$ . Given  $q_j^*$ , the conditional value function  $v_j \in \mathcal{V}$  solves:

$$v_j(x, I, \xi) = u_j(q_j^*, x, I, \xi; \theta) + \beta \mathbb{E}_{\xi'|\xi} \mathbb{E}_{x'|x, j} \left[ \log \left( \sum_{j' \in \mathcal{J}} \exp(v_{j'}(x', I', \xi')) \right) \right] + \beta \gamma \quad (25)$$

The model-implied conditional choice probability is  $p(j|x, I, \xi) = \frac{\exp(v_j(x, I, \xi))}{\sum_{j' \in \mathcal{J}} \exp(v_{j'}(x, I, \xi))}$ .

We assume that the optimal continuous choice is attained at an interior point and the dominated convergence theorem holds. Therefore, the first-order condition for (25) holds:

$$\frac{\partial u_j(q_j^*, x, I, \xi; \theta)}{\partial q} + \beta \mathbb{E}_{\xi'|\xi} \mathbb{E}_{x'|x, j} \left[ \sum_{j' \in \mathcal{J}} p(j'|x', I', \xi') \frac{\partial v_{j'}(x', I', \xi')}{\partial I'} \right] \frac{\partial L(x, I, q_j^*, j)}{\partial q} = 0$$

By the envelope theorem, we have:

$$\frac{\partial v_{j'}(x', I', \xi')}{\partial I'} = \frac{\partial u_{j'}(q_{j'}^*, x', I', \xi'; \theta)}{\partial q}$$

Therefore, the Euler equation for the optimal continuous choice is:

$$\frac{\partial u_j(q_j^*, x, I, \xi; \theta)}{\partial q} + \beta \mathbb{E}_{\xi'|\xi} \mathbb{E}_{x'|x, j} \left[ \sum_{j' \in \mathcal{J}} p(j'|x', I', \xi') \frac{\partial u_{j'}(q_{j'}^*, x', I', \xi'; \theta)}{\partial q} \right] \frac{\partial L(x, I, q_j^*, j)}{\partial q} = 0 \quad (26)$$

Let  $U := (\xi, \xi')$  be a vector of current and future consumption shocks. The serial independence is often imposed on consumption shocks, which implies a reference distribution:

$$dF_0(U) := \nu_0(d\xi) \nu_0(d\xi')$$

whose perturbation set is defined as:

$$\mathcal{F} := \{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu_0, \nu_0), D_{KL}(F \| F_0) \leq \delta\}$$

Suppose the scalar parameter of interest is the social welfare, defined as:

$$\mathbb{E}_{\nu_0} \mathbb{E}_{x,I} \left[ \log \left( \sum_{j \in \mathcal{J}} \exp(v_j(x, I, \xi)) \right) \right]$$

We convert the Bellman equation (25) and the Euler equation (26) into restrictions that depend on the joint distribution  $F \in \mathcal{F}$ . We assume there exists a class of test functions  $\mathcal{G}$  such that  $v := (v_1, \dots, v_J, q_1^*, \dots, q_J^*)$  solves the Bellman equation (25), and the Euler equation (26) if and only if:

$$\begin{aligned} \sup_{g^1 \in \mathcal{G}} \mathbb{E}_F \mathbb{E}_{x,I,j,x'} \left[ g_j^1(x, I, \xi) \left( v_j(x, I, \xi) - u_j(q_j^*, x, I, \xi; \theta) - \beta \log \left( \sum_{j' \in \mathcal{J}} \exp(v_{j'}(x', I', \xi')) \right) - \beta \gamma \right) \right] &= 0 \\ \sup_{g^2 \in \mathcal{G}} \mathbb{E}_F \mathbb{E}_{x,I,j,x'} \left[ g_j^2(x, I, \xi) \left( \frac{\partial u_j(q_j^*, x, I, \xi; \theta)}{\partial q} + \beta \sum_{j' \in \mathcal{J}} p(j'|x', I', \xi') \frac{\partial u_{j'}(q_{j'}^*, x', I', \xi'; \theta)}{\partial q} \frac{\partial I'(x, I, q_j^*, j)}{\partial q} \right) \right] &= 0 \end{aligned}$$

We rewrite the structural constraints as  $\sup_{g \in \mathcal{G}} \mathbb{E}_F [\psi(U; \theta, v, g)] = 0$  where  $\psi$  is the sum of the two expressions inside the expectations and  $g := (g^1, g^2)$ .

We consider the following moment conditions for estimation:  $\forall (j, x, I) \in \mathcal{J} \times \mathcal{X} \times \mathcal{I}$

$$\mathbb{E}_F [p(j|x, I, \xi)] = P_0(j|x, I), \quad \mathbb{E}_F [q_j^*(x, I, \xi)] = q_0(j, x, I)$$

where  $P_0(j|x, I)$  is the population CCP and  $q_0(j, x, I)$  is the population continuous choice function. We discretize  $\mathcal{X} \times \mathcal{I}$  for estimation, and rewrite the moment conditions as  $\mathbb{E}_F [m(U; v)] = P_0 \mathbb{E}_F [m(U; v)] = P_0$  where  $m(P_0)$  stacks the model-implied (population) CCPs and continuous choice functions.

Then, the lower bound on the social welfare is given by:

$$\begin{aligned} \inf_{(\theta, v, F) \in \Theta \times \mathcal{V} \times \mathcal{F}} \quad & \mathbb{E}_{\nu_0} \mathbb{E}_{x,I} \log \left( \sum_{j \in \mathcal{J}} \exp(v_j(x, I, \xi)) \right) \\ \text{s.t.} \quad & \mathbb{E}_F [m(U; v)] = P_0 \\ & \sup_{g \in \mathcal{G}} \mathbb{E}_F [\psi(U; \theta, v, g)] = 0 \end{aligned}$$

## B Proofs

### B.1 Supporting Lemmas

**Lemma 4** (Fan's Minimax Theory). *Suppose the following conditions hold:*

1.  $X$  be a compact Hausdorff space and  $Y$  a nonempty set (not necessarily topologized).
2. Let  $f : X \times Y \rightarrow \mathbb{R}$  be a real-valued function.
3. For  $\forall y \in Y$ ,  $f(\cdot, y)$  is convexlike on  $X$ , i.e., for all  $x_1, x_2 \in X$  and  $\lambda \in [0, 1]$ , there exists  $x_0 \in X$  such that  $f(x_0, y) \leq \lambda f(x_1, y) + (1 - \lambda)f(x_2, y)$ .
4. For  $\forall x \in X$ ,  $f(x, \cdot)$  is concavelike on  $Y$ , i.e., for all  $y_1, y_2 \in Y$  and  $\lambda \in [0, 1]$ , there exists  $y_0 \in Y$  such that  $f(x, y_0) \geq \lambda f(x, y_1) + (1 - \lambda)f(x, y_2)$ .
5. For  $\forall y \in Y$ ,  $f(\cdot, y)$  is lower semicontinuous on  $X$ .

Then, we have:

$$\sup_Y \inf_X f(x, y) = \inf_X \sup_Y f(x, y)$$

*Proof.* See [Riccieri and Simons \(2013\)](#) Theorem 1.3. □

**Lemma 5.** *Let  $\{\mathcal{A}_n\}$  be a sequence of compact sets such that  $d_H(\mathcal{A}_n, \mathcal{A}) = o_p(1)$  where  $\mathcal{A}$  is compact. Then, the following holds:*

- (Consistency) *If  $f : \mathcal{A} \rightarrow \mathbb{R}$  is continuous, then:  $|\inf_{\mathcal{A}_n} f - \inf_{\mathcal{A}} f| = o_p(1)$ .*
- (Convergence Rate) *If  $d_H(\mathcal{A}_n, \mathcal{A}) = O_p(c_n)$  for some  $c_n \rightarrow 0$ , and  $f$  is Lipschitz continuous, then:  $|\inf_{\mathcal{A}_n} f - \inf_{\mathcal{A}} f| = O_p(c_n)$ .*

*Proof.* As  $\mathcal{A}_n$  and  $\mathcal{A}$  are compact and  $f$  is continuous, the infimum is achieved by the extreme value theorem. Denote minimizers as  $a_n^* \in \mathcal{A}_n$  and  $a^* \in \mathcal{A}$ .

1. As  $d_H(\mathcal{A}_n, \mathcal{A}) = o_p(1)$ , there exists a sequence  $a_n \in \mathcal{A}$  such that  $d(a_n^*, a_n) = o_p(1)$ . By the continuity of  $f$ , we have:  $|f(a_n^*) - f(a_n)| = o_p(1)$ , which implies:  $f(a_n^*) = f(a_n) + o_p(1) \geq f(a^*) + o_p(1)$ . Similarly, there exists a sequence  $b_n \in \mathcal{A}_n$  such that  $d(b_n, a^*) = o_p(1)$ . By the continuity of  $f$ , we have:  $|f(b_n) - f(a^*)| = o_p(1)$ , which implies:  $f(a^*) = f(b_n) + o_p(1) \geq f(a_n^*) + o_p(1)$ . Combining both inequalities, we have  $f(a_n^*) \geq f(a^*) + o_p(1) \geq f(a_n^*) + o_p(1) + o_p(1)$ , which implies:  $|\inf_{\mathcal{A}_n} f - \inf_{\mathcal{A}} f| = |f(a_n^*) - f(a^*)| = o_p(1)$ .

2. For the second part, let  $L$  be the Lipschitz constant. There exists a sequence  $a_n \in \mathcal{A}$  such that  $d(a_n^*, a_n) = O_p(c_n)$ , by the Lipschitz continuity of  $f$ , we have:  $|f(a_n^*) - f(a_n)| \leq Ld(a_n^*, a_n) = O_p(c_n)$ , which implies:  $f(a_n^*) \geq f(a_n) - Ld(a_n^*, a_n) \geq f(a^*) - Ld(a_n^*, a_n)$ . Similarly, there exists a sequence  $b_n \in \mathcal{A}_n$  such that  $d(b_n, a^*) = O_p(c_n)$ . By the Lipschitz continuity of  $f$ , we have:  $|f(b_n) - f(a^*)| \leq Ld(b_n, a^*) = O_p(c_n)$ , which implies:  $f(a^*) \geq f(b_n) - Ld(b_n, a^*) \geq f(a_n^*) - Ld(b_n, a^*)$ . Combining both inequalities, we have:  $f(a_n^*) \geq f(a^*) - Ld(b_n, a^*) \geq f(a_n^*) - 2Ld(b_n, a^*)$ , which implies:  $|f(a_n^*) - f(a^*)| \leq 2Ld(b_n, a^*) = O_p(c_n)$ .

□

**Lemma 6.** *Let  $\delta > 0$  and  $F_0 \in \mathcal{P}(\mathcal{U})$ . Let  $\mathcal{F}_{KL} := \{F \in \mathcal{P}(\mathcal{U}) \mid D_{KL}(F\|F_0) \leq \delta\}$ .*

- (i)  $\mathcal{F}_{KL}$  is compact in the topology of weak convergence.
- (ii)  $\mathcal{F}_{KL}$  is closed in the topology of weak convergence.

*Proof.* 1. See [Pinski et al. \(2015\)](#) Proposition 2.1.

2. By [Nutz \(2021\)](#) Lemma 1.3,  $D_{KL}(F\|F_0)$  is lower-semicontinuous in the topology of weak convergence, i.e., for  $F_n \rightarrow F$  weakly, we have  $\liminf_{n \rightarrow \infty} D_{KL}(F_n\|F_0) \geq D_{KL}(F\|F_0)$ . Since  $F_n \in \mathcal{F}_{KL}$ , we have  $D_{KL}(F\|F_0) \leq \delta$ .

□

**Lemma 7.** *Let  $\nu_i \in \mathcal{P}(\mathcal{U}_i)$  for  $i \in \{1, \dots, k\}$ , then:*

- (i)  $\Pi(\nu_1, \dots, \nu_k)$  is closed in the topology of weak convergence.
- (ii)  $\Pi(\nu_1, \dots, \nu_k)$  is compact and convex in the topology of weak convergence.
- (iii)  $\Pi(\nu_1, \dots, \nu_k)$  is a Hausdorff topological space.
- (iv) Under Assumption 2(i),  $\Pi(\nu_1, \nu_k)$  is closed in the topology of weak convergence.
- (v)  $\Pi(\nu_1, \nu_k)$  is convex in the topology of weak convergence.
- (vi) Under Assumption 2(i),  $\Pi(\nu_1, \nu_k)$  is compact in the topology of weak convergence.
- (vii)  $\Pi(\nu_1, \nu_k)$  is a Hausdorff topological space.
- (viii) Suppose  $\mathcal{N}$  is convex and closed and Assumption 2(i) holds,  $\{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu, \nu_k), \nu \in \mathcal{N}\}$  is closed in the topology of weak convergence.

- (ix) Suppose  $\mathcal{N}$  is convex and closed and Assumption 2(i) holds, then  $\{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu, \nu_k), \nu \in \mathcal{N}\}$  is convex in the topology of weak convergence.
- (x) Suppose  $\mathcal{N}$  is convex and closed and Assumption 2(i) holds,  $\{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu, \nu_k), \nu \in \mathcal{N}\}$  is compact in the topology of weak convergence.
- (xi) Suppose  $\mathcal{N}$  is convex and closed and Assumption 2(i) holds,  $\{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu, \nu_k), \nu \in \mathcal{N}\}$  is a Hausdorff topological space.
- (xii) Lemmas 7(viii)-7(xi) also hold for  $\{F \in \mathcal{P}(\Xi^2) \mid F \in \Pi(\nu, \nu), \nu \in \mathcal{N}\}$ .

*Proof.* 1. **Closedness and compactness of  $\Pi(\nu_1, \dots, \nu_k)$ :** see the proof of [Villani et al. \(2009\)](#) Theorem 4.1.

- 2. **Closedness and compactness of  $\Pi(\nu_1, \nu_k)$ :** By Assumption 2(i),  $\Pi(\nu_1, \nu_k)$  is tight, and by Prokhorov's theorem it has a compact closure. By passing to the limit in the equation for marginals,  $\Pi(\nu_1, \nu_k)$  is closed. Therefore, it is compact.
- 3. **Closedness and compactness of  $\{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu, \nu_k), \nu \in \mathcal{N}\}$ :** By Assumption 2(i), the set  $\{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu, \nu_k), \nu \in \mathcal{N}\}$  is tight, and by Prokhorov's theorem it has a compact closure. By passing to the limit in the equation for marginals, it is closed. Therefore, it is compact.
- 4. **Closedness and compactness of  $\{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu, \nu), \nu \in \mathcal{N}\}$ :** The proof is similar to the previous part.
- 5. **Convexity:** It is straightforward.
- 6. **Hausdorff:** By [Billingsley \(2013\)](#) Page 72(i), the Prohorov distance is a metric on the space of probability measures. Metrizable topological spaces are Hausdorff.

□

**Lemma 8.** Let  $\delta \geq 0$ ,  $F_0 \in \mathcal{P}(\mathcal{U})$ , and  $\nu_i \in \mathcal{P}(\mathcal{U}_i)$  for  $i \in \{1, \dots, k\}$ . Define:

$$\mathcal{F} := \{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu_1, \dots, \nu_k), D_{KL}(F \| F_0) \leq \delta\}$$

- (i)  $\mathcal{F}$  is closed and compact in the topology of weak convergence.
- (ii)  $\mathcal{F}$  is convex in the topology of weak convergence.
- (iii)  $\mathcal{F}$  is a Hausdorff topological space.



- Proof.* 1. By Lemmas 6(ii) and 7(i),  $\mathcal{F}$  is the intersection of two closed sets. Therefore, it is closed. By [Rudin et al. \(1976\)](#) Theorem 2.35,  $\mathcal{F}$  is compact.
2. Since KL divergence is jointly convex (see [Nutz \(2021\)](#) Lemma 1.3), we have  $D_{KL}(\lambda F_1 + (1 - \lambda)F_2 \| F_0) \leq \lambda D_{KL}(F_1 \| F_0) + (1 - \lambda)D_{KL}(F_2 \| F_0) \leq \delta$  for  $\lambda \in [0, 1]$  and  $F_1, F_2 \in \mathcal{F}$ . Moreover,  $\lambda F_1 + (1 - \lambda)F_2 \in \Pi(\nu_1, \dots, \nu_k)$ . Therefore,  $\mathcal{F}$  is convex.
3. By [Billingsley \(2013\)](#) Page 72(i), the Prohorov distance is a metric on the space of probability measures. Metrizable topological spaces are Hausdorff.

□

**Lemma 9.** *Suppose Assumption 2(i) holds. Let  $\delta \geq 0$ ,  $F_0 \in \mathcal{P}(\mathcal{U})$ , and  $\nu_i \in \mathcal{P}(\mathcal{U}_i)$  for  $i \in \{1, k\}$ . Let  $\mathcal{F}_{\text{relaxed}} := \{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu_1, \nu_k), D_{KL}(F \| F_0) \leq \delta\}$ .*

- (i)  $\mathcal{F}_{\text{relaxed}}$  is closed and compact in the topology of weak convergence.
- (ii)  $\mathcal{F}_{\text{relaxed}}$  is convex in the topology of weak convergence.
- (iii)  $\mathcal{F}_{\text{relaxed}}$  is a Hausdorff topological space.

- Proof.* 1. By Lemmas 6(ii) and 7(iv),  $\mathcal{F}_{\text{relaxed}}$  is the intersection of two closed sets. Therefore, it is closed. By [Rudin et al. \(1976\)](#) Theorem 2.35,  $\mathcal{F}_{\text{relaxed}}$  is compact.
2. The proof is identical to the proof of Lemma 8(ii).
3. The proof is identical to the proof of Lemma 8(iii).

□

**Lemma 10.** *Suppose Assumption 2(i) holds. Let  $\delta \geq 0$ ,  $F_0 \in \mathcal{P}(\mathcal{U})$  and  $\nu_k \in \mathcal{P}(\mathcal{U}_k)$ . Suppose  $\mathcal{N} \subseteq \mathcal{P}(\mathcal{U}_1)$  is convex. Let  $\mathcal{F}_{\mathcal{N}, \text{Relaxed}} := \{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi(\nu, \nu_k), \nu \in \mathcal{N}, D_{KL}(F \| F_0) \leq \delta\}$ .*

- (i)  $\mathcal{F}_{\mathcal{N}, \text{Relaxed}}$  is closed and compact in the topology of weak convergence.
- (ii)  $\mathcal{F}_{\mathcal{N}, \text{Relaxed}}$  is convex in the topology of weak convergence.
- (iii)  $\mathcal{F}_{\mathcal{N}, \text{Relaxed}}$  is a Hausdorff topological space.

- Proof.* 1. By Lemma 7(viii), and [Rudin et al. \(1976\)](#) Theorem 2.35,  $\mathcal{F}_{\mathcal{N}, \text{Relaxed}}$  is compact.
2. Note that  $\mathcal{N}$  is convex. For any  $F_1, F_2 \in \mathcal{F}_{\mathcal{N}, \text{Relaxed}}$ , we have  $\nu_1, \nu_2 \in \mathcal{N}$ . By the convexity of  $\mathcal{N}$ , we have  $\lambda \nu_1 + (1 - \lambda) \nu_2 \in \mathcal{N}$  for  $\lambda \in [0, 1]$ . Therefore,  $\lambda F_1 + (1 - \lambda) F_2 \in \Pi(\lambda \nu_1 + (1 - \lambda) \nu_2, \nu_k)$  and  $D_{KL}(\lambda F_1 + (1 - \lambda) F_2 \| F_0) \leq \delta$ . Thus,  $\mathcal{F}_{\mathcal{N}, \text{Relaxed}}$  is convex.

3. The proof is identical to the proof of Lemma 8(iii).

□

## B.2 Proofs in Section 2

### B.2.1 Proof of Theorems 1 and 10

We only prove Theorem 1. The proof for Theorem 10 is similar. First, we prove the minimax part. The Lagrangian of the [Primal](#) problem is:

$$\kappa(\delta, P) = \inf_{\substack{(\theta, v) \in \Theta \times \mathcal{V} \\ F \in \Pi(\nu_1, \dots, \nu_k)}} \sup_{\substack{\lambda \in \mathbb{R}^{d_P} \\ \lambda_{KL} \geq 0, g \in \mathcal{G}}} \mathbb{E}_F [c(U; \theta, v, g, \lambda)] + \lambda_{KL} D_{KL}(F \| F_0) - \lambda_{KL} \delta - \lambda^T P$$

**Lemma 11** (Minimax). *Under Assumption 1, we have:*

$$\kappa(\delta, P) = \inf_{(\theta, v) \in \Theta \times \mathcal{V}} \sup_{\substack{\lambda \in \mathbb{R}^{d_P} \\ \lambda_{KL} \geq 0, g \in \mathcal{G}}} \inf_{F \in \Pi(\nu_1, \dots, \nu_k)} \mathbb{E}_F [c(U; \theta, v, g, \lambda)] + \lambda_{KL} D_{KL}(F \| F_0) - \lambda_{KL} \delta - \lambda^T P$$

*Proof.* In the proof, we show that the minimax theorem holds for a given  $(\theta, v)$  by verifying the conditions in Lemma 4. For notational simplicity, define:

$$\mathcal{L}(F, g, \lambda, \lambda_{KL}) := \mathbb{E}_F [c(U; \theta, v, g, \lambda)] + \lambda_{KL} D_{KL}(F \| F_0) - \lambda_{KL} \delta - \lambda^T P$$

- **Compactness:** By Lemma 7(ii),  $\Pi(\nu_1, \dots, \nu_k)$  is compact.
- **Hausdorff:** By Lemma 7(iii),  $\Pi(\nu_1, \dots, \nu_k)$  is Hausdorff.
- **Concavelike:** Note that  $\mathcal{L}(F, g, \lambda, \lambda_{KL})$  is linear in  $(\lambda, \lambda_{KL}, g)$ . Therefore, the concavelike condition is satisfied.
- **Convexlike:** By Lemma 7(ii),  $\Pi(\nu_1, \dots, \nu_k)$  is a convex space. Therefore, convex combinations of elements in  $\Pi(\nu_1, \dots, \nu_k)$  are also in  $\Pi(\nu_1, \dots, \nu_k)$ . Since  $D_{KL}(F \| F_0)$  is jointly convex (see [Nutz \(2021\)](#) Lemma 1.3), and the expectation is linear in  $F$ , the convexlike condition also holds.
- **Lower-semicontinuity:** Let  $h(U) := -(1 + \|\lambda\|_1)C_{\theta, v, g}(1 + d(U, \hat{U}))$ . Under Assumption 1(v), we have  $h \in L^1(F)$  for all  $F \in \Pi(\nu_1, \dots, \nu_k)$ . Therefore, for a given  $(\lambda, \lambda_{KL}, g)$ ,  $\mathbb{E}_F [c(U; \theta, v, g, \lambda)]$  is lower-semicontinuous in  $F$  by [Villani et al. \(2009\)](#) Lemma 4.3. By [Nutz \(2021\)](#) Lemma 1.3,  $D_{KL}(F \| F_0)$  is lower-semicontinuous in  $F$ .

By the superadditivity of  $\liminf$ ,  $\mathcal{L}(F, g, \lambda, \lambda_{KL})$  is also lower-semicontinuous in  $F$  for a given  $(\lambda, \lambda_{KL}, g)$ .

□

**Lemma 12.** *Define:*

$$\mathcal{L}(\delta, \theta, v, g, \lambda) := \sup_{\lambda_{KL} \geq 0} \inf_{F \in \Pi(\nu_1, \dots, \nu_k)} \mathbb{E}_F [c(U; \theta, v, g, \lambda)] + \lambda_{KL} D_{KL}(F \| F_0) - \lambda_{KL} \delta - \lambda^T P$$

*Under Assumption 1, we have:*

$$\mathcal{L}(\delta, \theta, v, g, \lambda) = \sup_{\lambda_{KL} \geq 0} \inf_{F \in \Pi(\nu_1, \dots, \nu_k)} \mathbb{E}_F [c(U; \theta, v, g, \lambda) + \lambda_{KL} \rho(U)] + \lambda_{KL} (D_{KL}(F \| F_\otimes) - \delta) - \lambda^T P$$

*Proof.* By Assumption 1(ii), we have:

$$D_{KL}(F \| F_0) = \int \log\left(\frac{dF}{dF_\otimes} \frac{dF_\otimes}{dF_0}\right) dF = \int \log\left(\frac{dF}{dF_\otimes}\right) dF + \int \log\left(\frac{dF_\otimes}{dF_0}\right) dF = D_{KL}(F \| F_\otimes) + \int \rho(U) dF$$

Moreover, by Assumption 1(v), we have  $c(U; \theta, v, g, \lambda), \rho \in L^1(F)$  for  $\forall F \in \Pi(\nu_1, \dots, \nu_k)$ .

Therefore, the sum of two expectations is also well-defined. □

**Lemma 13.** *Suppose Assumption 1 holds. If  $c(U; \theta, v, g, \lambda)$  and  $\rho(U)$  are continuous in  $U$ , then optimizing over  $\lambda_{KL} > 0$  gives the same value as optimizing over  $\lambda_{KL} \geq 0$ , i.e.,*

$$\mathcal{L}(\delta, \theta, v, g, \lambda) = \sup_{\lambda_{KL} > 0} \inf_{F \in \Pi(\nu_1, \dots, \nu_k)} \mathbb{E}_F [c(U; \theta, v, g, \lambda) + \lambda_{KL} \rho(U)] + \lambda_{KL} (D_{KL}(F \| F_\otimes) - \delta) - \lambda^T P$$

*Proof.* By [Eckstein and Nutz \(2023\)](#) Proposition 3.1, we have:

$$\lim_{\lambda_{KL} \downarrow 0} \inf_{F \in \Pi(\nu_1, \dots, \nu_k)} \mathbb{E}_F [c(U; \theta, v, g, \lambda) + \lambda_{KL} \rho(U)] + \lambda_{KL} (D_{KL}(F \| F_\otimes) - \delta) = \inf_{F \in \Pi(\nu_1, \dots, \nu_k)} \mathbb{E}_F [c(U; \theta, v, g, \lambda)]$$

Therefore, for any sequence  $\lambda_{KL}^i \rightarrow 0$  as  $i \rightarrow \infty$ , we have:

$$\lim_{i \rightarrow \infty} \inf_{F \in \Pi(\nu_1, \dots, \nu_k)} \mathbb{E}_F [c(U; \theta, v, g, \lambda) + \lambda_{KL}^i \rho(U)] + \lambda_{KL}^i (D_{KL}(F \| F_\otimes) - \delta) = \inf_{F \in \Pi(\nu_1, \dots, \nu_k)} \mathbb{E}_F [c(U; \theta, v, g, \lambda)]$$

□

Then, we show the EOT duality part. For  $\lambda_{KL} > 0$ , consider the following EOT problem:

$$\mathcal{C}(\theta, v, g, \lambda, \lambda_{KL}) := \inf_{F \in \Pi(\nu_1, \dots, \nu_k)} \mathbb{E}_F [c(U; \theta, v, g, \lambda) + \lambda_{KL} \rho(U)] + \lambda_{KL} D_{KL}(F \| F_\otimes)$$

**Theorem 12** (EOT Duality). *Under Assumption 1, for  $\lambda_{KL} > 0$ , the following holds:*

$$\mathcal{C}(\theta, v, g, \lambda, \lambda_{KL}) = \sup_{\{\phi_i \in L^1(\nu_i)\}_{i=1}^k} \sum_{i=1}^k \mathbb{E}_{\nu_i} \phi_i(U_i) - \lambda_{KL} \mathbb{E}_{F_0} \exp\left(\frac{\sum_{i=1}^k \phi_i(U_i) - c(U; \theta, v, g, \lambda)}{\lambda_{KL}}\right) + \lambda_{KL}$$

Moreover, the worst-case distribution is given by:

$$\frac{dF^*(U)}{dF_0(U)} = \exp\left(\frac{\sum_{i=1}^k \phi_i^*(U_i) - c(U; \theta, v, g, \lambda)}{\lambda_{KL}}\right) \quad F_0\text{-a.s.}$$

where  $\{\phi_i^*\}_{i=1}^k$  are unique maximizers up to additive constants  $F_0$ -almost surely.

*Proof.* Under Assumption 1(v),  $c(U; \theta, v, g, \lambda), \rho(U) \in L^1(F_\otimes)$ . For two marginals ( $k = 2$ ), see [Nutz \(2021\)](#) Theorem 4.7 and Remark 4.8(a). The results generalize directly to the multi-marginal case (see [Nutz and Wiesel \(2022\)](#) Section 6). Their results hold  $F_\otimes$ -a.s. Since  $F_0 \ll F_\otimes$ , the results also hold  $F_0$ -a.s. Moreover, note that:

$$\mathbb{E}_{F_\otimes} \exp\left(\frac{\sum_{i=1}^k \phi_i(U_i) - c(U; \theta, v, g, \lambda) - \lambda_{KL} \rho(U)}{\lambda_{KL}}\right) = \mathbb{E}_{F_0} \exp\left(\frac{\sum_{i=1}^k \phi_i(U_i) - c(U; \theta, v, g, \lambda)}{\lambda_{KL}}\right)$$

□

### B.2.2 Proof of Proposition 1

**Proof of Proposition 1.** As shown in [Nutz \(2021\)](#) equation 4.11, the fact that  $F^*$  in Theorem 12 is a probability measure with marginals  $(\nu_1, \dots, \nu_k)$  implies the Schrödinger equations: for  $\forall j \leq k$ :

$$\phi_i^*(U_j) = -\lambda_{KL} \log \mathbb{E}_{F_{\otimes, -j}} \exp\left(\frac{\sum_{i \neq j} \phi_i^*(U_i) - c(U; \theta, v, g, \lambda)}{\lambda_{KL}}\right) \quad \nu_j\text{-a.s.}$$

Since  $c$  is  $k$ -times continuously differentiable in  $U$ , the right-hand side is also  $k$ -times continuously differentiable in  $U_j$ . Therefore,  $\phi_j^*(U_j)$  is  $k$ -times continuously differentiable in  $U_j$  for  $\forall j \leq k$ . □

### B.2.3 Proof of Theorem 2

**Proof of Theorem 2.** In the proof, we first swap the order of infimum over  $F$  and the supremum over  $(\lambda, \lambda_{KL}, g)$  by the minimax theorem. Then, we swap the order of infimum

over  $\nu$  and the supremum over  $(\phi_1, \phi_2)$  by the minimax theorem. Define:

$$\mathcal{L}(F, g, \lambda, \lambda_{KL}) := \mathbb{E}_F [c(\xi, \xi'; \theta, v, g, \lambda)] + \lambda_{KL} D_{KL}(F \| F_0) - \lambda_{KL} \delta - \lambda^T P$$

- **Compactness and Hausdorff:** By Lemma 7(xii),  $\{F \in \mathcal{P}(\Xi^2) \mid F \in \Pi(\nu, \nu), \nu \in \mathcal{N}\}$  is compact and Hausdorff.
- **Concavelike:** Note that  $\mathcal{L}(F, g, \lambda, \lambda_{KL})$  is linear in  $(\lambda, \lambda_{KL}, g)$ . Therefore, the concavelike condition is satisfied.
- **Convexlike:** By Lemma 7(xii),  $\{F \in \mathcal{P}(\Xi^2) \mid F \in \Pi(\nu, \nu), \nu \in \mathcal{N}\}$  is convex. Since  $D_{KL}(F \| F_0)$  is jointly convex (see Nutz (2021) Lemma 1.3), and the expectation is linear in  $F$ , the convexlike condition also holds.
- **Lower-semicontinuity:** Let  $h(\xi, \xi') := -(1 + \|\lambda\|_1) C_{\theta, v, g}(1 + d(\xi, \xi', \hat{\xi}, \hat{\xi}'))$ . Under Assumption 1(v) and as  $\Xi$  is compact, we have  $h \in L^\infty(\Xi^2)$ . Therefore, for a given  $(\lambda, \lambda_{KL}, g)$ ,  $\mathbb{E}_F [c(\xi, \xi'; \theta, v, g, \lambda)]$  is lower-semicontinuous in  $F$  by Villani et al. (2009) Lemma 4.3. By Nutz (2021) Lemma 1.3,  $D_{KL}(F \| F_0)$  is lower-semicontinuous in  $F$ . By the superadditivity of  $\liminf$ ,  $\mathcal{L}(F, g, \lambda, \lambda_{KL})$  is also lower-semicontinuous in  $F$  for a given  $(\lambda, \lambda_{KL}, g)$ .

Therefore, by Lemma 4 and the EOT duality in Theorem 1, we have:

$$\begin{aligned} \kappa_{stationary}(\delta_1, \delta, P) &= \inf_{(\theta, v) \in \Theta \times \mathcal{V}} \sup_{\lambda \in \mathbb{R}^{d_P}, \lambda_{KL} \geq 0, g \in \mathcal{G}} \inf_{\nu \in \mathcal{N}} \mathcal{C}(\theta, v, g, \lambda, \lambda_{KL}, \nu) - \lambda_{KL} \delta - \lambda^T P \\ \mathcal{C}(\theta, v, g, \lambda, \lambda_{KL}, \nu) &= \sup_{\phi_1, \phi_2 \in L^1(\nu)} \mathbb{E}_\nu [\phi_1(\xi) + \phi_2(\xi')] - \lambda_{KL} \mathbb{E}_{F_0} \exp \left( \frac{\phi_1(\xi) + \phi_2(\xi') - c(\xi, \xi'; \theta, v, g, \lambda)}{\lambda_{KL}} \right) + \lambda_{KL} \delta \end{aligned}$$

Furthermore, as  $\Xi$  is compact, we can replace  $L^1(\nu)$  with  $L^\infty(\Xi)$ . Moreover,  $\phi_1^*$  and  $\phi_2^*$  are bounded (see Nutz (2021) Lemma 4.9) Moreover, by the lower semicontinuity of  $c(\xi, \xi'; \theta, v, g, \lambda)$ , the Fatou's lemma, and the proof of Theorem 12,  $\phi_1^*$  and  $\phi_2^*$  are lower semicontinuous.

Next, we swap the order of infimum over  $\nu$  and the supremum over  $(\phi_1, \phi_2)$ :

- **Compactness and Hausdorff:** By Lemma 6,  $\mathcal{N}$  is compact. Note that metrizable spaces are Hausdorff.
- **Convexlike:** As the expectation is linear in  $\nu$  and  $\mathcal{N}$  is convex, the convexlike condition is satisfied.
- **Concavelike:** It is straightforward to see that the objective function is concave in  $(\phi_1, \phi_2)$  as  $L^\infty(\Xi)$  is convex.

- **Lower-semicontinuity:** For a given  $(\phi_1, \phi_2)$ ,  $\mathbb{E}_\nu [\phi_1(\xi) + \phi_2(\xi')]$  is lower-semicontinuous in  $\nu$  by Villani et al. (2009) Lemma 4.3 as  $\phi_1$  and  $\phi_2$  are bounded.

The KL divergence DRO duality holds by Hu and Hong (2013) Theorem 1.  $\square$

## B.3 Proofs in Section 3

We only prove Theorem 3. The proof for Theorem 11 is the same.

### B.3.1 Proof of Theorem 3

- Proof of Theorem 3.**
1. **Minimax Part:** The proof of minimax part is similar to Lemma 11. Convexity and compactness of  $\Pi(\nu_1, \nu_T)$  are given by Lemmas 7(v) and 7(vi). Moreover, the bound on the cost function is replaced by the finite constant in Assumption 2(iii). Finally, Lemma 7(iii) is replaced by Lemma 7(vii).
  2. **Duality Part:** For notational simplicity, let  $c(U) := c(U; \theta, v, g, \lambda, \lambda_s)$ . Then,  $S_{\text{dyn}}$  can be rewritten as:

$$\inf_{F \in \Pi(\nu_1, \nu_T)} \mathbb{E}_F [c(U)] + \lambda_{KL} D_{KL}(F \| F_0) \quad (27)$$

By Assumption 2(iii),  $\exp(\frac{-c(U)}{\lambda_{KL}})$  is bounded and in particular,  $\exp(\frac{-c(U)}{\lambda_{KL}}) \in L^1(F_0)$ . Therefore, we can define the auxiliary reference measure  $R$  as follows:

$$dR(U) := \exp(\frac{-c(U)}{\lambda_{KL}}) dF_0(U)$$

Note that  $R \sim F_0$ . Then, (27) is equivalent to:<sup>38</sup>

$$\inf_{F \in \Pi(\nu_1, \nu_T)} \lambda_{KL} D_{KL}(F \| R) \quad (28)$$

By Léonard (2014) Theorem 2.4, we have:

$$D_{KL}(F \| R) = D_{KL}(F_{1,T} \| R_{1,T}) + \mathbb{E}_{F_{1,T}} [D_{KL}(F_{|1,T} \| R_{|1,T})]$$

where  $F_{1,T}$  and  $R_{1,T}$  are the two-period marginals of  $F$  and  $R$ , respectively.  $F_{|1,T}, R_{|1,T}$

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<sup>38</sup>This is the Schrödinger bridge problem, see Léonard (2013) for continuous time setting and De Bortoli et al. (2021) for discrete time setting.

are the conditional distribution given  $(\xi_1, \xi_T)$ . In particular, we have:

$$dR_{1,T}(\xi_1, \xi_T) := \int_{\xi_2 \dots \xi_{T-1}} dR(\xi_1, \dots, \xi_T) = \int_{\xi_2, \dots, \xi_{T-1}} \exp\left(\frac{-c(\xi_1, \dots, \xi_T)}{\lambda_{KL}}\right) dF_0(\xi_1, \dots, \xi_T)$$

The second term is minimized at  $F_{1,T}^* = R_{1,T}$ . Therefore, (28) can be reduced to the static Schrödinger bridge problem:

$$\inf_{F_{1,T} \in \tilde{\Pi}(\nu_1, \nu_T)} D_{KL}(F_{1,T} \| R_{1,T}) \quad (29)$$

where  $\tilde{\Pi}(\nu_1, \nu_T)$  is the set of all joint distributions of  $(\xi_1, \xi_T)$  with marginals  $\nu_1$  and  $\nu_T$ .

Note that:

$$dR_{1,T}(\xi_1, \xi_T) = \left( \int_{\xi_2, \dots, \xi_{T-1}} \exp\left(\frac{-c(\xi_1, \dots, \xi_T)}{\lambda_{KL}}\right) dF_0(\xi_2, \dots, \xi_{T-1} | \xi_1, \xi_T) \right) dF_0^{1,T}(\xi_1, \xi_T)$$

By Assumption 2(iii), we have  $R_{1,T} \sim F_0^{1,T}$ . By Assumption 2(ii), we have  $R_{1,T} \sim \nu_1 \otimes \nu_T$ . Therefore,  $\Pi_{\text{fin}}(\nu_1, \nu_T) := \{F \in \tilde{\Pi}(\nu_1, \nu_T) \mid D_{KL}(F \| R_{1,T}) < +\infty\} \neq \emptyset$ . By Assumption 2(ii) and Nutz (2021) Theorem 2.1, the unique solution to (29) has the form:

$$\frac{dF_{1,T}^*(\xi_1, \xi_T)}{dR_{1,T}(\xi_1, \xi_T)} = \exp(\phi_1^*(\xi_1) + \phi_T^*(\xi_T)) \quad R_{1,T}\text{-a.s.}$$

and  $\phi_1^* \in L^1(\nu_1)$ ,  $\phi_T^* \in L^1(\nu_T)$ . By Nutz (2021) Theorem 3.2, we have:

$$\inf_{F_{1,T} \in \tilde{\Pi}(\nu_1, \nu_T)} D_{KL}(F_{1,T} \| R_{1,T}) = \sup_{\phi_1 \in L^1(\nu_1), \phi_T \in L^1(\nu_T)} \mathbb{E}_{\nu_1} \phi_1 + \mathbb{E}_{\nu_T} \phi_T - \int \exp(\phi_1 + \phi_T) dR_{1,T} + 1$$

Multiplying both sides by  $\lambda_{KL}$  and letting  $\phi_1 := \lambda_{KL} \phi_1$  and  $\phi_T := \lambda_{KL} \phi_T$ , we have:

$$\inf_{F_{1,T} \in \tilde{\Pi}(\nu_1, \nu_T)} \lambda_{KL} D_{KL}(F_{1,T} \| R_{1,T}) = \sup_{\phi_1 \in L^1(\nu_1), \phi_T \in L^1(\nu_T)} \mathbb{E}_{\nu_1} \phi_1 + \mathbb{E}_{\nu_T} \phi_T - \lambda_{KL} \mathbb{E}_{R_{1,T}} \exp\left(\frac{\phi_1 + \phi_T}{\lambda_{KL}}\right) + \lambda_{KL}$$

Then, we have:

$$dF^*(U) = dF_{1,T}^* dF_{1,T}^* = \exp\left(\frac{\phi_1^*(\xi_1) + \phi_T^*(\xi_T) - c(U)}{\lambda_{KL}}\right) dF_0(U)$$

where  $\phi_1^*$  and  $\phi_T^*$  are the unique maximizers up to an additive constant.

**3. Markov Property:** Assumption 2(iv) implies that  $c(U)$  is also pairwise additive, i.e.,

$c(U) = \sum_{t=1}^{T-1} c_t(\xi_t, \xi_{t+1})$  for some  $c_t(\xi_t, \xi_{t+1})$ . Therefore, we can write:

$$dF^*(U) = \exp \left( \sum_{t=1}^{T-1} \frac{-c_t(\xi_t, \xi_{t+1})}{\lambda_{KL}} + \frac{\phi_1^*(\xi_1) + \phi_T^*(\xi_T)}{\lambda_{KL}} \right) dF_0(U)$$

which has Markov property as  $F_0$  has Markov property.<sup>39</sup>

4. Since  $\kappa_{\text{TI}}(\delta, P) \geq \tilde{\kappa}_{\text{TI}}(\delta, P)$  and the solution to  $\tilde{\kappa}_{\text{TI}}(\delta, P)$  corresponding to  $\lambda_{KL}^* > 0$  has the Markov property, we have:  $\kappa_{\text{TI}}(\delta, P) = \tilde{\kappa}_{\text{TI}}(\delta, P)$ .

□

### B.3.2 Proof of Theorem 4

**Proof of Theorem 4.** The proof of minimax part is similar to Lemma 11. Lemma 7(ii) is replaced by Lemmas 7(x) and 7(ix). Lemma 7(iii) is replaced by Lemma 7(xi). Finally, the bound on the cost function is replaced by the finite constant in Assumption 2(iii). □

### B.3.3 Proof of Lemma 1

**Proof of Lemma 1.** Let  $\pi_1, \pi_2$  be the unique solutions to the EOT problem with respect to  $\nu_1, \nu_2$ , respectively. Then,  $\pi := \lambda\pi_1 + (1 - \lambda)\pi_2 \in \tilde{\Pi}(\lambda\nu_1 + (1 - \lambda)\nu_2, \nu_T)$  for all  $\lambda \in [0, 1]$ . By [Nutz \(2021\)](#) Lemma 1.3, the KL divergence is jointly convex. Therefore, we have:

$$D_{KL}(\pi \| (\lambda\nu_1 + (1 - \lambda)\nu_2) \otimes \nu_T) \leq \lambda D_{KL}(\pi_1 \| \nu_1 \otimes \nu_T) + (1 - \lambda) D_{KL}(\pi_2 \| \nu_2 \otimes \nu_T)$$

Therefore, we have:

$$\begin{aligned} \text{EOT}(\lambda\nu_1 + (1 - \lambda)\nu_2, \nu_T) &\leq \int \tilde{c}(\xi_1, \xi_T) d\pi(\xi_1, \xi_T) + \lambda_{KL} D_{KL}(\pi \| (\lambda\nu_1 + (1 - \lambda)\nu_2) \otimes \nu_T) \\ &\leq \lambda \int \tilde{c}(\xi_1, \xi_T) d\pi_1(\xi_1, \xi_T) + (1 - \lambda) \int \tilde{c}(\xi_1, \xi_T) d\pi_2(\xi_1, \xi_T) \\ &\quad + \lambda_{KL} (\lambda D_{KL}(\pi_1 \| \nu_1 \otimes \nu_T) + (1 - \lambda) D_{KL}(\pi_2 \| \nu_2 \otimes \nu_T)) \\ &= \lambda \text{EOT}(\nu_1, \nu_T) + (1 - \lambda) \text{EOT}(\nu_2, \nu_T) \end{aligned}$$

See [Goldfeld et al. \(2024\)](#) Lemma E.23 for the directional derivative. □

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<sup>39</sup>The left part is the (unnormalized) pairwise Markov random field [Wainwright et al. \(2008\)](#).



## B.4 Proofs in Section 4

### B.4.1 Proof of Lemma 2

**Proof of Lemma 2.** By the continuity of  $\mathbb{E}_F[m(U; \theta, v(\alpha))]$ ,  $\mathcal{A}_I, \hat{\mathcal{A}}_I$  are closed. By [Rudin et al. \(1976\)](#) Theorem 2.35, a closed subset of a compact set is compact. Therefore,  $\mathcal{A}_I, \hat{\mathcal{A}}_I$  are compact as  $\mathcal{A}$  is compact (Lemmas 8(i) and 9(i)). For the nonempty part, see the proof of Theorem 5.  $\square$

### B.4.2 Proof of Theorem 5

**Proof of Theorem 5(i).** 1. Since  $\epsilon_n \geq \|P_0 - P_n\|_\infty$ , we have  $\mathcal{A}_I \subseteq \hat{\mathcal{A}}_I$ .

2. If  $\mathcal{A}_I = \mathcal{A}$ , the result is trivial.

3. Suppose  $\mathcal{A}_I \neq \mathcal{A}$ . By Assumption 3(v), we have for some  $\delta(\varepsilon) > 0$ :

$$\inf_{d(\alpha, \mathcal{A}_I) > \varepsilon} \|\mathbb{E}_F[m(U; \theta, v(\alpha))] - P_n\|_\infty \geq \inf_{d(\alpha, \mathcal{A}_I) > \varepsilon} \|\mathbb{E}_F[m(U; \theta, v(\alpha))] - P_0 + o_p(1)\|_\infty \geq \delta(\varepsilon) + o_p(1)$$

Similarly, we have:

$$\sup_{\hat{\mathcal{A}}_I} \|\mathbb{E}_F[m(U; \theta, v(\alpha))] - P_0\|_\infty \leq \sup_{\hat{\mathcal{A}}_I} \|\mathbb{E}_F[m(U; \theta, v(\alpha))] - P_n + o_p(1)\|_\infty \leq \frac{c_n}{\sqrt{n}} + o_p(1) = o_p(1)$$

Therefore, with probability approaching 1:

$$\sup_{\hat{\mathcal{A}}_I} \|\mathbb{E}_F[m(U; \theta, v(\alpha))] - P_0\|_\infty < \delta(\varepsilon) \leq \inf_{d(\alpha, \mathcal{A}_I) > \varepsilon} \|\mathbb{E}_F[m(U; \theta, v(\alpha))] - P_n\|_\infty$$

which implies that: with probability approaching 1,  $\hat{\mathcal{A}}_I \cap \{\alpha : d(\alpha, \mathcal{A}_I) > \varepsilon\} = \emptyset$ . Thus,  $\hat{\mathcal{A}}_I \subseteq \{\alpha : d(\alpha, \mathcal{A}_I) \leq \varepsilon\}$  with probability approaching 1. Therefore, we have with probability approaching 1,  $d_H(\hat{\mathcal{A}}_I, \mathcal{A}_I) \leq \varepsilon$ . As  $\varepsilon$  is arbitrary,  $d_H(\hat{\mathcal{A}}_I, \mathcal{A}_I) = o_p(1)$ .  $\square$

**Lemma 14** (Existence of a Polynomial Minorant). *Under Assumptions 3, 4, and 5, we have for  $\forall \varepsilon \in (0, 1)$  there exists  $(\kappa_\varepsilon, n_\varepsilon)$  such that for all  $n \geq n_\varepsilon$ , we have:*

$$\|\mathbb{E}_F[m(U; \theta, v(\alpha))] - P_n\|_\infty \geq \frac{C_1}{2} \min\{C_2, d(\alpha, \mathcal{A}_I)\}$$

*uniformly on  $\{\alpha : d(\alpha, \mathcal{A}_I) \geq \frac{\kappa_\varepsilon}{\sqrt{n}}\}$  with probability at least  $1 - \varepsilon$ .*

*Proof.* Note that:

$$\begin{aligned}
\sqrt{n}\|\mathbb{E}_F[m(U; \theta, v(\alpha))] - P_n\|_\infty &= \|\sqrt{n}(\mathbb{E}_F[m(U; \theta, v(\alpha))] - P_0) + \sqrt{n}(P_0 - P_n)\|_\infty \\
&\geq \left| \|\sqrt{n}(\mathbb{E}_F[m(U; \theta, v(\alpha))] - P_0)\|_\infty - \|\sqrt{n}(P_0 - P_n)\|_\infty \right| \\
&= \left| \|\sqrt{n}(\mathbb{E}_F[m(U; \theta, v(\alpha))] - P_0)\|_\infty - O_p(1) \right|
\end{aligned}$$

where we used  $\|x+y\| \geq \left| \|x\| - \|y\| \right|$  and  $\sqrt{n}(P_0 - P_n) = O_p(1)$ . Therefore, for  $\forall \varepsilon$  there exists  $M_\varepsilon > 0$  and  $n_{\varepsilon,1}$  such that for all  $n \geq n_{\varepsilon,1}$ , with probability at least  $1 - \varepsilon$ :  $|O_p(1)| \leq M_\varepsilon$ . Choose  $(\kappa_\varepsilon, n_\varepsilon)$  such that  $n_\varepsilon \geq n_{\varepsilon,1}$ ,  $C_1\kappa_\varepsilon \geq 2M_\varepsilon$ , and  $\sqrt{n}C_1C_2 \geq 2M_\varepsilon$  for all  $n \geq n_\varepsilon$ . Then, with probability at least  $1 - \varepsilon$ : uniformly on  $\{\alpha : d(\alpha, \mathcal{A}_I) \geq \frac{\kappa_\varepsilon}{\sqrt{n}}\}$ :

$$\begin{aligned}
\|\sqrt{n}(\mathbb{E}_F[m(U; \theta, v(\alpha))] - P_0)\|_\infty - O_p(1) &\geq \|\sqrt{n}(\mathbb{E}_F[m(U; \theta, v(\alpha))] - P_0)\|_\infty - M_\varepsilon \\
&\geq \sqrt{n}C_1 \min\{C_2, d(\alpha, \mathcal{A}_I)\} - M_\varepsilon \\
&\geq \frac{1}{2}\sqrt{n}C_1 \min\{C_2, d(\alpha, \mathcal{A}_I)\} + \frac{1}{2}\sqrt{n}C_1 \min\{C_2, \frac{\kappa_\varepsilon}{\sqrt{n}}\} - M_\varepsilon \\
&\geq \frac{1}{2}\sqrt{n}C_1 \min\{C_2, d(\alpha, \mathcal{A}_I)\}
\end{aligned}$$

Therefore, with probability at least  $1 - \varepsilon$ : uniformly on  $\{\alpha : d(\alpha, \mathcal{A}_I) \geq \frac{\kappa_\varepsilon}{\sqrt{n}}\}$ :

$$\|\mathbb{E}_F[m(U; \theta, v(\alpha))] - P_n\|_\infty \geq \frac{1}{2}C_1 \min\{C_2, d(\alpha, \mathcal{A}_I)\}$$

□

**Proof of Theorem 5(ii).** For  $\forall \varepsilon > 0$ , let the positive constants  $(\kappa_\varepsilon, n_\varepsilon)$  be as specified in Lemma 14. Let  $\bar{c} := \max\{\frac{C_1}{2}\kappa_\varepsilon, c_n\}$ . Furthermore, there exists  $n_{\varepsilon'} \geq n_\varepsilon$  such that for all  $n \geq n_{\varepsilon'}$ , with probability at least  $1 - \varepsilon$ :  $\varepsilon_n := \frac{\bar{c}}{\frac{C_1}{2}\sqrt{n}} \leq C_2$  as  $\frac{c_n}{\sqrt{n}} = o_p(1)$ , and  $\varepsilon_n \geq \frac{\kappa_\varepsilon}{\sqrt{n}}$ . Therefore, with probability at least  $1 - \varepsilon$ , for all  $n \geq n_{\varepsilon'}$ :

$$\inf_{\alpha, d(\alpha, \mathcal{A}_I) \geq \varepsilon_n} \|\mathbb{E}_F[m(U; \theta, v(\alpha))] - P_n\|_\infty \geq \frac{C_1}{2} \min\{C_2, d(\alpha, \mathcal{A}_I)\} \geq \frac{C_1}{2} \min\{C_2, \varepsilon_n\} \geq \frac{C_1}{2} \varepsilon_n = \frac{\bar{c}}{\sqrt{n}} \geq \frac{c_n}{\sqrt{n}}$$

Therefore, combining the first part in Theorem 5(i) we have: with probability at least  $1 - \varepsilon$ :  $\mathcal{A}_I \subseteq \hat{\mathcal{A}}_I \subseteq \{\alpha | d(\alpha, \mathcal{A}_I) \leq \varepsilon_n\}$ . Thus,  $d_H(\hat{\mathcal{A}}_I, \mathcal{A}_I) \leq \varepsilon_n$ . Therefore, we have: for  $\forall \varepsilon > 0$ , there exists  $n_{\varepsilon'}$  such that for all  $n \geq n_{\varepsilon'}$ , we have: with probability at least  $1 - \varepsilon$ :  $d_H(\hat{\mathcal{A}}_I, \mathcal{A}_I) \leq \varepsilon_n$ . As  $\varepsilon$  is arbitrary, we have  $\left| \frac{d_H(\hat{\mathcal{A}}_I, \mathcal{A}_I)}{\frac{\max\{1, c_n\}}{\sqrt{n}}} \right| \leq \frac{\max\{\frac{C_1}{2}\kappa_\varepsilon, c_n\}}{\max\{1, c_n\} \frac{C_1}{2}} := M_{1,\varepsilon}$  with probability at least  $1 - \varepsilon$ . Therefore, we have  $d_H(\hat{\mathcal{A}}_I, \mathcal{A}_I) = O_p(\frac{\max\{1, c_n\}}{\sqrt{n}})$ . □

### B.4.3 Proof of Theorem 6

*Proof.* It is a direct consequence of Lemma 5 and Theorem 5 □

### B.4.4 Proof of Theorem 7

**Proof of Theorem 7.** We verify conditions in [Bonnans and Shapiro \(2013\)](#) Theorem 4.25.

We first verify the setting in [Bonnans and Shapiro \(2013\)](#) Page 260. By [Bogachev and Ruas \(2007\)](#) Theorem 4.6.1, all real countably additive measures on  $(\mathcal{U}, \mathcal{B}(\mathcal{U}))$  with the variation norm forms a Banach space. Therefore, the product of  $\mathbb{R}^{d_\theta}$  and all real countably additive measures with the variation norm plus the Euclidean norm also forms a Banach space. Therefore, the setting is satisfied.

Moreover, by Lemmas 8(i), 8(ii), 9(i), and 9(ii), the sets  $\mathcal{F}$  and  $\mathcal{F}_{\text{relaxed}}$  are closed and convex. By Assumption 3(i),  $\Theta$  is also closed (as it is compact) and convex. Therefore,  $\mathcal{A}$  is closed and convex. According to [Bonnans and Shapiro \(2013\)](#) equation (2.194), the Robinson's constraint qualification (defined in their equation (2.163)) is equivalent to Assumption 8(i). In their notation,  $G(\alpha, P) := P(\alpha) - P$  and  $f(\alpha, P) = s(\alpha)$ . Thus, the Robinson's constraint qualification holds.

By [Bonnans and Shapiro \(2013\)](#) Theorem 4.9. The Robinson's constraint qualification implies the directional regularity condition for any direction.

Therefore, by [Bonnans and Shapiro \(2013\)](#) Theorem 4.25, the maps  $\kappa(\delta, P)$  and  $\tilde{\kappa}_{\text{TI}}(\delta, P)$  are Hadamard directionally differentiable at  $P_0$ , and note that  $D_P \mathcal{L}(\alpha, \lambda, P_0)h = -\lambda^T h$  where  $\mathcal{L}(\alpha, \lambda, P) := s(\alpha) + \lambda^T (P(\alpha) - P)$ .

The asymptotic distribution follows [Fang and Santos \(2019\)](#) Theorem 2.1. □

## B.5 Proofs in Section 5

### B.5.1 Proof of Theorem 8

**Proof of Theorem 8.** The quantization rate is in [Eckstein and Nutz \(2024\)](#) Remark 2.1. Other parts follow directly from [Eckstein and Nutz \(2024\)](#) Theorem 3.1(i). □

### B.5.2 Proof of Theorem 9

**Lemma 15.** *Under Assumptions 3 and 9(ii),  $\mathcal{A}_{I, \text{Lip}}^\delta$  is compact.*

*Proof.* By Assumption 9(ii),  $\Pi_{\text{TH}}$  and  $\Pi_{\text{TI}}$  are both tight, and by Prokhorov's theorem, they have compact closure. By passing to the limit in the equation for marginals,  $\Pi_{\text{TH}}$  and  $\Pi_{\text{TI}}$  are closed, which implies  $\mathcal{A}_{\text{Lip}}^\delta = \{F \in \mathcal{P}(\mathcal{U}) \mid F \in \Pi, D_{KL}(F\|F_0) \leq \delta\}$  is closed where  $\Pi$  is either  $\Pi_{\text{TH}}$  or  $\Pi_{\text{TI}}$ . By Lemma 6(i) and [Rudin et al. \(1976\)](#) Theorem 2.35,  $\mathcal{A}_{\text{Lip}}^\delta$  is compact. By Assumption 3(v),  $\mathcal{A}_{\text{Lip}}^\delta$  is closed. Therefore, it is compact.  $\square$

**Lemma 16.** *Let  $\Pi$  be either  $\Pi_{\text{TH}}$  or  $\Pi_{\text{TI}}$ . Under Assumption 9,  $D_{KL}(F\|F_0)$  is continuously differentiable on  $\{F \in \Pi \mid D_{KL}(F\|F_0) \leq \delta\}$ .*

*Proof.* The directional derivative<sup>40</sup> of  $D_{KL}(F\|F_0)$  in the direction  $F_1$  is:

$$D_F D_{KL}(F\|F_0)(F_1 - F) = -D_{KL}(F\|F_0) + \int \log \frac{dF}{dF_0} dF_1$$

which is linear in  $F_1$ . Under Assumption 9, we have:

$$\left| \log \frac{dF(U)}{dF_0(U)} - \log \frac{dF(U')}{dF_0(U')} \right| \leq \frac{1}{C_3} (|dF(U') - dF(U)| + |dF_0(U') - dF_0(U)|) \leq \frac{2L}{C_3} \|U' - U\|$$

Therefore,  $\log \frac{dF(U)}{dF_0(U)}$  is Lipschitz continuous in  $U$ . Moreover, as  $\log \frac{dF(U)}{dF_0(U)}$  is bounded from above, we have  $\log \frac{dF}{dF_0} \in L^1(F_2)$  for any  $F_2 \in \Pi$ . By the Kantorovich-Rubinstein duality<sup>41</sup>, it holds that:

$$\left| \int \log \frac{dF}{dF_0} (dF_2 - dF_3) \right| \leq \frac{2L}{C_3} W_1(F_2, F_3)$$

which implies the directional derivative is continuous in  $F$  as  $W_1$  is a metric on  $\mathcal{P}(\mathcal{U})$ , (see [Villani \(2021\)](#) Remark 7.13(iii)), and thus Gâteaux differentiable.

Moreover, for any  $F_1, F_2, F \in \Pi$ , let  $\|\cdot\|_{TV}$  be the total variation norm, we have:

$$\left| \int \log \frac{dF_1}{dF_2} dF \right| \leq \frac{C_4}{C_3} \int |dF_1(U) - dF_2(U)| dU = \frac{2C_4}{C_3} \|F_1 - F_2\|_{TV}$$

Furthermore, note that:

$$\begin{aligned} & |D_{KL}(F_1\|F_0) - D_{KL}(F_2\|F_0)| \\ & \leq \left| \int dF_1 \log dF_1 - \int dF_2 \log dF_2 \right| + \left| \int (dF_1 - dF_2) \log dF_0 \right| \\ & = \left| \int dF_1 (\log dF_1 - \log dF_2) + \int (dF_1 - dF_2) \log dF_2 \right| + \left| \int (dF_1 - dF_2) \log dF_0 \right| \end{aligned}$$

<sup>40</sup>See [Nutz \(2021\)](#) equation 1.10.

<sup>41</sup>See [Villani \(2021\)](#). The cost function is  $c(U, U') = \|U' - U\|$ .

$$\leq \frac{2C_4}{C_3} \|F_1 - F_2\|_{TV} + 4|\log C_4| \|F_1 - F_2\|_{TV} = \left( \frac{2C_4}{C_3} + 4|\log C_4| \right) \|F_1 - F_2\|_{TV}$$

Therefore, we have:

$$\|D_{F_1} D_{KL}(F_1 \| F_0) - D_{F_2} D_{KL}(F_2 \| F_0)\|_{op} \leq \left( \frac{4C_4}{C_3} + 4|\log C_4| \right) \|F_1 - F_2\|_{TV}$$

which implies its Gâteaux derivative is also continuous in  $F$  in the operator norm topology. Therefore,  $D_{KL}(F \| F_0)$  is continuously differentiable on  $\{F \in \Pi | D_{KL}(F \| F_0) \leq \delta\}$ .  $\square$

**Proof of Theorem 9.** Consider the following optimization problem:

$$\kappa(\delta, P_0) = \inf_{\alpha \in \Theta \times \Pi_{TH}} s(\alpha) \quad \text{s.t.} \quad P(\alpha) = P_0, \quad D_{KL}(F \| F_0) \leq \delta$$

Note that  $\Pi_{TH}$  is convex and closed,  $\mathcal{A}_{I,Lip}^\delta$  is compact by Lemma 15, and  $s(\alpha)$  is continuous. By the extreme value theorem, the infimum is achieved. Therefore,  $\mathcal{A}_{I,Lip}^{\delta,*}$  is nonempty.

We aim to show the Hadamard directionally differentiability of  $\kappa(\delta, P_0)$  in the directional  $d := (1, 0^{d_P})$ . We will verify the conditions in [Bonnans and Shapiro \(2013\)](#) Theorem 4.25. In particular, we verify the directional regularity condition in the direction  $d := (1, 0^{d_P})$  for all  $\alpha^* \in \mathcal{A}_{I,Lip}^{\delta,*}$ . By Lemma 16,  $D_{KL}(F \| F_0)$  is continuously differentiable on  $\Pi_{TH}$ . Therefore, the setting in [Bonnans and Shapiro \(2013\)](#) Page 260 is satisfied.

For  $\alpha^* = (\theta^*, F^*) \in \mathcal{A}_{I,Lip}^{\delta,*}$ , by [Bonnans and Shapiro \(2013\)](#) Theorem 4.9, the directional regularity condition in the direction  $d$  is equivalent to:

$$0 \in \text{int} \left\{ \begin{pmatrix} D_{KL}(F^* \| F_0) - \delta \\ P(\alpha^*) - P_0 \end{pmatrix} + \begin{pmatrix} D_{F^*} D_{KL}(F^* \| F_0)(\Pi - F^*) - [0, +\infty) \\ DP(\alpha^*)(\Theta \times \Pi - \alpha^*) - 0^{d_P} \end{pmatrix} - \begin{pmatrix} (-\infty, 0] \\ 0^{d_P} \end{pmatrix} \right\}$$

By Assumption 9(iv), the second part is satisfied. The first part is straightforward as  $-[0, +\infty) - (-\infty, 0] = \mathbb{R}$ .

Assumption 9(v) is the same as Assumption (iii) of Theorem 4.25 in [Bonnans and Shapiro \(2013\)](#). Moreover, Assumption 9(vi) is the same as Assumption (iv) of Theorem 4.25 in [Bonnans and Shapiro \(2013\)](#). Therefore, [Bonnans and Shapiro \(2013\)](#) Theorem 4.25 shows the right differentiability of  $\kappa(\delta, P_0)$ .

The proof for  $\tilde{\kappa}_{TI}(\delta, P_0)$  is the same.  $\square$

## B.6 Proof in Section 7

### B.6.1 Proof of Lemma 3

*Proof of Lemma 3.* (i) See [Rust et al. \(2002\)](#) Theorem 1.

- (ii) • ( $\Leftarrow$ ) Suppose  $V^*(\omega)$  is the unique fixed point of (9).

**(Existence)** Note that  $s_0(\omega) := 1 - \exp(\omega - V^*(\omega))$  is a fixed point of (15).

**(Uniqueness)** Suppose  $s_0(\omega)$  is a fixed point of (15). Let  $V(\omega) := \omega - \log(1 - s_0(\omega))$ . Then, we show that  $V(\omega)$  is a fixed point of (9). It suffices to show that:

$$\beta \mathbb{E}[V(\omega')|\omega] = \log(\exp(V(\omega)) - \exp(\omega))$$

where the right-hand Side equals to  $\omega + \log\left(\frac{s_0(\omega)}{1-s_0(\omega)}\right)$ , and the left-hand side equals to  $\beta \mathbb{E}[\omega' - \log(1 - s_0(\omega'))|\omega]$ . Since  $s_0(\omega)$  is a fixed point of (15), we have RHS = LHS. Therefore,  $V(\omega)$  is a fixed point of (9).

Prove by contradiction. Suppose there are two fixed points  $s_0(\omega)$  and  $\tilde{s}_0(\omega)$  of (15). Then, we have:  $V(\omega) = \omega - \log(1 - s_0(\omega))$ ,  $\tilde{V}(\omega) = \omega - \log(1 - \tilde{s}_0(\omega))$  are both the fixed points to (9) as shown above. Since (9) has a unique fixed point, we have  $V(\omega) = \tilde{V}(\omega)$  for all  $\omega$ . Therefore, we have  $s_0(\omega) = \tilde{s}_0(\omega)$  for all  $\omega$ . Contradiction.

- ( $\Rightarrow$ ) Suppose (15) has a unique fixed point  $s_0^*(\omega)$ .

**(Existence)** Note that  $V(\omega) := \omega - \log(1 - s_0^*(\omega))$  is a fixed point of (9).

**(Uniqueness)** Suppose  $V(\omega)$  is a fixed point. Then, we show that  $s_0(\omega) := 1 - \exp(\omega - V(\omega))$  is a fixed point of (15). As  $V(\omega) = \omega - \log(1 - s_0(\omega))$  is a fixed point of (9), we have:  $\exp(\omega - \log(1 - s_0(\omega))) = \exp(\omega) + \exp(\beta \mathbb{E}[\omega' - \log(1 - s_0(\omega'))|\omega])$ . It implies  $\frac{1}{1-s_0(\omega)} = 1 + \exp(\beta \mathbb{E}[\omega' - \log(1 - s_0(\omega'))|\omega] - \omega)$ . Rearranging it shows that  $s_0(\omega)$  is a fixed point of (15).

Prove by contradiction. Suppose there are two fixed points  $V_1(\omega)$  and  $V_2(\omega)$  of (9). Then, we have:  $s_0(\omega) = 1 - \exp(\omega - V_1(\omega))$ ,  $\tilde{s}_0(\omega) = 1 - \exp(\omega - V_2(\omega))$  are both the fixed points to (15) as shown above. Since (15) has a unique fixed point, we have  $s_0(\omega) = \tilde{s}_0(\omega)$  for all  $\omega$ . Therefore, we have  $V_1(\omega) = V_2(\omega)$  for all  $\omega$ . Contradiction.

- (iii) This is a direct consequence of the above argument.

□